Linear Algebra Tutorials

Questions and Complete Solutions

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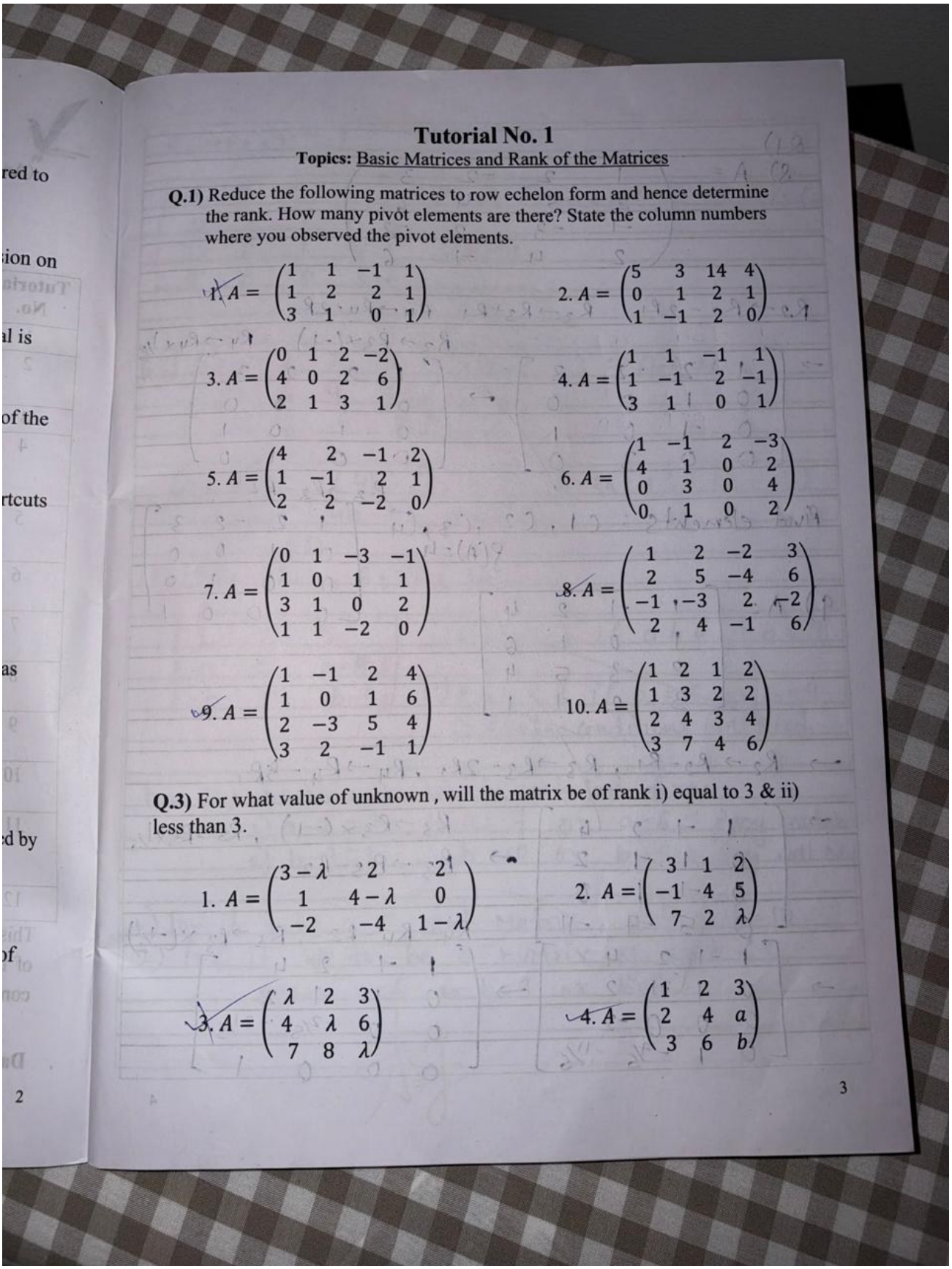
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Tutorial No. 1: Basic Matrices and Rank of the Matrices

Questions



Tutorial No. 1

Topics: Basic Matrices and Rank of the Matrices

Q.1) Reduce the following matrices to row echelon form and hence determine the rank. How many pivot elements are there? State the column numbers where you observed the pivot elements.

1. $A = \begin{pmatrix} 1 & 1 & -1 & 1 \\ 1 & 2 & 2 & 1 \\ 3 & 1 & 0 & 1 \end{pmatrix}$

2. $A = \begin{pmatrix} 5 & 3 & 14 & 4 \\ 0 & 1 & 2 & 1 \\ 1 & -1 & 2 & 0 \end{pmatrix}$

3. $A = \begin{pmatrix} 0 & 1 & 2 & -2 \\ 4 & 0 & 2 & 6 \\ 2 & 1 & 3 & 1 \end{pmatrix}$

4. $A = \begin{pmatrix} 1 & 1 & -1 & 1 \\ 1 & -1 & 2 & -1 \\ 3 & 1 & 0 & 1 \end{pmatrix}$

5. $A = \begin{pmatrix} 4 & 2 & -1 & 2 \\ 1 & -1 & 2 & 1 \\ 2 & 2 & -2 & 0 \end{pmatrix}$

6. $A = \begin{pmatrix} 1 & -1 & 2 & -3 \\ 4 & 1 & 0 & 2 \\ 0 & 3 & 0 & 4 \\ 0 & 1 & 0 & 2 \end{pmatrix}$

7. $A = \begin{pmatrix} 0 & 1 & -3 & -1 \\ 1 & 0 & 1 & 1 \\ 3 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 \end{pmatrix}$

8. $A = \begin{pmatrix} 1 & 2 & -2 & 3 \\ 2 & 5 & -4 & 6 \\ -1 & -3 & 2 & -2 \\ 2 & 4 & -1 & 6 \end{pmatrix}$

9. $A = \begin{pmatrix} 1 & -1 & 2 & 4 \\ 1 & 0 & 1 & 6 \\ 2 & -3 & 5 & 4 \\ 3 & 2 & -1 & 1 \end{pmatrix}$

10. $A = \begin{pmatrix} 1 & 2 & 1 & 2 \\ 1 & 3 & 2 & 2 \\ 2 & 4 & 3 & 4 \\ 3 & 7 & 4 & 6 \end{pmatrix}$

Q.3) For what value of unknown , will the matrix be of rank i) equal to 3 & ii) less than 3.

1. $A = \begin{pmatrix} 3-\lambda & 2 & 2 \\ 1 & 4-\lambda & 0 \\ -2 & -4 & 1-\lambda \end{pmatrix}$

2. $A = \begin{pmatrix} 3 & 1 & 2 \\ -1 & 4 & 5 \\ 7 & 2 & \lambda \end{pmatrix}$

3. $A = \begin{pmatrix} \lambda & 2 & 3 \\ 4 & \lambda & 6 \\ 7 & 8 & \lambda \end{pmatrix}$

4. $A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & a \\ 3 & 6 & b \end{pmatrix}$

Solutions — Tutorial No. 1

Q.1 Row echelon form, rank, pivots

(1)

$$A = \begin{pmatrix} 1 & 1 & -1 & 1 \\ 1 & 2 & 2 & 1 \\ 3 & 1 & 0 & 1 \end{pmatrix}$$
$$\xrightarrow[R_3 \to R_3 - 3R_1]{R_2 \to R_2 - R_1} \begin{pmatrix} 1 & 1 & -1 & 1 \\ 0 & 1 & 3 & 0 \\ 0 & -2 & 3 & -2 \end{pmatrix}$$
$$\xrightarrow{R_3 \to R_3 + 2R_2} \begin{pmatrix} 1 & 1 & -1 & 1 \\ 0 & 1 & 3 & 0 \\ 0 & 0 & 9 & -2 \end{pmatrix}$$

Row echelon form has 3 non-zero row(s): $\mathbf{rank}(A) = 3$; number of pivots = 3, located in column(s) 1, 2, 3.

(2)

$$A = \begin{pmatrix} 5 & 3 & 14 & 4 \\ 0 & 1 & 2 & 1 \\ 1 & -1 & 2 & 0 \end{pmatrix}$$
$$\xrightarrow{R_3 \to 5R_3 - R_1} \begin{pmatrix} 5 & 3 & 14 & 4 \\ 0 & 1 & 2 & 1 \\ 0 & -8 & -4 & -4 \end{pmatrix}$$
$$\xrightarrow{R_3 \to R_3 + 8R_2} \begin{pmatrix} 5 & 3 & 14 & 4 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 12 & 4 \end{pmatrix}$$

Row echelon form has 3 non-zero row(s): $\mathbf{rank}(A) = 3$; number of pivots = 3, located in column(s) 1, 2, 3.

(3)

$$A = \begin{pmatrix} 0 & 1 & 2 & -2 \\ 4 & 0 & 2 & 6 \\ 2 & 1 & 3 & 1 \end{pmatrix}$$
$$\xrightarrow[R_3 \to 2R_3 - R_1]{R_1 \leftrightarrow R_2} \begin{pmatrix} 4 & 0 & 2 & 6 \\ 0 & 1 & 2 & -2 \\ 0 & 2 & 4 & -4 \end{pmatrix}$$
$$\xrightarrow{R_3 \to R_3 - 2R_2} \begin{pmatrix} 4 & 0 & 2 & 6 \\ 0 & 1 & 2 & -2 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Row echelon form has 2 non-zero row(s): $\mathbf{rank}(A) = 2$; number of pivots = 2, located in column(s) 1, 2.

(4)

$$A = \begin{pmatrix} 1 & 1 & -1 & 1 \\ 1 & -1 & 2 & -1 \\ 3 & 1 & 0 & 1 \end{pmatrix}$$
$$\xrightarrow[R_3 \to R_3 - 3R_1]{R_2 \to R_2 - R_1} \begin{pmatrix} 1 & 1 & -1 & 1 \\ 0 & -2 & 3 & -2 \\ 0 & -2 & 3 & -2 \end{pmatrix}$$

$$\xrightarrow{R_3 \to R_3 - R_2} \begin{pmatrix} 1 & 1 & -1 & 1 \\ 0 & -2 & 3 & -2 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Row echelon form has 2 non-zero row(s): $\mathbf{rank}(A) = 2$; number of pivots = 2, located in column(s) 1, 2.

(5)

$$A = \begin{pmatrix} 4 & 2 & -1 & 2 \\ 1 & -1 & 2 & 1 \\ 2 & 2 & -2 & 0 \end{pmatrix}$$
$$\xrightarrow[R_3 \to 2R_3 - R_1]{R_2 \to 4R_2 - R_1} \begin{pmatrix} 4 & 2 & -1 & 2 \\ 0 & -6 & 9 & 2 \\ 0 & 2 & -3 & -2 \end{pmatrix}$$
$$\xrightarrow{R_3 \to 3R_3 + R_2} \begin{pmatrix} 4 & 2 & -1 & 2 \\ 0 & -6 & 9 & 2 \\ 0 & 0 & 0 & -4 \end{pmatrix}$$

Row echelon form has 3 non-zero row(s): $\mathbf{rank}(A) = 3$; number of pivots = 3, located in column(s) 1, 2, 4.

(6)

$$A = \begin{pmatrix} 1 & -1 & 2 & -3 \\ 4 & 1 & 0 & 2 \\ 0 & 3 & 0 & 4 \\ 0 & 1 & 0 & 2 \end{pmatrix}$$
$$\xrightarrow{R_2 \to R_2 - 4R_1} \begin{pmatrix} 1 & -1 & 2 & -3 \\ 0 & 5 & -8 & 14 \\ 0 & 3 & 0 & 4 \\ 0 & 1 & 0 & 2 \end{pmatrix}$$
$$\xrightarrow[R_4 \to 5R_4 - R_2]{R_3 \to 5R_3 - 3R_2} \begin{pmatrix} 1 & -1 & 2 & -3 \\ 0 & 5 & -8 & 14 \\ 0 & 0 & 24 & -22 \\ 0 & 0 & 8 & -4 \end{pmatrix}$$
$$\xrightarrow{R_4 \to 3R_4 - R_3} \begin{pmatrix} 1 & -1 & 2 & -3 \\ 0 & 5 & -8 & 14 \\ 0 & 0 & 24 & -22 \\ 0 & 0 & 0 & 10 \end{pmatrix}$$

Row echelon form has 4 non-zero row(s): $\mathbf{rank}(A) = 4$; number of pivots = 4, located in column(s) 1, 2, 3, 4.

(7)

$$A = \begin{pmatrix} 0 & 1 & -3 & -1 \\ 1 & 0 & 1 & 1 \\ 3 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 \end{pmatrix}$$
$$\xrightarrow[\substack{R_3 \to R_3 - 3R_1 \\ R_4 \to R_4 - R_1}]{R_1 \leftrightarrow R_2} \begin{pmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & -3 & -1 \\ 0 & 1 & -3 & -1 \\ 0 & 1 & -3 & -1 \end{pmatrix}$$
$$\xrightarrow[R_4 \to R_4 - R_2]{R_3 \to R_3 - R_2} \begin{pmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & -3 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Row echelon form has 2 non-zero row(s): $\mathbf{rank}(A) = 2$; number of pivots = 2, located in column(s) 1, 2.

(8)

$$A = \begin{pmatrix} 1 & 2 & -2 & 3 \\ 2 & 5 & -4 & 6 \\ -1 & -3 & 2 & -2 \\ 2 & 4 & -1 & 6 \end{pmatrix}$$
$$\xrightarrow[R_4 \to R_4 - 2R_1]{\substack{R_2 \to R_2 - 2R_1 \\ R_3 \to R_3 + R_1}} \begin{pmatrix} 1 & 2 & -2 & 3 \\ 0 & 1 & 0 & 0 \\ 0 & -1 & 0 & 1 \\ 0 & 0 & 3 & 0 \end{pmatrix}$$
$$\xrightarrow{R_3 \to R_3 + R_2} \begin{pmatrix} 1 & 2 & -2 & 3 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 3 & 0 \end{pmatrix}$$
$$\xrightarrow{R_3 \leftrightarrow R_4} \begin{pmatrix} 1 & 2 & -2 & 3 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Row echelon form has 4 non-zero row(s): $\mathbf{rank}(A) = 4$; number of pivots = 4, located in column(s) 1, 2, 3, 4.

(9)

$$A = \begin{pmatrix} 1 & -1 & 2 & 4 \\ 1 & 0 & 1 & 6 \\ 2 & -3 & 5 & 4 \\ 3 & 2 & -1 & 1 \end{pmatrix}$$
$$\xrightarrow[R_4 \to R_4 - 3R_1]{\substack{R_2 \to R_2 - R_1 \\ R_3 \to R_3 - 2R_1}} \begin{pmatrix} 1 & -1 & 2 & 4 \\ 0 & 1 & -1 & 2 \\ 0 & -1 & 1 & -4 \\ 0 & 5 & -7 & -11 \end{pmatrix}$$
$$\xrightarrow[R_4 \to R_4 - 5R_2]{\substack{R_3 \to R_3 + R_2}} \begin{pmatrix} 1 & -1 & 2 & 4 \\ 0 & 1 & -1 & 2 \\ 0 & 0 & 0 & -2 \\ 0 & 0 & -2 & -21 \end{pmatrix}$$
$$\xrightarrow{R_3 \leftrightarrow R_4} \begin{pmatrix} 1 & -1 & 2 & 4 \\ 0 & 1 & -1 & 2 \\ 0 & 0 & -2 & -21 \\ 0 & 0 & 0 & -2 \end{pmatrix}$$

Row echelon form has 4 non-zero row(s): $\mathbf{rank}(A) = 4$; number of pivots = 4, located in column(s) 1, 2, 3, 4.

(10)

$$A = \begin{pmatrix} 1 & 2 & 1 & 2 \\ 1 & 3 & 2 & 2 \\ 2 & 4 & 3 & 4 \\ 3 & 7 & 4 & 6 \end{pmatrix}$$
$$\xrightarrow[R_4 \to R_4 - 3R_1]{\substack{R_2 \to R_2 - R_1 \\ R_3 \to R_3 - 2R_1}} \begin{pmatrix} 1 & 2 & 1 & 2 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \end{pmatrix}$$

$$\xrightarrow{R_4 \to R_4 - R_2} \begin{pmatrix} 1 & 2 & 1 & 2 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Row echelon form has 3 non-zero row(s): **rank**$(A) = 3$; number of pivots = 3, located in column(s) 1, 2, 3. **Q.3 Values of the unknown for which the rank is 3 / less than 3**

A 3×3 matrix has rank 3 iff $\det \neq 0$.

(1) $\det A = -(\lambda - 3)^2 (\lambda - 2)$.

- (i) rank = 3 for $\lambda \neq 2, 3$.
- (ii) rank < 3 for $\lambda = 2$ (then rank = 2); $\lambda = 3$ (then rank = 2).

(2) $\det A = 13\lambda - 55$.

- (i) rank = 3 for $\lambda \neq \frac{55}{13}$.
- (ii) rank < 3 for $\lambda = \frac{55}{13}$ (then rank = 2).

(3) $\det A = \lambda^3 - 77\lambda + 180$. The cubic has no rational root; its real roots are $\lambda \approx -9.769, 2.554, 7.215$.

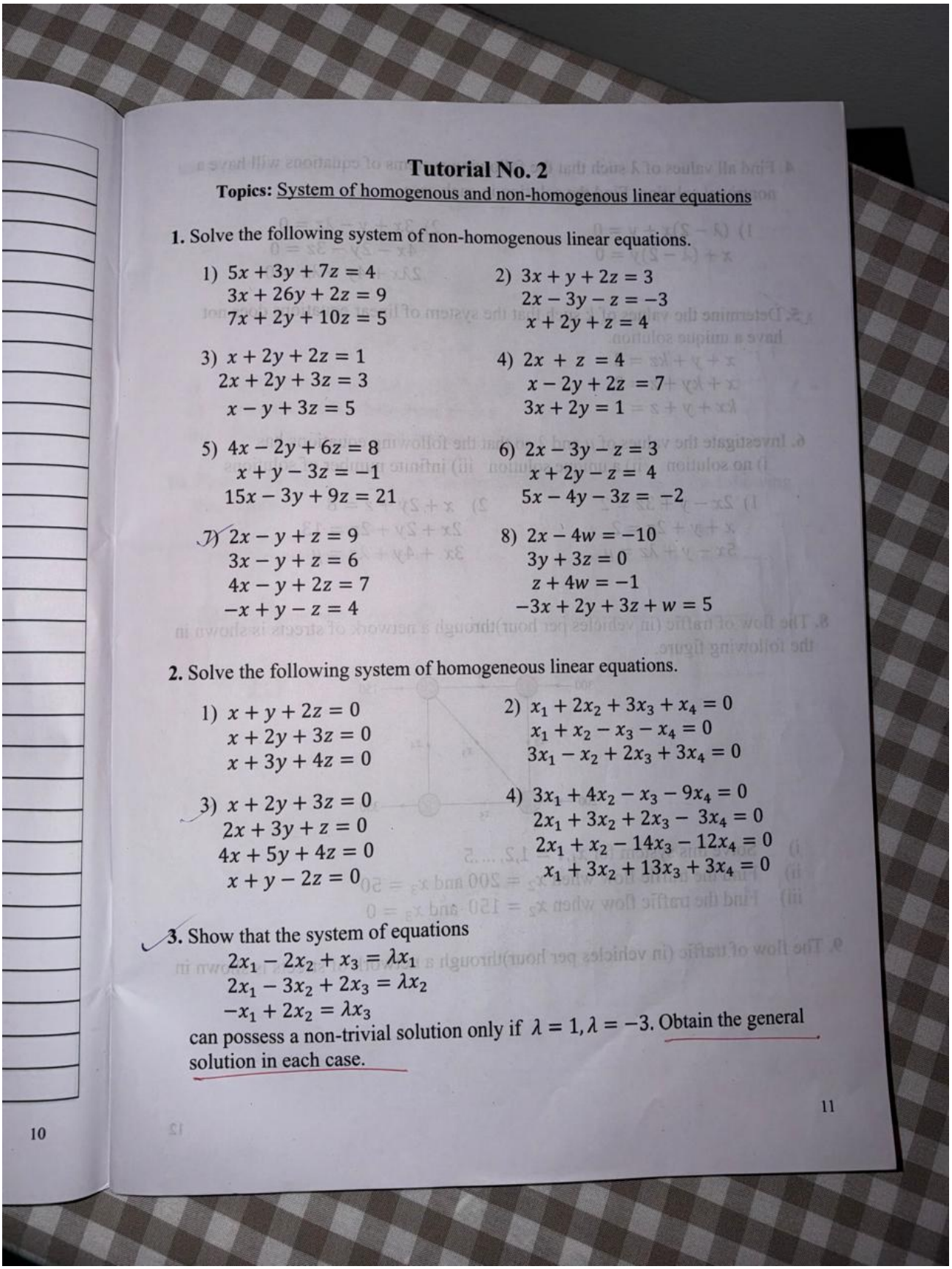
- (i) rank = 3 for every λ except these three values.
- (ii) rank < 3 exactly for these three values; there rank = 2 (e.g. the minor $\lambda^2 - 8 \neq 0$ there).

(4) $A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & a \\ 3 & 6 & b \end{pmatrix}$. $R_2 \to R_2 - 2R_1$, $R_3 \to R_3 - 3R_1$ gives $\begin{pmatrix} 1 & 2 & 3 \\ 0 & 0 & a-6 \\ 0 & 0 & b-9 \end{pmatrix}$.

- (i) rank = 3: **never** (columns 1 and 2 are always proportional, $\det A = 0$ for all a, b).
- (ii) rank < 3: for all a, b. In particular rank = 1 if $a = 6$ and $b = 9$, and rank = 2 otherwise.

Tutorial No. 2: System of Homogeneous and Non-homogeneous Linear Equations

Questions



Tutorial No. 2

Topics: System of homogenous and non-homogenous linear equations

1. Solve the following system of non-homogenous linear equations.

1) $5x + 3y + 7z = 4$
$3x + 26y + 2z = 9$
$7x + 2y + 10z = 5$

2) $3x + y + 2z = 3$
$2x - 3y - z = -3$
$x + 2y + z = 4$

3) $x + 2y + 2z = 1$
$2x + 2y + 3z = 3$
$x - y + 3z = 5$

4) $2x + z = 4$
$x - 2y + 2z = 7$
$3x + 2y = 1$

5) $4x - 2y + 6z = 8$
$x + y - 3z = -1$
$15x - 3y + 9z = 21$

6) $2x - 3y - z = 3$
$x + 2y - z = 4$
$5x - 4y - 3z = -2$

7) $2x - y + z = 9$
$3x - y + z = 6$
$4x - y + 2z = 7$
$-x + y - z = 4$

8) $2x - 4w = -10$
$3y + 3z = 0$
$z + 4w = -1$
$-3x + 2y + 3z + w = 5$

2. Solve the following system of homogeneous linear equations.

1) $x + y + 2z = 0$
$x + 2y + 3z = 0$
$x + 3y + 4z = 0$

2) $x_1 + 2x_2 + 3x_3 + x_4 = 0$
$x_1 + x_2 - x_3 - x_4 = 0$
$3x_1 - x_2 + 2x_3 + 3x_4 = 0$

3) $x + 2y + 3z = 0$
$2x + 3y + z = 0$
$4x + 5y + 4z = 0$
$x + y - 2z = 0$

4) $3x_1 + 4x_2 - x_3 - 9x_4 = 0$
$2x_1 + 3x_2 + 2x_3 - 3x_4 = 0$
$2x_1 + x_2 - 14x_3 - 12x_4 = 0$
$x_1 + 3x_2 + 13x_3 + 3x_4 = 0$

3. Show that the system of equations

$2x_1 - 2x_2 + x_3 = \lambda x_1$
$2x_1 - 3x_2 + 2x_3 = \lambda x_2$
$-x_1 + 2x_2 = \lambda x_3$

can possess a non-trivial solution only if $\lambda = 1, \lambda = -3$. Obtain the general solution in each case.

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4. Find all values of λ such that the following systems of equations will have a nontrivial solution. Find the solution in each case.

1) $(\lambda-2)x+y=0$
$x+(\lambda-2)y=0$

2) $3x+y-\lambda z=0$
$4x-2y-3z=0$
$2\lambda x+4y-\lambda z=0$

5. Determine the values of k such that the system of linear equations does not have a unique solution.

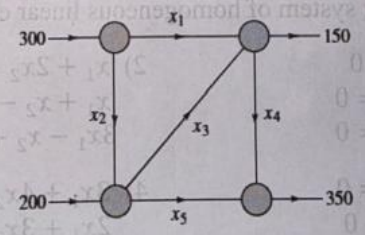
$x+y+kz=3$
$x+ky+z=2$
$kx+y+z=1$

6. Investigate the values of μ and λ so that the following equations has
i) no solution ii) a unique solution iii) infinite number of solutions

1) $2x-y+3z=2$
$x+y+2z=2$
$5x-y+\lambda z=\mu$

2) $x+2y+z=8$
$2x+2y+2z=13$
$3x+4y+\lambda z=\mu$

8. The flow of traffic (in vehicles per hour)through a network of streets is shown in the following figure.

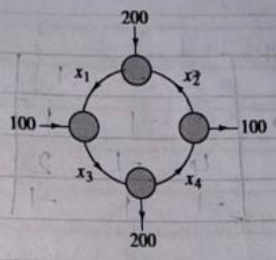


i) Solve this system for $x_i, i = 1,2,\ldots.5$
ii) Find the traffic flow when $x_2 = 200$ and $x_3 = 50$
iii) Find the traffic flow when $x_2 = 150$ and $x_3 = 0$

9. The flow of traffic (in vehicles per hour)through a network of streets is shown in

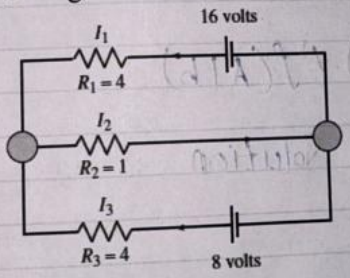
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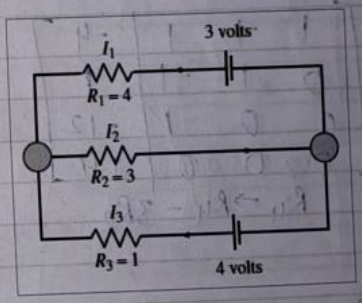
the following figure.



i) Solve this system for $x_i, i = 1,2,3,4$
ii) Find the traffic flow when $x_4 = 0$
iii) Find the traffic flow when $x_4 = 100$

10. Find the currents I_1, I_2, I_3 for the electrical networks shown in the following two figures.





Solutions — Tutorial No. 2

Q.1 Non-homogeneous systems

(1) Augmented matrix:

$$[A|b] = \left(\begin{array}{ccc|c} 5 & 3 & 7 & 4 \\ 3 & 26 & 2 & 9 \\ 7 & 2 & 10 & 5 \end{array}\right)$$
$$\xrightarrow[R_3 \to 5R_3 - 7R_1]{R_2 \to 5R_2 - 3R_1} \left(\begin{array}{ccc|c} 5 & 3 & 7 & 4 \\ 0 & 121 & -11 & 33 \\ 0 & -11 & 1 & -3 \end{array}\right)$$
$$\xrightarrow{R_3 \to 11R_3 + R_2} \left(\begin{array}{ccc|c} 5 & 3 & 7 & 4 \\ 0 & 121 & -11 & 33 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

$\text{rank}(A) = \text{rank}[A|b] = 2$, unknowns $= 3 \Rightarrow$ infinitely many solutions. Solution: $x = \frac{7}{11} - \frac{16t}{11}, \quad y = \frac{t}{11} + \frac{3}{11}, \quad z = t$, i.e.

$$\mathbf{x} = \begin{pmatrix} \frac{7}{11} \\ \frac{3}{11} \\ 0 \end{pmatrix} + t \begin{pmatrix} -\frac{16}{11} \\ \frac{1}{11} \\ 1 \end{pmatrix}, \quad t \in \mathbb{R}.$$

(2) Augmented matrix:

$$[A|b] = \left(\begin{array}{ccc|c} 3 & 1 & 2 & 3 \\ 2 & -3 & -1 & -3 \\ 1 & 2 & 1 & 4 \end{array}\right)$$
$$\xrightarrow[R_3 \to 3R_3 - R_1]{R_2 \to 3R_2 - 2R_1} \left(\begin{array}{ccc|c} 3 & 1 & 2 & 3 \\ 0 & -11 & -7 & -15 \\ 0 & 5 & 1 & 9 \end{array}\right)$$
$$\xrightarrow{R_3 \to 11R_3 + 5R_2} \left(\begin{array}{ccc|c} 3 & 1 & 2 & 3 \\ 0 & -11 & -7 & -15 \\ 0 & 0 & -24 & 24 \end{array}\right)$$

$\text{rank}(A) = \text{rank}[A|b] = 3$, unknowns $= 3 \Rightarrow$ unique solution. Solution: $x = 1, \quad y = 2, \quad z = -1$ (unique solution)

(3) Augmented matrix:

$$[A|b] = \left(\begin{array}{ccc|c} 1 & 2 & 2 & 1 \\ 2 & 2 & 3 & 3 \\ 1 & -1 & 3 & 5 \end{array}\right)$$
$$\xrightarrow[R_3 \to R_3 - R_1]{R_2 \to R_2 - 2R_1} \left(\begin{array}{ccc|c} 1 & 2 & 2 & 1 \\ 0 & -2 & -1 & 1 \\ 0 & -3 & 1 & 4 \end{array}\right)$$
$$\xrightarrow{R_3 \to 2R_3 - 3R_2} \left(\begin{array}{ccc|c} 1 & 2 & 2 & 1 \\ 0 & -2 & -1 & 1 \\ 0 & 0 & 5 & 5 \end{array}\right)$$

$\text{rank}(A) = \text{rank}[A|b] = 3$, unknowns $= 3 \Rightarrow$ unique solution. Solution: $x = 1, \quad y = -1, \quad z = 1$ (unique solution)

(4) Augmented matrix:

$$[A|b] = \left(\begin{array}{ccc|c} 2 & 0 & 1 & 4 \\ 1 & -2 & 2 & 7 \\ 3 & 2 & 0 & 1 \end{array}\right)$$

$$\xrightarrow[R_3\to 2R_3-3R_1]{R_2\to 2R_2-R_1} \left(\begin{array}{ccc|c} 2 & 0 & 1 & 4 \\ 0 & -4 & 3 & 10 \\ 0 & 4 & -3 & -10 \end{array}\right)$$

$$\xrightarrow{R_3\to R_3+R_2} \left(\begin{array}{ccc|c} 2 & 0 & 1 & 4 \\ 0 & -4 & 3 & 10 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

$\operatorname{rank}(A) = \operatorname{rank}[A|b] = 2$, unknowns $= 3 \Rightarrow$ infinitely many solutions. Solution: $x = 2 - \frac{t}{2}, \quad y = \frac{3t}{4} - \frac{5}{2}, \quad z = t$, i.e.

$$\mathbf{x} = \begin{pmatrix} 2 \\ -\frac{5}{2} \\ 0 \end{pmatrix} + t \begin{pmatrix} -\frac{1}{2} \\ \frac{3}{4} \\ 1 \end{pmatrix}, \quad t \in \mathbb{R}.$$

(5) Augmented matrix:

$$[A|b] = \left(\begin{array}{ccc|c} 4 & -2 & 6 & 8 \\ 1 & 1 & -3 & -1 \\ 15 & -3 & 9 & 21 \end{array}\right)$$

$$\xrightarrow[R_3\to 4R_3-15R_1]{R_2\to 4R_2-R_1} \left(\begin{array}{ccc|c} 4 & -2 & 6 & 8 \\ 0 & 6 & -18 & -12 \\ 0 & 18 & -54 & -36 \end{array}\right)$$

$$\xrightarrow{R_3\to R_3-3R_2} \left(\begin{array}{ccc|c} 4 & -2 & 6 & 8 \\ 0 & 6 & -18 & -12 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

$\operatorname{rank}(A) = \operatorname{rank}[A|b] = 2$, unknowns $= 3 \Rightarrow$ infinitely many solutions. Solution: $x = 1, \quad y = 3t - 2, \quad z = t$, i.e.

$$\mathbf{x} = \begin{pmatrix} 1 \\ -2 \\ 0 \end{pmatrix} + t \begin{pmatrix} 0 \\ 3 \\ 1 \end{pmatrix}, \quad t \in \mathbb{R}.$$

(6) Augmented matrix:

$$[A|b] = \left(\begin{array}{ccc|c} 2 & -3 & -1 & 3 \\ 1 & 2 & -1 & 4 \\ 5 & -4 & -3 & -2 \end{array}\right)$$

$$\xrightarrow[R_3\to 2R_3-5R_1]{R_2\to 2R_2-R_1} \left(\begin{array}{ccc|c} 2 & -3 & -1 & 3 \\ 0 & 7 & -1 & 5 \\ 0 & 7 & -1 & -19 \end{array}\right)$$

$$\xrightarrow{R_3\to R_3-R_2} \left(\begin{array}{ccc|c} 2 & -3 & -1 & 3 \\ 0 & 7 & -1 & 5 \\ 0 & 0 & 0 & -24 \end{array}\right)$$

$\operatorname{rank}(A) = 2 \neq \operatorname{rank}[A|b] = 3 \Rightarrow$ the system is **inconsistent (no solution)**.

(7) Augmented matrix:

$$[A|b] = \left(\begin{array}{ccc|c} 2 & -1 & 1 & 9 \\ 3 & -1 & 1 & 6 \\ 4 & -1 & 2 & 7 \\ -1 & 1 & -1 & 4 \end{array}\right)$$

$$\xrightarrow[\substack{R_3\to R_3-2R_1 \\ R_4\to 2R_4+R_1}]{R_2\to 2R_2-3R_1} \left(\begin{array}{ccc|c} 2 & -1 & 1 & 9 \\ 0 & 1 & -1 & -15 \\ 0 & 1 & 0 & -11 \\ 0 & 1 & -1 & 17 \end{array}\right)$$

$$\xrightarrow[R_4\to R_4-R_2]{R_3\to R_3-R_2} \left(\begin{array}{cccc|c} 2 & -1 & 1 & & 9 \\ 0 & 1 & -1 & & -15 \\ 0 & 0 & 1 & & 4 \\ 0 & 0 & 0 & & 32 \end{array}\right)$$

$\text{rank}(A) = 3 \neq \text{rank}[A|b] = 4 \Rightarrow$ the system is **inconsistent (no solution)**.

(8) Augmented matrix:

$$[A|b] = \left(\begin{array}{cccc|c} 2 & 0 & 0 & -4 & -10 \\ 0 & 3 & 3 & 0 & 0 \\ 0 & 0 & 1 & 4 & -1 \\ -3 & 2 & 3 & 1 & 5 \end{array}\right)$$

$$\xrightarrow{R_4\to 2R_4+3R_1} \left(\begin{array}{cccc|c} 2 & 0 & 0 & -4 & -10 \\ 0 & 3 & 3 & 0 & 0 \\ 0 & 0 & 1 & 4 & -1 \\ 0 & 4 & 6 & -10 & -20 \end{array}\right)$$

$$\xrightarrow{R_4\to 3R_4-4R_2} \left(\begin{array}{cccc|c} 2 & 0 & 0 & -4 & -10 \\ 0 & 3 & 3 & 0 & 0 \\ 0 & 0 & 1 & 4 & -1 \\ 0 & 0 & 6 & -30 & -60 \end{array}\right)$$

$$\xrightarrow{R_4\to R_4-6R_3} \left(\begin{array}{cccc|c} 2 & 0 & 0 & -4 & -10 \\ 0 & 3 & 3 & 0 & 0 \\ 0 & 0 & 1 & 4 & -1 \\ 0 & 0 & 0 & -54 & -54 \end{array}\right)$$

$\text{rank}(A) = \text{rank}[A|b] = 4$, unknowns $= 4 \Rightarrow$ unique solution. Solution: $x = -3, \quad y = 5, \quad z = -5, \quad w = 1$ (unique solution) **Q.2 Homogeneous systems**

(1)

$$A = \begin{pmatrix} 1 & 1 & 2 \\ 1 & 2 & 3 \\ 1 & 3 & 4 \end{pmatrix}$$

$$\xrightarrow[R_3\to R_3-R_1]{R_2\to R_2-R_1} \begin{pmatrix} 1 & 1 & 2 \\ 0 & 1 & 1 \\ 0 & 2 & 2 \end{pmatrix}$$

$$\xrightarrow{R_3\to R_3-2R_2} \begin{pmatrix} 1 & 1 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

$\text{rank} = 2 < 3 \Rightarrow$ non-trivial solutions exist (1 free variable(s)): $x = -t, \quad y = -t, \quad z = t$, i.e.

$$\mathbf{x} = t \begin{pmatrix} -1 \\ -1 \\ 1 \end{pmatrix}, \quad t \in \mathbb{R}.$$

(2)

$$A = \begin{pmatrix} 1 & 2 & 3 & 1 \\ 1 & 1 & -1 & -1 \\ 3 & -1 & 2 & 3 \end{pmatrix}$$

$$\xrightarrow[R_3\to R_3-3R_1]{R_2\to R_2-R_1} \begin{pmatrix} 1 & 2 & 3 & 1 \\ 0 & -1 & -4 & -2 \\ 0 & -7 & -7 & 0 \end{pmatrix}$$

$$\xrightarrow{R_3 \to R_3 - 7R_2} \begin{pmatrix} 1 & 2 & 3 & 1 \\ 0 & -1 & -4 & -2 \\ 0 & 0 & 21 & 14 \end{pmatrix}$$

rank $= 3 < 4 \Rightarrow$ non-trivial solutions exist (1 free variable(s)): $x_1 = -\frac{t}{3}, \quad x_2 = \frac{2t}{3}, \quad x_3 = -\frac{2t}{3}, \quad x_4 = t$, i.e.

$$\mathbf{x} = t \begin{pmatrix} -\frac{1}{3} \\ \frac{2}{3} \\ -\frac{2}{3} \\ 1 \end{pmatrix}, \quad t \in \mathbb{R}.$$

(3)

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \\ 4 & 5 & 4 \\ 1 & 1 & -2 \end{pmatrix}$$

$$\xrightarrow[R_4 \to R_4 - R_1]{\substack{R_2 \to R_2 - 2R_1 \\ R_3 \to R_3 - 4R_1}} \begin{pmatrix} 1 & 2 & 3 \\ 0 & -1 & -5 \\ 0 & -3 & -8 \\ 0 & -1 & -5 \end{pmatrix}$$

$$\xrightarrow[R_4 \to R_4 - R_2]{R_3 \to R_3 - 3R_2} \begin{pmatrix} 1 & 2 & 3 \\ 0 & -1 & -5 \\ 0 & 0 & 7 \\ 0 & 0 & 0 \end{pmatrix}$$

rank $=$ number of unknowns $\Rightarrow$ only the trivial solution: all unknowns $= 0$.

(4)

$$A = \begin{pmatrix} 3 & 4 & -1 & -9 \\ 2 & 3 & 2 & -3 \\ 2 & 1 & -14 & -12 \\ 1 & 3 & 13 & 3 \end{pmatrix}$$

$$\xrightarrow[R_4 \to 3R_4 - R_1]{\substack{R_2 \to 3R_2 - 2R_1 \\ R_3 \to 3R_3 - 2R_1}} \begin{pmatrix} 3 & 4 & -1 & -9 \\ 0 & 1 & 8 & 9 \\ 0 & -5 & -40 & -18 \\ 0 & 5 & 40 & 18 \end{pmatrix}$$

$$\xrightarrow[R_4 \to R_4 - 5R_2]{R_3 \to R_3 + 5R_2} \begin{pmatrix} 3 & 4 & -1 & -9 \\ 0 & 1 & 8 & 9 \\ 0 & 0 & 0 & 27 \\ 0 & 0 & 0 & -27 \end{pmatrix}$$

$$\xrightarrow{R_4 \to R_4 + R_3} \begin{pmatrix} 3 & 4 & -1 & -9 \\ 0 & 1 & 8 & 9 \\ 0 & 0 & 0 & 27 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

rank $= 3 < 4 \Rightarrow$ non-trivial solutions exist (1 free variable(s)): $x_1 = 11t, \quad x_2 = -8t, \quad x_3 = t, \quad x_4 = 0$, i.e.

$$\mathbf{x} = t \begin{pmatrix} 11 \\ -8 \\ 1 \\ 0 \end{pmatrix}, \quad t \in \mathbb{R}.$$

Q.3 Eigen-type system

The system is $(A-\lambda I)\mathbf{x}=\mathbf{0}$ with $A=\begin{pmatrix}2&-2&1\\2&-3&2\\-1&2&0\end{pmatrix}$. A non-trivial solution exists iff $\det(A-\lambda I)=0$.

$$\det(A-\lambda I)=-(\lambda-1)^2(\lambda+3)=0$$

Expanded: $-\lambda^3-\lambda^2+5\lambda-3=0$, hence $\lambda=1$ or $\lambda=-3$ only. $\lambda=1$*:*

$$A-\lambda I=\begin{pmatrix}1&-2&1\\2&-4&2\\-1&2&-1\end{pmatrix}$$
$$\xrightarrow[R_3\to R_3+R_1]{R_2\to R_2-2R_1}\begin{pmatrix}1&-2&1\\0&0&0\\0&0&0\end{pmatrix}$$

General solution: $x_1=-s+2t,\quad x_2=t,\quad x_3=s$, i.e.

$$\mathbf{x}=t\begin{pmatrix}2\\1\\0\end{pmatrix}+s\begin{pmatrix}-1\\0\\1\end{pmatrix},\quad t,s\in\mathbb{R}.$$

$\lambda=-3$*:*

$$A-\lambda I=\begin{pmatrix}5&-2&1\\2&0&2\\-1&2&3\end{pmatrix}$$
$$\xrightarrow[R_3\to 5R_3+R_1]{R_2\to 5R_2-2R_1}\begin{pmatrix}5&-2&1\\0&4&8\\0&8&16\end{pmatrix}$$
$$\xrightarrow{R_3\to R_3-2R_2}\begin{pmatrix}5&-2&1\\0&4&8\\0&0&0\end{pmatrix}$$

General solution: $x_1=-t,\quad x_2=-2t,\quad x_3=t$, i.e.

$$\mathbf{x}=t\begin{pmatrix}-1\\-2\\1\end{pmatrix},\quad t\in\mathbb{R}.$$

Q.4 Non-trivial solutions

(1) $\det\begin{pmatrix}\lambda-2&1\\1&\lambda-2\end{pmatrix}=(\lambda-3)(\lambda-1)=0\Rightarrow\lambda=1,\,3.$
$\lambda=1$: $x=t,\quad y=t$, i.e.

$$\mathbf{x}=t\begin{pmatrix}1\\1\end{pmatrix},\quad t\in\mathbb{R}.$$

$\lambda=3$: $x=-t,\quad y=t$, i.e.

$$\mathbf{x}=t\begin{pmatrix}-1\\1\end{pmatrix},\quad t\in\mathbb{R}.$$

(2) Coefficient matrix $\begin{pmatrix}3&1&-\lambda\\4&-2&-3\\2\lambda&4&-\lambda\end{pmatrix}$; $\det=-4\left(\lambda^2+3\lambda-9\right)=0\Rightarrow\lambda=-\frac{3}{2}+\frac{3\sqrt{5}}{2},-\frac{3\sqrt{5}}{2}-\frac{3}{2}$.

$\lambda=-\frac{3}{2}+\frac{3\sqrt{5}}{2}$ *(1.854):* (row reduction involves surds, so the null space is given directly; rank = 2)
Solution: $x=\frac{3\sqrt{5}t}{10},\quad y=t\left(-\frac{3}{2}+\frac{3\sqrt{5}}{5}\right),\quad z=t$, i.e.

$$\mathbf{x}=t\begin{pmatrix}\frac{3\sqrt{5}}{10}\\-\frac{3}{2}+\frac{3\sqrt{5}}{5}\\1\end{pmatrix},\quad t\in\mathbb{R}.$$

$\lambda = -\frac{3\sqrt{5}}{2} - \frac{3}{2}$ *(−4.854):* (row reduction involves surds, so the null space is given directly; rank = 2) Solution: $x = -\frac{3\sqrt{5}t}{10}, \quad y = t\left(-\frac{3}{2} - \frac{3\sqrt{5}}{5}\right), \quad z = t$, i.e.

$$\mathbf{x} = t\begin{pmatrix} -\frac{3\sqrt{5}}{10} \\ -\frac{3}{2} - \frac{3\sqrt{5}}{5} \\ 1 \end{pmatrix}, \quad t \in \mathbb{R}.$$

Q.5 Values of k with no unique solution

$\det A = -(k-1)^2(k+2)$. For $k \neq -2, k \neq 1$: $\det A \neq 0$, rank = rank$[A|b] = 3 \Rightarrow$ **unique solution**. *Case $k = -2$:*

$$[A|b] = \left(\begin{array}{ccc|c} 1 & 1 & -2 & 3 \\ 1 & -2 & 1 & 2 \\ -2 & 1 & 1 & 1 \end{array}\right)$$
$$\xrightarrow[R_3 \to R_3 + 2R_1]{R_2 \to R_2 - R_1} \left(\begin{array}{ccc|c} 1 & 1 & -2 & 3 \\ 0 & -3 & 3 & -1 \\ 0 & 3 & -3 & 7 \end{array}\right)$$
$$\xrightarrow{R_3 \to R_3 + R_2} \left(\begin{array}{ccc|c} 1 & 1 & -2 & 3 \\ 0 & -3 & 3 & -1 \\ 0 & 0 & 0 & 6 \end{array}\right)$$

Here rank$A = 2 <$ rank$[A|b] \Rightarrow$ **no solution**. *Case $k = 1$:*

$$[A|b] = \left(\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 \end{array}\right)$$
$$\xrightarrow[R_3 \to R_3 - R_1]{R_2 \to R_2 - R_1} \left(\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & -2 \end{array}\right)$$

Here rank$A = 1 <$ rank$[A|b] \Rightarrow$ **no solution**. So the system fails to have a unique solution exactly for $k = 1$ and $k = -2$ (in both cases it has no solution). **Q.6 Investigate μ and λ**

(1) $\det A = 3(\lambda - 8)$. For $\lambda \neq 8$: $\det A \neq 0$, rank = rank$[A|b] = 3 \Rightarrow$ **unique solution**. *Case $\lambda = 8$:*

$$[A|b] = \left(\begin{array}{ccc|c} 2 & -1 & 3 & 2 \\ 1 & 1 & 2 & 2 \\ 5 & -1 & 8 & \mu \end{array}\right)$$
$$\xrightarrow[R_3 \to 2R_3 - 5R_1]{R_2 \to 2R_2 - R_1} \left(\begin{array}{ccc|c} 2 & -1 & 3 & 2 \\ 0 & 3 & 1 & 2 \\ 0 & 3 & 1 & 2\mu - 10 \end{array}\right)$$
$$\xrightarrow{R_3 \to R_3 - R_2} \left(\begin{array}{ccc|c} 2 & -1 & 3 & 2 \\ 0 & 3 & 1 & 2 \\ 0 & 0 & 0 & 2\mu - 12 \end{array}\right)$$

Here rank$A = 2 < 3$. The last row gives $2\mu - 12 = 0$. So if $\mu = 6$: rank$A =$rank$[A|b] = 2 \Rightarrow$ **infinitely many solutions**; if $\mu \neq 6$: rank$A <$rank$[A|b] \Rightarrow$ **no solution**.

(i) no solution: $\lambda = 8,\ \mu \neq 6$; (ii) unique: $\lambda \neq 8$; (iii) infinitely many: $\lambda = 8,\ \mu = 6$

(2) $\det A = -2(\lambda - 3)$. For $\lambda \neq 3$: $\det A \neq 0$, rank = rank$[A|b] = 3 \Rightarrow$ **unique solution**. *Case $\lambda = 3$:*

$$[A|b] = \left(\begin{array}{ccc|c} 1 & 2 & 1 & 8 \\ 2 & 2 & 2 & 13 \\ 3 & 4 & 3 & \mu \end{array}\right)$$

$$\xrightarrow[R_3\to R_3-3R_1]{R_2\to R_2-2R_1} \left(\begin{array}{ccc|c} 1 & 2 & 1 & 8 \\ 0 & -2 & 0 & -3 \\ 0 & -2 & 0 & \mu-24 \end{array}\right)$$

$$\xrightarrow{R_3\to R_3-R_2} \left(\begin{array}{ccc|c} 1 & 2 & 1 & 8 \\ 0 & -2 & 0 & -3 \\ 0 & 0 & 0 & \mu-21 \end{array}\right)$$

Here rank$A = 2 < 3$. The last row gives $\mu - 21 = 0$. So if $\mu = 21$: rank$A$ =rank$[A|b] = 2 \Rightarrow$ **infinitely many solutions**; if $\mu \neq 21$: rank$A$ <rank$[A|b] \Rightarrow$ **no solution**.

(i) no solution: $\lambda = 3,\ \mu \neq 21$; (ii) unique: $\lambda \neq 3$; (iii) infinitely many: $\lambda = 3,\ \mu = 21$

Q.8 Traffic flow (first network)

Assume the directions: x_1 from top-left to top-right, x_2 down the left side, x_3 along the diagonal from top-right to bottom-left, x_4 down the right side, x_5 along the bottom to the right. Inflow = outflow at each junction:

$$\begin{aligned} \text{top-left: } & x_1 + x_2 = 300 \\ \text{top-right: } & x_1 = x_3 + x_4 + 150 \\ \text{bottom-left: } & x_2 + x_3 + 200 = x_5 \\ \text{bottom-right: } & x_4 + x_5 = 350 \end{aligned}$$

$$[A|b] = \left(\begin{array}{ccccc|c} 1 & 1 & 0 & 0 & 0 & 300 \\ 1 & 0 & -1 & -1 & 0 & 150 \\ 0 & 1 & 1 & 0 & -1 & -200 \\ 0 & 0 & 0 & 1 & 1 & 350 \end{array}\right)$$

$$\xrightarrow{R_2\to R_2-R_1} \left(\begin{array}{ccccc|c} 1 & 1 & 0 & 0 & 0 & 300 \\ 0 & -1 & -1 & -1 & 0 & -150 \\ 0 & 1 & 1 & 0 & -1 & -200 \\ 0 & 0 & 0 & 1 & 1 & 350 \end{array}\right)$$

$$\xrightarrow{R_3\to R_3+R_2} \left(\begin{array}{ccccc|c} 1 & 1 & 0 & 0 & 0 & 300 \\ 0 & -1 & -1 & -1 & 0 & -150 \\ 0 & 0 & 0 & -1 & -1 & -350 \\ 0 & 0 & 0 & 1 & 1 & 350 \end{array}\right)$$

$$\xrightarrow{R_4\to R_4+R_3} \left(\begin{array}{ccccc|c} 1 & 1 & 0 & 0 & 0 & 300 \\ 0 & -1 & -1 & -1 & 0 & -150 \\ 0 & 0 & 0 & -1 & -1 & -350 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right)$$

(i) Free variables: x_2, x_3. General solution: $x_1 = 300 - x_2,\ x_4 = 150 - x_2 - x_3,\ x_5 = 200 + x_2 + x_3$ with x_2, x_3 free (consistent: total inflow 500 = total outflow). **(ii)** $x_2 = 200,\ x_3 = 50$: $x_1 = 100,\ x_4 = -100,\ x_5 = 450$. (The negative x_4 means traffic would have to flow *opposite* to the assumed direction on that street, so this is not feasible if the street is one-way.) **(iii)** $x_2 = 150,\ x_3 = 0$: $x_1 = 150,\ x_4 = 0,\ x_5 = 350$. **Q.9 Traffic flow (second network)**

Assume: x_1 left→top, x_2 top→right, x_3 left→bottom, x_4 right→bottom. Junction equations (in = out):

$$\begin{aligned} \text{left: } & x_1 + x_3 = 100 \\ \text{top: } & x_1 + 200 = x_2 \\ \text{right: } & x_2 = x_4 + 100 \\ \text{bottom: } & x_3 + x_4 = 200 \end{aligned}$$

$$[A|b] = \left(\begin{array}{cccc|c} 1 & 0 & 1 & 0 & 100 \\ 1 & -1 & 0 & 0 & -200 \\ 0 & 1 & 0 & -1 & 100 \\ 0 & 0 & 1 & 1 & 200 \end{array}\right)$$

$$\xrightarrow{R_2 \to R_2 - R_1} \left(\begin{array}{cccc|c} 1 & 0 & 1 & 0 & 100 \\ 0 & -1 & -1 & 0 & -300 \\ 0 & 1 & 0 & -1 & 100 \\ 0 & 0 & 1 & 1 & 200 \end{array}\right)$$

$$\xrightarrow{R_3 \to R_3 + R_2} \left(\begin{array}{cccc|c} 1 & 0 & 1 & 0 & 100 \\ 0 & -1 & -1 & 0 & -300 \\ 0 & 0 & -1 & -1 & -200 \\ 0 & 0 & 1 & 1 & 200 \end{array}\right)$$

$$\xrightarrow{R_4 \to R_4 + R_3} \left(\begin{array}{cccc|c} 1 & 0 & 1 & 0 & 100 \\ 0 & -1 & -1 & 0 & -300 \\ 0 & 0 & -1 & -1 & -200 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right)$$

(i) One free variable ($x_4 = t$): $x_1 = t - 100,\ x_2 = t + 100,\ x_3 = 200 - t,\ x_4 = t$. **(ii)** $x_4 = 0$: $x_1 = -100,\ x_2 = 100,\ x_3 = 200$ (the negative x_1 means the flow is actually reversed on that street). **(iii)** $x_4 = 100$: $x_1 = 0,\ x_2 = 200,\ x_3 = 100$. **Q.10 Electrical networks (Kirchhoff's laws)**

Take I_1 (top branch) and I_3 (bottom branch) leaving the left node, with both batteries driving current in the direction of I_1, I_3, and I_2 (middle) returning. Node law: $I_2 = I_1 + I_3$. Loop law (sum of RI drops = emf) round each loop containing the middle branch. *First network*:

$$\begin{aligned} I_1 - I_2 + I_3 &= 0 \\ 4I_1 + 1I_2 &= 16 \\ 4I_3 + 1I_2 &= 8 \end{aligned}$$

$$[A|b] = \left(\begin{array}{ccc|c} 1 & -1 & 1 & 0 \\ 4 & 1 & 0 & 16 \\ 0 & 1 & 4 & 8 \end{array}\right)$$

$$\xrightarrow{R_2 \to R_2 - 4R_1} \left(\begin{array}{ccc|c} 1 & -1 & 1 & 0 \\ 0 & 5 & -4 & 16 \\ 0 & 1 & 4 & 8 \end{array}\right)$$

$$\xrightarrow{R_3 \to 5R_3 - R_2} \left(\begin{array}{ccc|c} 1 & -1 & 1 & 0 \\ 0 & 5 & -4 & 16 \\ 0 & 0 & 24 & 24 \end{array}\right)$$

$$I_1 = 3 \text{ A}, \quad I_2 = 4 \text{ A}, \quad I_3 = 1 \text{ A}$$

Second network:

$$\begin{aligned} I_1 - I_2 + I_3 &= 0 \\ 4I_1 + 3I_2 &= 3 \\ 1I_3 + 3I_2 &= 4 \end{aligned}$$

$$[A|b] = \left(\begin{array}{ccc|c} 1 & -1 & 1 & 0 \\ 4 & 3 & 0 & 3 \\ 0 & 3 & 1 & 4 \end{array}\right)$$

$$\xrightarrow{R_2 \to R_2 - 4R_1} \left(\begin{array}{ccc|c} 1 & -1 & 1 & 0 \\ 0 & 7 & -4 & 3 \\ 0 & 3 & 1 & 4 \end{array}\right)$$

$$\xrightarrow{R_3 \to 7R_3 - 3R_2} \left(\begin{array}{ccc|c} 1 & -1 & 1 & 0 \\ 0 & 7 & -4 & 3 \\ 0 & 0 & 19 & 19 \end{array} \right)$$

$$I_1 = 0 \text{ A}, \quad I_2 = 1 \text{ A}, \quad I_3 = 1 \text{ A}$$

Tutorial No. 3: Subspace of Vector Space

Questions

Tutorial No.3

Topics: Subspace of vector space

Q.1) Which of the following are subspaces of the vector space V where $V = \mathbb{R}^2 = \left\{\begin{pmatrix} x \\ y \end{pmatrix} \middle| x, y \in \mathbb{R}\right\}$. Justify your answer.

1. $W = \left\{\begin{pmatrix} x \\ 0 \end{pmatrix} \middle| x \in \mathbb{R}\right\}$
2. $W = \left\{\begin{pmatrix} x \\ -x \end{pmatrix} \middle| x \in \mathbb{R}\right\}$
3. $W = \left\{\begin{pmatrix} x \\ 1 \end{pmatrix} \middle| x \in \mathbb{R}\right\}$
4. $W = \left\{\begin{pmatrix} x \\ 3x+2 \end{pmatrix} \middle| x \in \mathbb{R}\right\}$
5. $W = \left\{\begin{pmatrix} x \\ y \end{pmatrix} \middle| x^2 + y^2 = 0 \;\&\; x, y \in \mathbb{R}\right\}$
6. $W = \left\{\begin{pmatrix} x \\ y \end{pmatrix} \middle| x - 3y = 0, 2x - 6y = 0 \;\; \& x, y \in \mathbb{R}\right\}$
7. $W = \left\{\begin{pmatrix} x \\ y \end{pmatrix} \middle| x^2 - y^2 = 0 \;\&\; x, y \in \mathbb{R}\right\}$
8. $W = \left\{\begin{pmatrix} x \\ y \end{pmatrix} \middle| x^2 + y^2 = 1 \;\& x, y \in \mathbb{R}\right\}$
9. $W = \left\{\begin{pmatrix} x \\ y \end{pmatrix} \middle| x + 4y = 5 \;\& x, y \in \mathbb{R}\right\}$
10. $W = \left\{\begin{pmatrix} x \\ y \end{pmatrix} \middle| x - 3y = 2,\; 2x - 6y = 4 \;\; \& x, y \in \mathbb{R}\right\}$

Q.2) Which of the following are subspaces of the vector space V where $V = \mathbb{R}^3 = \left\{\begin{pmatrix} x \\ y \\ z \end{pmatrix} \middle| x, y, z \in \mathbb{R}\right\}$. Justify your answer.

1. $W = \left\{\begin{pmatrix} x \\ 0 \\ z \end{pmatrix} \middle| x, z \in \mathbb{R}\right\}$
2. $W = \left\{\begin{pmatrix} x \\ 0 \\ 0 \end{pmatrix} \middle| x \in \mathbb{R}\right\}$
3. $W = \left\{\begin{pmatrix} x \\ x \\ x \end{pmatrix} \middle| x \in \mathbb{R}\right\}$
4. $W = \left\{\begin{pmatrix} 0 \\ 0 \\ z \end{pmatrix} \middle| z \in \mathbb{R}\right\}$

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4. $W = \left\{ \begin{pmatrix} x \\ 0 \\ 5 \end{pmatrix} \middle| \, x \in \mathbb{R} \right\}$

6. $W = \left\{ \begin{pmatrix} -1 \\ 0 \\ -3 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 3 \end{pmatrix} \right\}$

7. $W = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \middle| \, x + 2y + 3z = 0 \;\&\; x, y, z \in \mathbb{R} \right\}$

8. $W = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \middle| \, x + 2y + 3z = 3 \;\&\; x, y, z \in \mathbb{R} \right\}$

Q.3) Which of the following are subspaces of the vector space V where $V = P_2(t) = \{a + bt + ct^2 | \, a, b, c \in \mathbb{R}\}$? Justify your answer.

1. $W = \{a + bt \mid a, b \in \mathbb{R}\}$

2. $W = \{bt \mid b \in \mathbb{R}\}$

3. $W = \{a \mid a \in \mathbb{R}\}$

4. $W = \{a + bt^2 | \, a, b \in \mathbb{R}\}$

5. $W = \{a + t \mid a \in \mathbb{R}\}$

6. $W = \{3 + bt \mid b \in \mathbb{R}\}$

7. $W = \{at + bt^2 | \, a, b \in \mathbb{R}\}$

8. $W = \{a + bt \mid a \geq 0, b \in \mathbb{R}\}$

Q.4) Which of the following are subspaces of the vector space V of all 2×2 matrices over reals i.e. $V = \left\{ \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \middle| \, a_{ij} \in \mathbb{R} \right\}$? Justify your answer.

1. $W = \left\{ \begin{pmatrix} a & 0 \\ b & c \end{pmatrix} \middle| \, a, b, c \in \mathbb{R} \right\}$

2. $W = \left\{ \begin{pmatrix} a & b \\ 0 & c \end{pmatrix} \middle| \, a, b, c \in \mathbb{R} \right\}$

3. $W = \left\{ \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix} \middle| \, a, b \in \mathbb{R} \right\}$

4. $W = \left\{ \begin{pmatrix} a & 0 \\ b & 0 \end{pmatrix} \middle| \, a, b \in \mathbb{R} \right\}$

5. $W = \left\{ \begin{pmatrix} a & b \\ b & c \end{pmatrix} \middle| \, a, b, c \in \mathbb{R} \right\}$

6. $W = \left\{ \begin{pmatrix} a & 0 \\ 0 & -a \end{pmatrix} \middle| \, a \in \mathbb{R} \right\}$

7. $W = \left\{ \begin{pmatrix} a & a \\ a & -a \end{pmatrix} \middle| \, a \in \mathbb{R} \right\}$

8. $W = \left\{ \begin{pmatrix} a & 0 \\ b & c \end{pmatrix} \middle| \, a, b, c \neq 1 \in \mathbb{R} \right\}$

Solutions — Tutorial No. 3

Q.1 Subspaces of $\mathbb{R}^2$

1. **Subspace.** Contains $(0,0)$; $(x_1,0)+(x_2,0)=(x_1+x_2,0)\in W$; $c(x,0)=(cx,0)\in W$. (The x-axis.)
2. **Subspace.** $(0,0)$ $(x=0)$; $(x_1,-x_1)+(x_2,-x_2)=(x_1+x_2,-(x_1+x_2))\in W$; $c(x,-x)=(cx,-cx)\in W$. (The line $y=-x$.)
3. **Not a subspace.** $(0,0)\notin W$ because the second entry is always 1. (Also $(x,1)+(z,1)=(x+z,2)\notin W$.)
4. **Not a subspace.** $(0,0)\notin W$: for $x=0$ the vector is $(0,2)$, and $y=3x+2$ is a line not through the origin.
5. **Subspace.** $x^2+y^2=0 \iff x=y=0$, so $W=\{(0,0)\}$, the zero subspace.
6. **Subspace.** Both equations reduce to $x=3y$, so $W=\{(3y,y)\}$ is a line through the origin; closed under addition and scalar multiplication.
7. **Not a subspace.** $x^2-y^2=0 \iff y=\pm x$ (two lines). $(1,1),(1,-1)\in W$ but $(1,1)+(1,-1)=(2,0)\notin W$: not closed under addition.
8. **Not a subspace.** $(0,0)\notin W$ since $0+0\neq 1$ (unit circle). Also not closed under addition or scaling.
9. **Not a subspace.** $(0,0)\notin W$ since $0+0\neq 5$ (a line not through the origin).
10. **Not a subspace.** Both equations say $x-3y=2$; $(0,0)$ does not satisfy it, so W is a line not through the origin.

Q.2 Subspaces of $\mathbb{R}^3$

1. **Subspace.** $W=\{(x,0,z)\}$ is the xz-plane: contains $\mathbf{0}$, and sums/scalar multiples keep the middle entry 0.
2. **Subspace.** $W=\{(x,0,0)\}$ is the x-axis; contains $\mathbf{0}$, closed under $+$ and scalar multiplication.
3. **Subspace.** $W=\{x(1,1,1)\}=\text{span}\{(1,1,1)\}$, a line through the origin.
4. **Subspace.** $W=\{(0,0,z)\}$ is the z-axis $=\text{span}\{(0,0,1)\}$.
5. **Not a subspace.** (Printed as a second "4".) $W=\{(x,0,5)\}$: the zero vector $(0,0,0)\notin W$ because the third entry is always 5.
6. **Not a subspace.** $W=\{(-1,0,-3),(0,0,0),(1,0,3)\}$ has only three vectors. $(1,0,3)+(1,0,3)=(2,0,6)\notin W$: not closed under addition (and $2(1,0,3)\notin W$).
7. **Subspace.** $x+2y+3z=0$ is a plane through the origin (null space of [1 2 3]): $\mathbf{0}$ satisfies it and it is closed under $+$ and scalar multiplication.
8. **Not a subspace.** $x+2y+3z=3$: $\mathbf{0}$ gives $0\neq 3$, so $\mathbf{0}\notin W$ (a plane not through the origin).

Q.3 Subspaces of $P_2(t)$

1. **Subspace.** $W=\text{span}\{1,t\}$: contains 0 and is closed under addition and scalar multiples (polynomials of degree ≤ 1).
2. **Subspace.** $W=\text{span}\{t\}$.
3. **Subspace.** $W=\text{span}\{1\}$ (constant polynomials).
4. **Subspace.** $W=\text{span}\{1,t^2\}$.

5. **Not a subspace.** $a + t$: the coefficient of t is always 1, so the zero polynomial is not in W (also $(a + t) + (b + t) = (a + b) + 2t \notin W$).

6. **Not a subspace.** $3 + bt$: the constant term is always $3 \neq 0$, so $0 \notin W$.

7. **Subspace.** $W = \text{span}\{t, t^2\}$.

8. **Not a subspace.** Not closed under scalar multiplication: $1 + t \in W$ ($a = 1 \geq 0$) but $(-1)(1 + t) = -1 - t \notin W$ since $a = -1 < 0$.

Q.4 Subspaces of $M_{2\times 2}(\mathbb{R})$

1. **Subspace.** Lower triangular matrices $\begin{pmatrix} a & 0 \\ b & c \end{pmatrix}$ (zero entry fixed): contains O, closed under + and scalar multiplication.

2. **Subspace.** Upper triangular matrices: same argument.

3. **Subspace.** Diagonal matrices $\begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}$.

4. **Subspace.** Matrices with second column zero, $\begin{pmatrix} a & 0 \\ b & 0 \end{pmatrix}$.

5. **Subspace.** Symmetric matrices $\begin{pmatrix} a & b \\ b & c \end{pmatrix}$: $A^T = A$, $B^T = B \Rightarrow (A + B)^T = A + B$, $(kA)^T = kA$; O is symmetric.

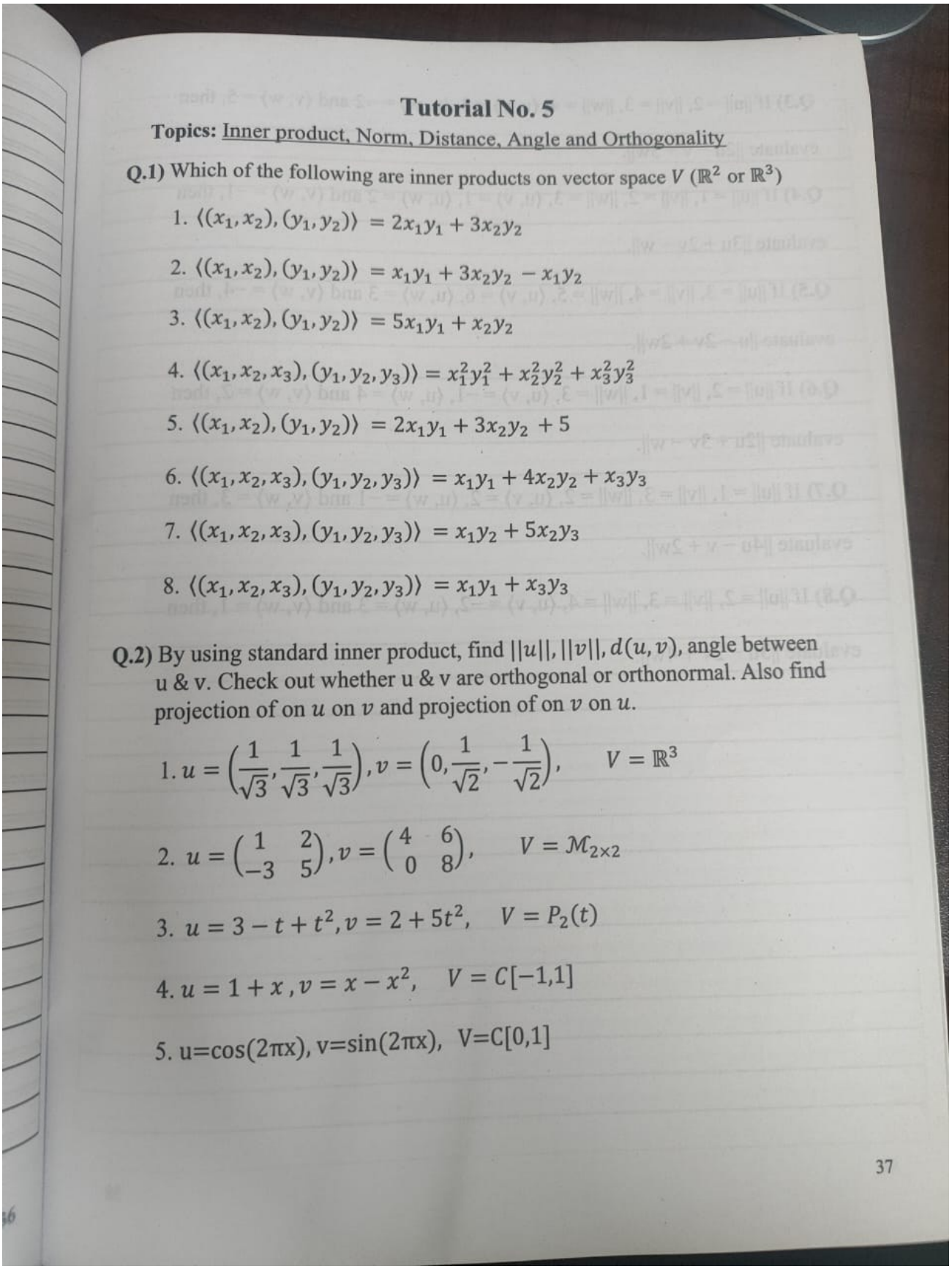
6. **Subspace.** $W = \{a\,\text{diag}(1, -1)\}$ is the span of one matrix.

7. **Subspace.** $W = \{a \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}\}$, the span of one matrix.

8. **Not a subspace.** Not closed under scalar multiplication: $A = \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & 0 \end{pmatrix} \in W$ but $2A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ has $a = 1$, so $2A \notin W$ (also $a = 2$, $a = -1$ add to 1).

Tutorial No. 5: Inner Product, Norm, Distance, Angle and Orthogonality

Questions



Tutorial No. 5

Topics: Inner product, Norm, Distance, Angle and Orthogonality

Q.1) Which of the following are inner products on vector space V ($\mathbb{R}^2$ or $\mathbb{R}^3$)

1. $\langle (x_1, x_2), (y_1, y_2) \rangle = 2x_1y_1 + 3x_2y_2$
2. $\langle (x_1, x_2), (y_1, y_2) \rangle = x_1y_1 + 3x_2y_2 - x_1y_2$
3. $\langle (x_1, x_2), (y_1, y_2) \rangle = 5x_1y_1 + x_2y_2$
4. $\langle (x_1, x_2, x_3), (y_1, y_2, y_3) \rangle = x_1^2y_1^2 + x_2^2y_2^2 + x_3^2y_3^2$
5. $\langle (x_1, x_2), (y_1, y_2) \rangle = 2x_1y_1 + 3x_2y_2 + 5$
6. $\langle (x_1, x_2, x_3), (y_1, y_2, y_3) \rangle = x_1y_1 + 4x_2y_2 + x_3y_3$
7. $\langle (x_1, x_2, x_3), (y_1, y_2, y_3) \rangle = x_1y_2 + 5x_2y_3$
8. $\langle (x_1, x_2, x_3), (y_1, y_2, y_3) \rangle = x_1y_1 + x_3y_3$

Q.2) By using standard inner product, find $||u||, ||v||, d(u, v)$, angle between u & v. Check out whether u & v are orthogonal or orthonormal. Also find projection of on u on v and projection of on v on u.

1. $u = \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right), v = \left(0, \frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}\right), \quad V = \mathbb{R}^3$
2. $u = \begin{pmatrix} 1 & 2 \\ -3 & 5 \end{pmatrix}, v = \begin{pmatrix} 4 & 6 \\ 0 & 8 \end{pmatrix}, \quad V = \mathcal{M}_{2\times 2}$
3. $u = 3 - t + t^2, v = 2 + 5t^2, \quad V = P_2(t)$
4. $u = 1 + x, v = x - x^2, \quad V = C[-1,1]$
5. u=cos(2πx), v=sin(2πx), V=C[0,1]

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Q.3) If $\|u\| = 2$, $\|v\| = 3$, $\|w\| = 4$, $\langle u, v\rangle = 3$, $\langle u, w\rangle = -2$ and $\langle v, w\rangle = 5$, then evaluate $\|2u - v + 3w\|$.

Q.4) If $\|u\| = 1$, $\|v\| = 2$, $\|w\| = 3$, $\langle u, v\rangle = 1$, $\langle u, w\rangle = 2$ and $\langle v, w\rangle = -1$, then evaluate $\|3u + 2v - w\|$.

Q.5) If $\|u\| = 3$, $\|v\| = 4$, $\|w\| = 5$, $\langle u, v\rangle = 6$, $\langle u, w\rangle = 3$ and $\langle v, w\rangle = -4$, then evaluate $\|u - 2v + 2w\|$.

Q.6) If $\|u\| = 2$, $\|v\| = 1$, $\|w\| = 3$, $\langle u, v\rangle = -1$, $\langle u, w\rangle = 4$ and $\langle v, w\rangle = 2$, then evaluate $\|2u + 3v - w\|$.

Q.7) If $\|u\| = 1$, $\|v\| = 3$, $\|w\| = 2$, $\langle u, v\rangle = 2$, $\langle u, w\rangle = -1$ and $\langle v, w\rangle = 3$, then evaluate $\|4u - v + 2w\|$.

Q.8) If $\|u\| = 2$, $\|v\| = 3$, $\|w\| = 4$, $\langle u, v\rangle = -2$, $\langle u, w\rangle = 3$ and $\langle v, w\rangle = 1$, then evaluate $\|3u - 2v + 4w\|$.

Solutions — Tutorial No. 5

Q.1 Which are inner products?

(An inner product must be symmetric, linear in the first slot, and positive definite: $\langle u, u\rangle > 0$ for $u \neq 0$.)

1. **Yes, inner product.** Symmetric, bilinear, and $2x_1^2 + 3x_2^2 > 0$ for $(x_1, x_2) \neq 0$ (weights $2, 3 > 0$).
2. **Not an inner product.** Not symmetric: $\langle u, v\rangle$ contains $-x_1y_2$, but $\langle v, u\rangle$ contains $-y_1x_2$; e.g. $\langle (1,0),(0,1)\rangle = -1$ while $\langle (0,1),(1,0)\rangle = 0$.
3. **Yes, inner product.** Weighted dot product with positive weights $5, 1$: symmetric, bilinear, positive definite.
4. **Not an inner product.** Not linear: $\langle 2u, v\rangle = 4\langle u, v\rangle \neq 2\langle u, v\rangle$ (squares of the entries).
5. **Not an inner product.** Not linear: $\langle \mathbf{0}, v\rangle = 5 \neq 0$ (the added constant breaks bilinearity).
6. **Yes, inner product.** Weighted dot product with positive weights $1, 4, 1$.
7. **Not an inner product.** Not symmetric ($x_1y_2 \neq y_1x_2$ in general) and not positive: for $u = (1, 0, 0)$ we get $\langle u, u\rangle = x_1x_2 + 5x_2x_3 = 0$ though $u \neq 0$.
8. **Not an inner product.** Not positive definite: for $u = (0, 1, 0) \neq \mathbf{0}$, $\langle u, u\rangle = 0 + 0 = 0$.

Q.2 Standard inner products

1. $V = \mathbb{R}^3$, $\langle u, v\rangle = u \cdot v$

$$\begin{aligned} \|u\| &= \sqrt{\langle u, u\rangle} = \sqrt{1} = 1 \\ \|v\| &= \sqrt{\langle v, v\rangle} = \sqrt{1} = 1 \\ \langle u, v\rangle &= 0 \\ d(u, v) &= \|u - v\| = \sqrt{2} \\ \cos\theta &= \frac{\langle u, v\rangle}{\|u\|\|v\|} = 0 \;\Rightarrow\; \theta = \frac{\pi}{2} \approx 90.00^\circ \end{aligned}$$

$\langle u, v\rangle = 0$, so u, v are **orthogonal** and, both being unit vectors, **orthonormal**.

$$\operatorname{proj}_v u = \frac{\langle u, v\rangle}{\langle v, v\rangle} v = 0\, v = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}, \qquad \operatorname{proj}_u v = \frac{\langle v, u\rangle}{\langle u, u\rangle} u = 0\, u = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

2. $V = M_{2\times 2}$, $\langle A, B\rangle = \operatorname{tr}(B^T A) = \sum a_{ij} b_{ij}$

$$\begin{aligned} \|u\| &= \sqrt{\langle u, u\rangle} = \sqrt{39} \\ \|v\| &= \sqrt{\langle v, v\rangle} = \sqrt{116} = 2\sqrt{29} \\ \langle u, v\rangle &= 56 \\ d(u, v) &= \|u - v\| = \sqrt{43} \\ \cos\theta &= \frac{\langle u, v\rangle}{\|u\|\|v\|} = \frac{28\sqrt{1131}}{1131} \;\Rightarrow\; \theta = \arccos\left(\frac{28\sqrt{1131}}{1131}\right) \approx 33.64^\circ \end{aligned}$$

$\langle u, v\rangle = 56 \neq 0$, so u, v are **not orthogonal** (hence not orthonormal).

$$\operatorname{proj}_v u = \frac{\langle u, v\rangle}{\langle v, v\rangle} v = \frac{14}{29} v = \begin{pmatrix} \frac{56}{29} & \frac{84}{29} \\ 0 & \frac{112}{29} \end{pmatrix}, \qquad \operatorname{proj}_u v = \frac{\langle v, u\rangle}{\langle u, u\rangle} u = \frac{56}{39} u = \begin{pmatrix} \frac{56}{39} & \frac{112}{39} \\ -\frac{56}{13} & \frac{280}{39} \end{pmatrix}$$

3. $V = P_2(t)$, $\langle a_0 + a_1 t + a_2 t^2,\, b_0 + b_1 t + b_2 t^2 \rangle = a_0 b_0 + a_1 b_1 + a_2 b_2$

$$\begin{aligned}
\|u\| &= \sqrt{\langle u, u\rangle} = \sqrt{11} \\
\|v\| &= \sqrt{\langle v, v\rangle} = \sqrt{29} \\
\langle u, v\rangle &= 11 \\
d(u, v) &= \|u - v\| = \sqrt{18} = 3\sqrt{2} \\
\cos\theta &= \frac{\langle u, v\rangle}{\|u\|\|v\|} = \frac{\sqrt{319}}{29} \;\Rightarrow\; \theta = \arccos\left(\frac{\sqrt{319}}{29}\right) \approx 51.98^\circ
\end{aligned}$$

$\langle u, v\rangle = 11 \neq 0$, so u, v are **not orthogonal** (hence not orthonormal).

$$\text{proj}_v u = \frac{\langle u, v\rangle}{\langle v, v\rangle} v = \frac{11}{29}\, v = \frac{55t^2}{29} + \frac{22}{29}, \qquad \text{proj}_u v = \frac{\langle v, u\rangle}{\langle u, u\rangle} u = 1\, u = t^2 - t + 3$$

4. $V = C[-1, 1]$, $\langle f, g\rangle = \int_{-1}^{1} f(x) g(x)\, dx$

$$\begin{aligned}
\|u\| &= \sqrt{\langle u, u\rangle} = \sqrt{\frac{8}{3}} = \frac{2\sqrt{6}}{3} \\
\|v\| &= \sqrt{\langle v, v\rangle} = \sqrt{\frac{16}{15}} = \frac{4\sqrt{15}}{15} \\
\langle u, v\rangle &= 0 \\
d(u, v) &= \|u - v\| = \sqrt{\frac{56}{15}} = \frac{2\sqrt{210}}{15} \\
\cos\theta &= \frac{\langle u, v\rangle}{\|u\|\|v\|} = 0 \;\Rightarrow\; \theta = \frac{\pi}{2} \approx 90.00^\circ
\end{aligned}$$

$\langle u, v\rangle = 0$, so u, v are **orthogonal**, but not orthonormal (not both unit vectors).

$$\text{proj}_v u = \frac{\langle u, v\rangle}{\langle v, v\rangle} v = 0\, v = 0, \qquad \text{proj}_u v = \frac{\langle v, u\rangle}{\langle u, u\rangle} u = 0\, u = 0$$

5. $V = C[0, 1]$, $\langle f, g\rangle = \int_{0}^{1} f(x) g(x)\, dx$

$$\begin{aligned}
\|u\| &= \sqrt{\langle u, u\rangle} = \sqrt{\frac{1}{2}} = \frac{\sqrt{2}}{2} \\
\|v\| &= \sqrt{\langle v, v\rangle} = \sqrt{\frac{1}{2}} = \frac{\sqrt{2}}{2} \\
\langle u, v\rangle &= 0 \\
d(u, v) &= \|u - v\| = \sqrt{1} = 1 \\
\cos\theta &= \frac{\langle u, v\rangle}{\|u\|\|v\|} = 0 \;\Rightarrow\; \theta = \frac{\pi}{2} \approx 90.00^\circ
\end{aligned}$$

$\langle u, v\rangle = 0$, so u, v are **orthogonal**, but not orthonormal (not both unit vectors).

$$\text{proj}_v u = \frac{\langle u, v\rangle}{\langle v, v\rangle} v = 0\, v = 0, \qquad \text{proj}_u v = \frac{\langle v, u\rangle}{\langle u, u\rangle} u = 0\, u = 0$$

Q.3–Q.8 Norms of linear combinations

Use $\|au + bv + cw\|^2 = a^2\|u\|^2 + b^2\|v\|^2 + c^2\|w\|^2 + 2ab\langle u, v\rangle + 2ac\langle u, w\rangle + 2bc\langle v, w\rangle$.

Q.3 $\|2u - v + 3w\|^2 = 4(4) + 1(9) + 9(16) + 2(-2)(3) + 2(6)(-2) + 2(-3)(5) = 103$, so $\|2u - v + 3w\| = \sqrt{103} = \sqrt{103}$.

Q.4 $\|3u+2v-w\|^2 = 9(1)+4(4)+1(9)+2(6)(1)+2(-3)(2)+2(-2)(-1) = 38$, so $\|3u+2v-w\| = \sqrt{38} = \sqrt{38}$.

Q.5 $\|u-2v+2w\|^2 = 1(9)+4(16)+4(25)+2(-2)(6)+2(2)(3)+2(-4)(-4) = 193$, so $\|u-2v+2w\| = \sqrt{193} = \sqrt{193}$.

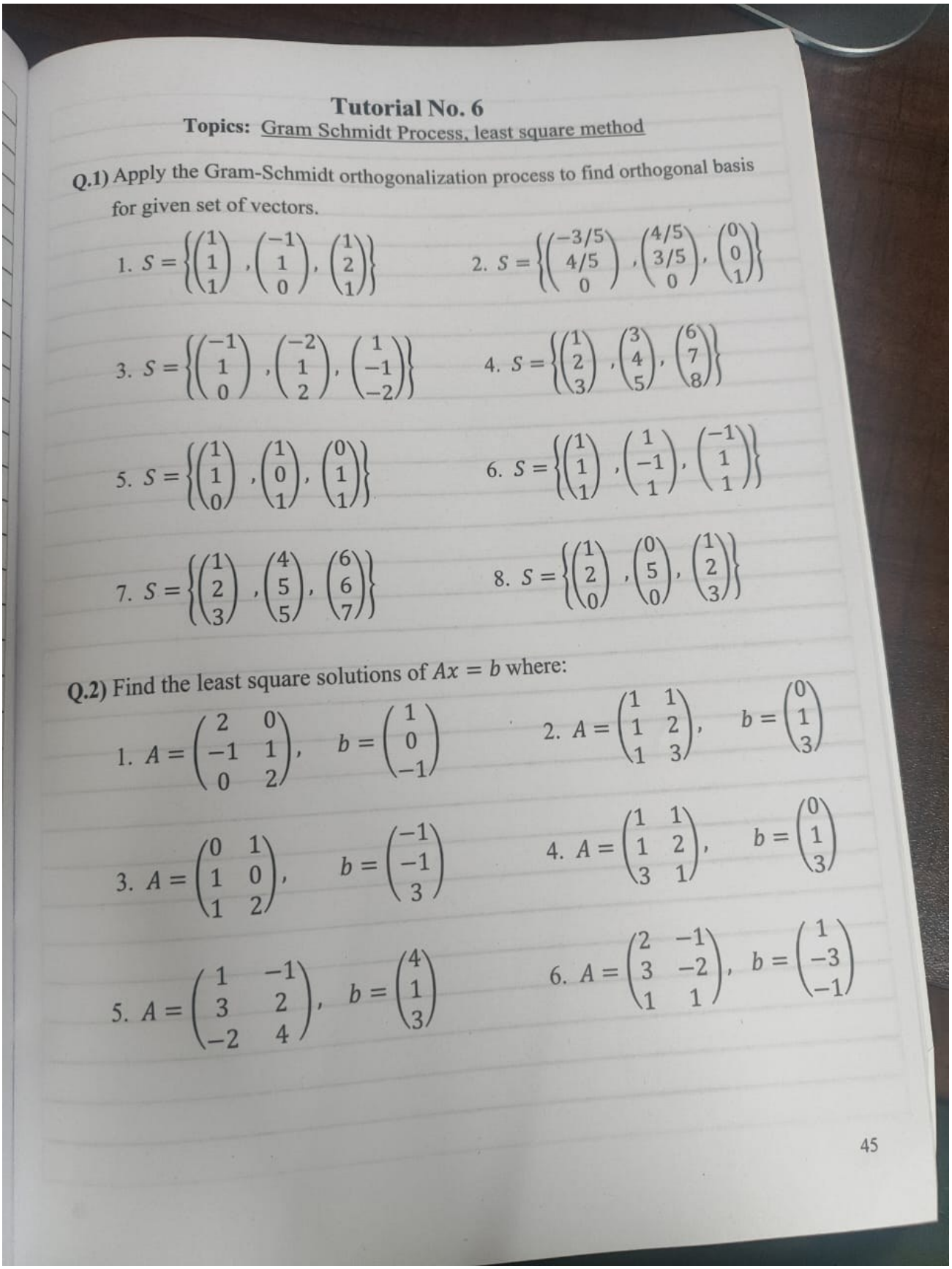
Q.6 $\|2u+3v-w\|^2 = 4(4)+9(1)+1(9)+2(6)(-1)+2(-2)(4)+2(-3)(2) = -6$, so $\|2u+3v-w\| = \sqrt{-6} = \sqrt{6}i$.

Q.7 $\|4u-v+2w\|^2 = 16(1)+1(9)+4(4)+2(-4)(2)+2(8)(-1)+2(-2)(3) = -3$, so $\|4u-v+2w\| = \sqrt{-3} = \sqrt{3}i$.

Q.8 $\|3u-2v+4w\|^2 = 9(4)+4(9)+16(16)+2(-6)(-2)+2(12)(3)+2(-8)(1) = 408$, so $\|3u-2v+4w\| = \sqrt{408} = 2\sqrt{102}$.

Tutorial No. 6: Gram–Schmidt Process, Least Square Method

Questions



Tutorial No. 6

Topics: Gram Schmidt Process, least square method

Q.1) Apply the Gram-Schmidt orthogonalization process to find orthogonal basis for given set of vectors.

1. $S = \left\{ \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} \right\}$

2. $S = \left\{ \begin{pmatrix} -3/5 \\ 4/5 \\ 0 \end{pmatrix}, \begin{pmatrix} 4/5 \\ 3/5 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right\}$

3. $S = \left\{ \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -2 \\ 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 1 \\ -1 \\ -2 \end{pmatrix} \right\}$

4. $S = \left\{ \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \begin{pmatrix} 3 \\ 4 \\ 5 \end{pmatrix}, \begin{pmatrix} 6 \\ 7 \\ 8 \end{pmatrix} \right\}$

5. $S = \left\{ \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \right\}$

6. $S = \left\{ \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}, \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix} \right\}$

7. $S = \left\{ \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \begin{pmatrix} 4 \\ 5 \\ 5 \end{pmatrix}, \begin{pmatrix} 6 \\ 6 \\ 7 \end{pmatrix} \right\}$

8. $S = \left\{ \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 5 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \right\}$

Q.2) Find the least square solutions of $Ax = b$ where:

1. $A = \begin{pmatrix} 2 & 0 \\ -1 & 1 \\ 0 & 2 \end{pmatrix}, \quad b = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$

2. $A = \begin{pmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{pmatrix}, \quad b = \begin{pmatrix} 0 \\ 1 \\ 3 \end{pmatrix}$

3. $A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \\ 1 & 2 \end{pmatrix}, \quad b = \begin{pmatrix} -1 \\ -1 \\ 3 \end{pmatrix}$

4. $A = \begin{pmatrix} 1 & 1 \\ 1 & 2 \\ 3 & 1 \end{pmatrix}, \quad b = \begin{pmatrix} 0 \\ 1 \\ 3 \end{pmatrix}$

5. $A = \begin{pmatrix} 1 & -1 \\ 3 & 2 \\ -2 & 4 \end{pmatrix}, \quad b = \begin{pmatrix} 4 \\ 1 \\ 3 \end{pmatrix}$

6. $A = \begin{pmatrix} 2 & -1 \\ 3 & -2 \\ 1 & 1 \end{pmatrix}, \quad b = \begin{pmatrix} 1 \\ -3 \\ -1 \end{pmatrix}$

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Solutions — Tutorial No. 6

Q.1 Gram–Schmidt

Formula: $v_1 = u_1$, $v_k = u_k - \sum_{j<k} \frac{\langle u_k, v_j\rangle}{\langle v_j, v_j\rangle} v_j$, and $e_k = v_k / \|v_k\|$.

(1) $u_1 = (1,\, 1,\, 1)^T,\ u_2 = (-1,\, 1,\, 0)^T,\ u_3 = (1,\, 2,\, 1)^T$

$$v_1 = u_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

$$v_2 = u_2 - \frac{\langle u_2, v_1\rangle}{\langle v_1, v_1\rangle} v_1 = \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}$$

$$v_3 = u_3 - \frac{\langle u_3, v_1\rangle}{\langle v_1, v_1\rangle} v_1 - \frac{\langle u_3, v_2\rangle}{\langle v_2, v_2\rangle} v_2 = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} - \frac{4}{3}\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} - \frac{1}{2}\begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} \frac{1}{6} \\ \frac{1}{6} \\ -\frac{1}{3} \end{pmatrix}$$

$$\text{Orthogonal basis: } \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix},\ \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix},\ \begin{pmatrix} \frac{1}{6} \\ \frac{1}{6} \\ -\frac{1}{3} \end{pmatrix}$$

$$\text{Orthonormal basis: } e_1 = \begin{pmatrix} \frac{\sqrt{3}}{3} \\ \frac{\sqrt{3}}{3} \\ \frac{\sqrt{3}}{3} \end{pmatrix},\ e_2 = \begin{pmatrix} -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} \\ 0 \end{pmatrix},\ e_3 = \begin{pmatrix} \frac{\sqrt{6}}{6} \\ \frac{\sqrt{6}}{6} \\ -\frac{\sqrt{6}}{3} \end{pmatrix}$$

(2) $u_1 = (-\frac{3}{5},\, \frac{4}{5},\, 0)^T,\ u_2 = (\frac{4}{5},\, \frac{3}{5},\, 0)^T,\ u_3 = (0,\, 0,\, 1)^T$

$$v_1 = u_1 = \begin{pmatrix} -\frac{3}{5} \\ \frac{4}{5} \\ 0 \end{pmatrix}$$

$$v_2 = u_2 - \frac{\langle u_2, v_1\rangle}{\langle v_1, v_1\rangle} v_1 = \begin{pmatrix} \frac{4}{5} \\ \frac{3}{5} \\ 0 \end{pmatrix} = \begin{pmatrix} \frac{4}{5} \\ \frac{3}{5} \\ 0 \end{pmatrix}$$

$$v_3 = u_3 - \frac{\langle u_3, v_1\rangle}{\langle v_1, v_1\rangle} v_1 - \frac{\langle u_3, v_2\rangle}{\langle v_2, v_2\rangle} v_2 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

$$\text{Orthogonal basis: } \begin{pmatrix} -\frac{3}{5} \\ \frac{4}{5} \\ 0 \end{pmatrix},\ \begin{pmatrix} \frac{4}{5} \\ \frac{3}{5} \\ 0 \end{pmatrix},\ \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

$$\text{Orthonormal basis: } e_1 = \begin{pmatrix} -\frac{3}{5} \\ \frac{4}{5} \\ 0 \end{pmatrix},\ e_2 = \begin{pmatrix} \frac{4}{5} \\ \frac{3}{5} \\ 0 \end{pmatrix},\ e_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

(3) $u_1 = (-1,\, 1,\, 0)^T,\ u_2 = (-2,\, 1,\, 2)^T,\ u_3 = (1,\, -1,\, -2)^T$

$$v_1 = u_1 = \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}$$

$$v_2 = u_2 - \frac{\langle u_2, v_1\rangle}{\langle v_1, v_1\rangle} v_1 = \begin{pmatrix} -2 \\ 1 \\ 2 \end{pmatrix} - \frac{3}{2}\begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} -\frac{1}{2} \\ -\frac{1}{2} \\ 2 \end{pmatrix}$$

$$v_3 = u_3 - \frac{\langle u_3, v_1\rangle}{\langle v_1, v_1\rangle}v_1 - \frac{\langle u_3, v_2\rangle}{\langle v_2, v_2\rangle}v_2 = \begin{pmatrix}1\\-1\\-2\end{pmatrix} + 1\begin{pmatrix}-1\\1\\0\end{pmatrix} + \frac{8}{9}\begin{pmatrix}-\frac{1}{2}\\-\frac{1}{2}\\2\end{pmatrix} = \begin{pmatrix}-\frac{4}{9}\\-\frac{4}{9}\\-\frac{2}{9}\end{pmatrix}$$

$$\text{Orthogonal basis: } \begin{pmatrix}-1\\1\\0\end{pmatrix},\ \begin{pmatrix}-\frac{1}{2}\\-\frac{1}{2}\\2\end{pmatrix},\ \begin{pmatrix}-\frac{4}{9}\\-\frac{4}{9}\\-\frac{2}{9}\end{pmatrix}$$

$$\text{Orthonormal basis: } e_1 = \begin{pmatrix}-\frac{\sqrt{2}}{2}\\\frac{\sqrt{2}}{2}\\0\end{pmatrix},\ e_2 = \begin{pmatrix}-\frac{\sqrt{2}}{6}\\-\frac{\sqrt{2}}{6}\\\frac{2\sqrt{2}}{3}\end{pmatrix},\ e_3 = \begin{pmatrix}-\frac{2}{3}\\-\frac{2}{3}\\-\frac{1}{3}\end{pmatrix}$$

(4) $u_1 = (1,\,2,\,3)^T,\ u_2 = (3,\,4,\,5)^T,\ u_3 = (6,\,7,\,8)^T$

$$v_1 = u_1 = \begin{pmatrix}1\\2\\3\end{pmatrix}$$

$$v_2 = u_2 - \frac{\langle u_2, v_1\rangle}{\langle v_1, v_1\rangle}v_1 = \begin{pmatrix}3\\4\\5\end{pmatrix} - \frac{13}{7}\begin{pmatrix}1\\2\\3\end{pmatrix} = \begin{pmatrix}\frac{8}{7}\\\frac{2}{7}\\-\frac{4}{7}\end{pmatrix}$$

$$v_3 = u_3 - \frac{\langle u_3, v_1\rangle}{\langle v_1, v_1\rangle}v_1 - \frac{\langle u_3, v_2\rangle}{\langle v_2, v_2\rangle}v_2 = \begin{pmatrix}6\\7\\8\end{pmatrix} - \frac{22}{7}\begin{pmatrix}1\\2\\3\end{pmatrix} - \frac{5}{2}\begin{pmatrix}\frac{8}{7}\\\frac{2}{7}\\-\frac{4}{7}\end{pmatrix} = \begin{pmatrix}0\\0\\0\end{pmatrix}$$

Since $v_3 = \mathbf{0}$, the given vectors are linearly dependent; the orthogonal basis of their span has 2 vectors.

$$\text{Orthogonal basis: } \begin{pmatrix}1\\2\\3\end{pmatrix},\ \begin{pmatrix}\frac{8}{7}\\\frac{2}{7}\\-\frac{4}{7}\end{pmatrix}$$

$$\text{Orthonormal basis: } e_1 = \begin{pmatrix}\frac{\sqrt{14}}{14}\\\frac{\sqrt{14}}{7}\\\frac{3\sqrt{14}}{14}\end{pmatrix},\ e_2 = \begin{pmatrix}\frac{4\sqrt{21}}{21}\\\frac{\sqrt{21}}{21}\\-\frac{2\sqrt{21}}{21}\end{pmatrix}$$

(5) $u_1 = (1,\,1,\,0)^T,\ u_2 = (1,\,0,\,1)^T,\ u_3 = (0,\,1,\,1)^T$

$$v_1 = u_1 = \begin{pmatrix}1\\1\\0\end{pmatrix}$$

$$v_2 = u_2 - \frac{\langle u_2, v_1\rangle}{\langle v_1, v_1\rangle}v_1 = \begin{pmatrix}1\\0\\1\end{pmatrix} - \frac{1}{2}\begin{pmatrix}1\\1\\0\end{pmatrix} = \begin{pmatrix}\frac{1}{2}\\-\frac{1}{2}\\1\end{pmatrix}$$

$$v_3 = u_3 - \frac{\langle u_3, v_1\rangle}{\langle v_1, v_1\rangle}v_1 - \frac{\langle u_3, v_2\rangle}{\langle v_2, v_2\rangle}v_2 = \begin{pmatrix}0\\1\\1\end{pmatrix} - \frac{1}{2}\begin{pmatrix}1\\1\\0\end{pmatrix} - \frac{1}{3}\begin{pmatrix}\frac{1}{2}\\-\frac{1}{2}\\1\end{pmatrix} = \begin{pmatrix}-\frac{2}{3}\\\frac{2}{3}\\\frac{2}{3}\end{pmatrix}$$

$$\text{Orthogonal basis: } \begin{pmatrix}1\\1\\0\end{pmatrix},\ \begin{pmatrix}\frac{1}{2}\\-\frac{1}{2}\\1\end{pmatrix},\ \begin{pmatrix}-\frac{2}{3}\\\frac{2}{3}\\\frac{2}{3}\end{pmatrix}$$

$$\text{Orthonormal basis: } e_1 = \begin{pmatrix}\frac{\sqrt{2}}{2}\\\frac{\sqrt{2}}{2}\\0\end{pmatrix},\ e_2 = \begin{pmatrix}\frac{\sqrt{6}}{6}\\-\frac{\sqrt{6}}{6}\\\frac{\sqrt{6}}{3}\end{pmatrix},\ e_3 = \begin{pmatrix}-\frac{\sqrt{3}}{3}\\\frac{\sqrt{3}}{3}\\\frac{\sqrt{3}}{3}\end{pmatrix}$$

(6) $u_1 = (1,\ 1,\ 1)^T,\ u_2 = (1,\ -1,\ 1)^T,\ u_3 = (-1,\ 1,\ 1)^T$

$$v_1 = u_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

$$v_2 = u_2 - \frac{\langle u_2, v_1 \rangle}{\langle v_1, v_1 \rangle} v_1 = \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} - \frac{1}{3} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} \frac{2}{3} \\ -\frac{4}{3} \\ \frac{2}{3} \end{pmatrix}$$

$$v_3 = u_3 - \frac{\langle u_3, v_1 \rangle}{\langle v_1, v_1 \rangle} v_1 - \frac{\langle u_3, v_2 \rangle}{\langle v_2, v_2 \rangle} v_2 = \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix} - \frac{1}{3} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + \frac{1}{2} \begin{pmatrix} \frac{2}{3} \\ -\frac{4}{3} \\ \frac{2}{3} \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$$

$$\text{Orthogonal basis: } \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix},\ \begin{pmatrix} \frac{2}{3} \\ -\frac{4}{3} \\ \frac{2}{3} \end{pmatrix},\ \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$$

$$\text{Orthonormal basis: } e_1 = \begin{pmatrix} \frac{\sqrt{3}}{3} \\ \frac{\sqrt{3}}{3} \\ \frac{\sqrt{3}}{3} \end{pmatrix},\ e_2 = \begin{pmatrix} \frac{\sqrt{6}}{6} \\ -\frac{\sqrt{6}}{3} \\ \frac{\sqrt{6}}{6} \end{pmatrix},\ e_3 = \begin{pmatrix} -\frac{\sqrt{2}}{2} \\ 0 \\ \frac{\sqrt{2}}{2} \end{pmatrix}$$

(7) $u_1 = (1,\ 2,\ 3)^T,\ u_2 = (4,\ 5,\ 5)^T,\ u_3 = (6,\ 6,\ 7)^T$

$$v_1 = u_1 = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$$

$$v_2 = u_2 - \frac{\langle u_2, v_1 \rangle}{\langle v_1, v_1 \rangle} v_1 = \begin{pmatrix} 4 \\ 5 \\ 5 \end{pmatrix} - \frac{29}{14} \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} = \begin{pmatrix} \frac{27}{14} \\ \frac{6}{7} \\ -\frac{17}{14} \end{pmatrix}$$

$$v_3 = u_3 - \frac{\langle u_3, v_1 \rangle}{\langle v_1, v_1 \rangle} v_1 - \frac{\langle u_3, v_2 \rangle}{\langle v_2, v_2 \rangle} v_2 = \begin{pmatrix} 6 \\ 6 \\ 7 \end{pmatrix} - \frac{39}{14} \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} - \frac{115}{83} \begin{pmatrix} \frac{27}{14} \\ \frac{6}{7} \\ -\frac{17}{14} \end{pmatrix} = \begin{pmatrix} \frac{45}{83} \\ -\frac{63}{83} \\ \frac{27}{83} \end{pmatrix}$$

$$\text{Orthogonal basis: } \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix},\ \begin{pmatrix} \frac{27}{14} \\ \frac{6}{7} \\ -\frac{17}{14} \end{pmatrix},\ \begin{pmatrix} \frac{45}{83} \\ -\frac{63}{83} \\ \frac{27}{83} \end{pmatrix}$$

$$\text{Orthonormal basis: } e_1 = \begin{pmatrix} \frac{\sqrt{14}}{14} \\ \frac{\sqrt{14}}{7} \\ \frac{3\sqrt{14}}{14} \end{pmatrix},\ e_2 = \begin{pmatrix} \frac{27\sqrt{1162}}{1162} \\ \frac{6\sqrt{1162}}{581} \\ -\frac{17\sqrt{1162}}{1162} \end{pmatrix},\ e_3 = \begin{pmatrix} \frac{5\sqrt{83}}{83} \\ -\frac{7\sqrt{83}}{83} \\ \frac{3\sqrt{83}}{83} \end{pmatrix}$$

(8) $u_1 = (1,\ 2,\ 0)^T,\ u_2 = (0,\ 5,\ 0)^T,\ u_3 = (1,\ 2,\ 3)^T$

$$v_1 = u_1 = \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}$$

$$v_2 = u_2 - \frac{\langle u_2, v_1 \rangle}{\langle v_1, v_1 \rangle} v_1 = \begin{pmatrix} 0 \\ 5 \\ 0 \end{pmatrix} - 2 \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} = \begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix}$$

$$v_3 = u_3 - \frac{\langle u_3, v_1 \rangle}{\langle v_1, v_1 \rangle} v_1 - \frac{\langle u_3, v_2 \rangle}{\langle v_2, v_2 \rangle} v_2 = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} - 1 \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 3 \end{pmatrix}$$

$$\text{Orthogonal basis: } \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix},\ \begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix},\ \begin{pmatrix} 0 \\ 0 \\ 3 \end{pmatrix}$$

$$\text{Orthonormal basis: } e_1 = \begin{pmatrix} \frac{\sqrt{5}}{5} \\ \frac{2\sqrt{5}}{5} \\ 0 \end{pmatrix},\ e_2 = \begin{pmatrix} -\frac{2\sqrt{5}}{5} \\ \frac{\sqrt{5}}{5} \\ 0 \end{pmatrix},\ e_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

Q.2 Least squares solutions of $A\mathbf{x} = \mathbf{b}$

Solve the normal equations $A^T A\,\hat{\mathbf{x}} = A^T\mathbf{b}$.

(1)

$$A^T A = \begin{pmatrix} 5 & -1 \\ -1 & 5 \end{pmatrix}, \quad A^T\mathbf{b} = \begin{pmatrix} 2 \\ -2 \end{pmatrix}$$
$$\hat{\mathbf{x}} = (A^T A)^{-1} A^T\mathbf{b} = \begin{pmatrix} \frac{1}{3} \\ -\frac{1}{3} \end{pmatrix}$$

Least-squares solution: $\hat{x}_1 = \frac{1}{3},\ \hat{x}_2 = -\frac{1}{3}$ (residual $\|\mathbf{b} - A\hat{\mathbf{x}}\| = \frac{\sqrt{6}}{3}$).

(2)

$$A^T A = \begin{pmatrix} 3 & 6 \\ 6 & 14 \end{pmatrix}, \quad A^T\mathbf{b} = \begin{pmatrix} 4 \\ 11 \end{pmatrix}$$
$$\hat{\mathbf{x}} = (A^T A)^{-1} A^T\mathbf{b} = \begin{pmatrix} -\frac{5}{3} \\ \frac{3}{2} \end{pmatrix}$$

Least-squares solution: $\hat{x}_1 = -\frac{5}{3},\ \hat{x}_2 = \frac{3}{2}$ (residual $\|\mathbf{b} - A\hat{\mathbf{x}}\| = \frac{\sqrt{6}}{6}$).

(3)

$$A^T A = \begin{pmatrix} 2 & 2 \\ 2 & 5 \end{pmatrix}, \quad A^T\mathbf{b} = \begin{pmatrix} 2 \\ 5 \end{pmatrix}$$
$$\hat{\mathbf{x}} = (A^T A)^{-1} A^T\mathbf{b} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

Least-squares solution: $\hat{x}_1 = 0,\ \hat{x}_2 = 1$ (residual $\|\mathbf{b} - A\hat{\mathbf{x}}\| = \sqrt{6}$).

(4)

$$A^T A = \begin{pmatrix} 11 & 6 \\ 6 & 6 \end{pmatrix}, \quad A^T\mathbf{b} = \begin{pmatrix} 10 \\ 5 \end{pmatrix}$$
$$\hat{\mathbf{x}} = (A^T A)^{-1} A^T\mathbf{b} = \begin{pmatrix} 1 \\ -\frac{1}{6} \end{pmatrix}$$

Least-squares solution: $\hat{x}_1 = 1,\ \hat{x}_2 = -\frac{1}{6}$ (residual $\|\mathbf{b} - A\hat{\mathbf{x}}\| = \frac{\sqrt{30}}{6}$).

(5)

$$A^T A = \begin{pmatrix} 14 & -3 \\ -3 & 21 \end{pmatrix}, \quad A^T\mathbf{b} = \begin{pmatrix} 1 \\ 10 \end{pmatrix}$$
$$\hat{\mathbf{x}} = (A^T A)^{-1} A^T\mathbf{b} = \begin{pmatrix} \frac{17}{95} \\ \frac{143}{285} \end{pmatrix}$$

Least-squares solution: $\hat{x}_1 = \frac{17}{95},\ \hat{x}_2 = \frac{143}{285}$ (residual $\|\mathbf{b} - A\hat{\mathbf{x}}\| = \frac{77\sqrt{285}}{285}$).

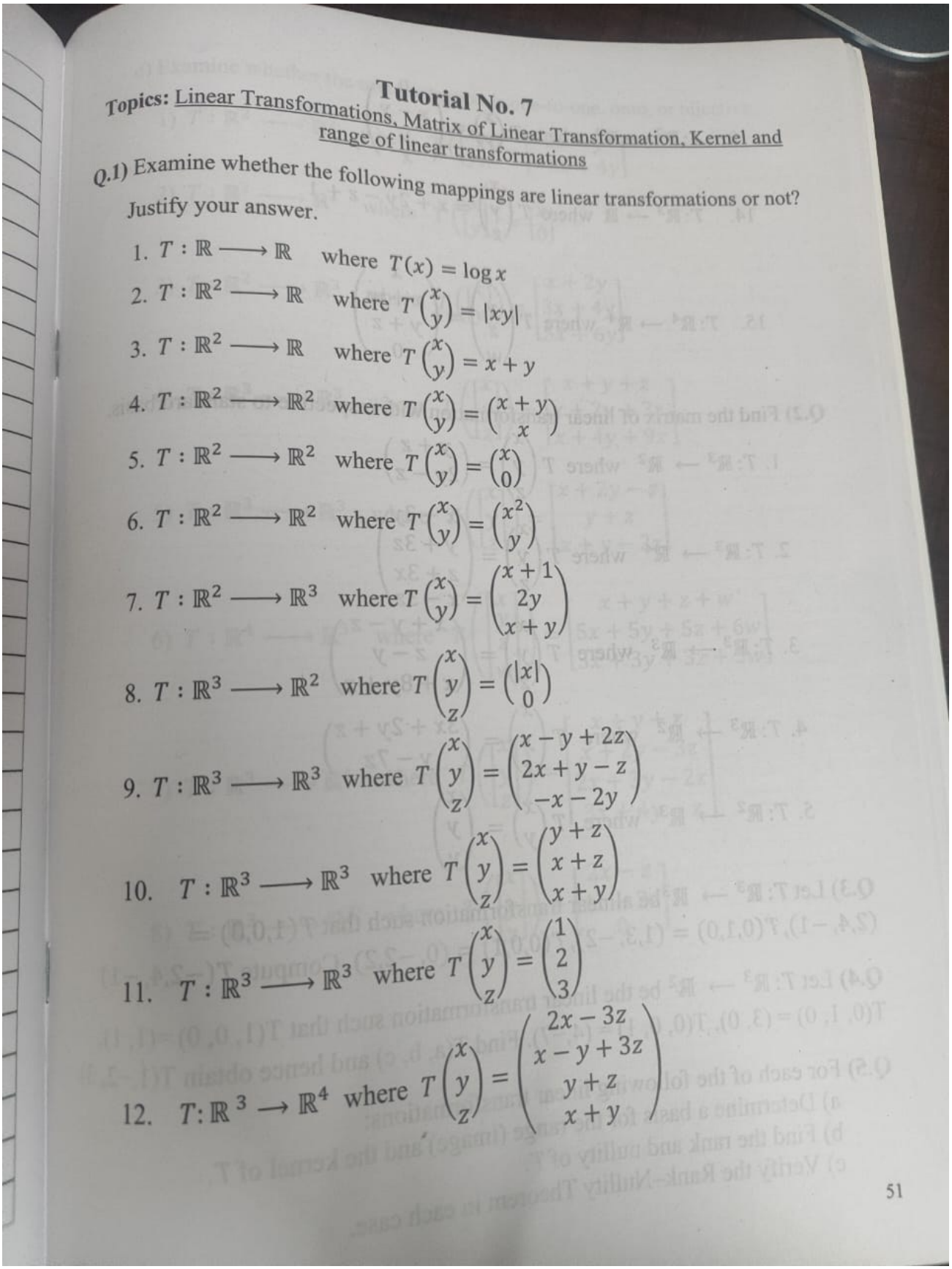
(6)

$$A^T A = \begin{pmatrix} 14 & -7 \\ -7 & 6 \end{pmatrix}, \quad A^T\mathbf{b} = \begin{pmatrix} -8 \\ 4 \end{pmatrix}$$
$$\hat{\mathbf{x}} = (A^T A)^{-1} A^T\mathbf{b} = \begin{pmatrix} -\frac{4}{7} \\ 0 \end{pmatrix}$$

Least-squares solution: $\hat{x}_1 = -\frac{4}{7},\ \hat{x}_2 = 0$ (residual $\|\mathbf{b} - A\hat{\mathbf{x}}\| = \frac{3\sqrt{35}}{7}$).

Tutorial No. 7: Linear Transformations, Matrix of a Linear Transformation, Kernel and Range

Questions



Tutorial No. 7

Topics: Linear Transformations, Matrix of Linear Transformation, Kernel and range of linear transformations

Q.1) Examine whether the following mappings are linear transformations or not? Justify your answer.

1. $T : \mathbb{R} \longrightarrow \mathbb{R}$ where $T(x) = \log x$
2. $T : \mathbb{R}^2 \longrightarrow \mathbb{R}$ where $T\begin{pmatrix} x \\ y \end{pmatrix} = |xy|$
3. $T : \mathbb{R}^2 \longrightarrow \mathbb{R}$ where $T\begin{pmatrix} x \\ y \end{pmatrix} = x + y$
4. $T : \mathbb{R}^2 \longrightarrow \mathbb{R}^2$ where $T\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x+y \\ x \end{pmatrix}$
5. $T : \mathbb{R}^2 \longrightarrow \mathbb{R}^2$ where $T\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ 0 \end{pmatrix}$
6. $T : \mathbb{R}^2 \longrightarrow \mathbb{R}^2$ where $T\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x^2 \\ y \end{pmatrix}$
7. $T : \mathbb{R}^2 \longrightarrow \mathbb{R}^3$ where $T\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x+1 \\ 2y \\ x+y \end{pmatrix}$
8. $T : \mathbb{R}^3 \longrightarrow \mathbb{R}^2$ where $T\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} |x| \\ 0 \end{pmatrix}$
9. $T : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ where $T\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x-y+2z \\ 2x+y-z \\ -x-2y \end{pmatrix}$
10. $T : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ where $T\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} y+z \\ x+z \\ x+y \end{pmatrix}$
11. $T : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ where $T\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$
12. $T : \mathbb{R}^3 \longrightarrow \mathbb{R}^4$ where $T\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 2x-3z \\ x-y+3z \\ y+z \\ x+y \end{pmatrix}$

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13. $T: \mathbb{R}^2 \to \mathbb{R}^4$ where $T\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ x-y \\ y \\ x \end{pmatrix}$

14. $T: \mathbb{R}^3 \to \mathbb{R}$ where $T\begin{pmatrix} x \\ y \\ z \end{pmatrix} = x + 2y - z + 1$

15. $T: \mathbb{R}^4 \to \mathbb{R}^4$ where $T\begin{pmatrix} x \\ y \\ z \\ w \end{pmatrix} = \begin{pmatrix} x \\ x+w \\ y+z \\ 0 \end{pmatrix}$

Q.2) Find the matrix of linear transformation with respective to standard basis.

1. $T: \mathbb{R}^3 \to \mathbb{R}^2$ where $T\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} y+z \\ 2y-z \end{pmatrix}$

2. $T: \mathbb{R}^3 \to \mathbb{R}^4$ where $T\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x-3y \\ y+3z \\ z+3x \\ x-3z \end{pmatrix}$

3. $T: \mathbb{R}^3 \to \mathbb{R}^3$ where $T\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x+y-z \\ z-y \\ x+8y+z \end{pmatrix}$

4. $T: \mathbb{R}^3 \to \mathbb{R}^2$ where $T\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 3x+2y+z \\ y-7z \end{pmatrix}$

5. $T: \mathbb{R}^2 \to \mathbb{R}^3$ where $T\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ y \\ y \end{pmatrix}$

Q.3) Let $T: \mathbb{R}^3 \to \mathbb{R}^3$ be alinear transformation such that $T(1,0,0) = (2,4,-1), T(0,1,0) = (1,3,-2), T(0,0,1) = (0,-2,2)$. Compute $T(-2,4,-1)$

Q.4) Let $T: \mathbb{R}^3 \to \mathbb{R}^2$ be the linear transformation such that $T(1, 0, 0) = (1, 1)$, $T(0, 1, 0) = (3, 0)$, $T(0, 0, 1) = (4,-7)$. Find $T(a, b, c)$ and hence obtain $T(1,-2, 3)$

Q.5) For each of the following linear transformations:

a) Determine a basis for the range (image) and the kernel of T.

b) Find the rank and nullity of T.

c) Verify the Rank–Nullity Theorem in each case.

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d) Examine whether the transformation is one-to-one, onto, or bijective.

1) $T : \mathbb{R}^2 \longrightarrow \mathbb{R}^2$ where $T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} 2x - y \\ -8x + 4y \end{bmatrix}$

2) $T : \mathbb{R}^2 \longrightarrow \mathbb{R}^2$ where $T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} x \\ 0 \end{bmatrix}$

3) $T : \mathbb{R}^2 \longrightarrow \mathbb{R}^3$ where $T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} x + 2y \\ 3x + 4y \\ 5x + 6y \end{bmatrix}$

4) $T : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ where $T\left(\begin{bmatrix} x \\ y \\ z \end{bmatrix}\right) = \begin{bmatrix} x + y + z \\ x + 2y + 3z \\ x + 4y + 9z \end{bmatrix}$

5) $T : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ where $T\left(\begin{bmatrix} x \\ y \\ z \end{bmatrix}\right) = \begin{bmatrix} x + 2y - z \\ y + z \\ x + y - 2z \end{bmatrix}$

6) $T : \mathbb{R}^4 \longrightarrow \mathbb{R}^3$ where $T\left(\begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix}\right) = \begin{bmatrix} x + y + z + w \\ 5x + 5y + 5z + 6w \\ 3x + 3y + 3z + 3w \end{bmatrix}$

7) $T : \mathbb{R}^3 \longrightarrow \mathbb{R}^4$ where $T\left(\begin{bmatrix} x \\ y \\ z \end{bmatrix}\right) = \begin{bmatrix} x + y + z \\ x + 2y - 3z \\ 2x + 3y - 2z \\ 3x + 4y - z \end{bmatrix}$

8) $T : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ where $T\left(\begin{bmatrix} x \\ y \\ z \end{bmatrix}\right) = \begin{bmatrix} 2x - z \\ 4x - 2z \\ 0 \end{bmatrix}$

Solutions — Tutorial No. 7

Q.1 Linear or not?

(Test: $T(u+v) = T(u) + T(v)$ and $T(cu) = cT(u)$; equivalently every component must be a linear homogeneous expression in the inputs.)

1. Not linear. $T(1+1) = \log 2 \neq T(1) + T(1) = 0$, so additivity fails (also $T(0)$ is undefined and $T(2 \cdot 1) \neq 2T(1)$).

2. Not linear. $T(c\mathbf{x}) = |c^2xy| = c^2|xy| \neq cT(\mathbf{x})$; e.g. $T(1,1) = 1$, $T(2,2) = 4 \neq 2$.

3. Linear. $T(\mathbf{u}+\mathbf{v}) = (x_1+x_2) + (y_1+y_2) = T(\mathbf{u}) + T(\mathbf{v})$, $T(c\mathbf{u}) = c(x+y)$. Matrix $\begin{bmatrix}1 & 1\end{bmatrix}$.

4. Linear. Components $x+y,\ x$ are linear: matrix $\begin{pmatrix}1 & 1\\ 1 & 0\end{pmatrix}$.

5. Linear. $T(x,y) = (x,0)$: matrix $\begin{pmatrix}1 & 0\\ 0 & 0\end{pmatrix}$ (projection on the x-axis).

6. Not linear. x^2 is not linear: $T(2,0) = (4,0) \neq 2T(1,0) = (2,0)$.

7. Not linear. $T(0,0) = (1,0,0) \neq \mathbf{0}$; a linear map must send $\mathbf{0}$ to $\mathbf{0}$.

8. Not linear. $|x|$ is not linear: $T(-1,0,0) = (1,0) \neq -T(1,0,0) = (-1,0)$.

9. Linear. All components linear homogeneous: matrix $\begin{pmatrix}1 & -1 & 2\\ 2 & 1 & -1\\ -1 & -2 & 0\end{pmatrix}$.

10. Linear. Matrix $\begin{pmatrix}0 & 1 & 1\\ 1 & 0 & 1\\ 1 & 1 & 0\end{pmatrix}$.

11. Not linear. Constant map: $T(\mathbf{0}) = (1,2,3) \neq \mathbf{0}$.

12. Linear. Matrix $\begin{pmatrix}2 & 0 & -3\\ 1 & -1 & 3\\ 0 & 1 & 1\\ 1 & 1 & 0\end{pmatrix}$ (all components linear homogeneous).

13. Linear. Matrix $\begin{pmatrix}0 & 0\\ 1 & -1\\ 0 & 1\\ 1 & 0\end{pmatrix}$.

14. Not linear. $T(\mathbf{0}) = 1 \neq 0$ (the "+1" makes it affine, not linear).

15. Linear. Matrix $\begin{pmatrix}1 & 0 & 0 & 0\\ 1 & 0 & 0 & 1\\ 0 & 1 & 1 & 0\\ 0 & 0 & 0 & 0\end{pmatrix}$.

Q.2 Standard matrices

The columns of $[T]$ are $T(e_1), T(e_2), \ldots$, i.e. the coefficient matrix of the formula.

(1) $[T] = \begin{pmatrix}0 & 1 & 1\\ 0 & 2 & -1\end{pmatrix}$ (size 2×3).

(2) $[T] = \begin{pmatrix}1 & -3 & 0\\ 0 & 1 & 3\\ 3 & 0 & 1\\ 1 & 0 & -3\end{pmatrix}$ (size 4×3).

(3) $[T] = \begin{pmatrix} 1 & 1 & -1 \\ 0 & -1 & 1 \\ 1 & 8 & 1 \end{pmatrix}$ (size 3×3).

(4) $[T] = \begin{pmatrix} 3 & 2 & 1 \\ 0 & 1 & -7 \end{pmatrix}$ (size 2×3).

(5) $[T] = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 1 \end{pmatrix}$ (size 3×2).

Q.3

$T(x, y, z) = xT(e_1) + yT(e_2) + zT(e_3)$, so $[T] = \begin{pmatrix} 2 & 1 & 0 \\ 4 & 3 & -2 \\ -1 & -2 & 2 \end{pmatrix}$ and

$$T(-2, 4, -1) = -2 \begin{pmatrix} 2 \\ 4 \\ -1 \end{pmatrix} + 4 \begin{pmatrix} 1 \\ 3 \\ -2 \end{pmatrix} - 1 \begin{pmatrix} 0 \\ -2 \\ 2 \end{pmatrix} = \begin{pmatrix} 0 \\ 6 \\ -8 \end{pmatrix}$$

Q.4

$T(a, b, c) = aT(e_1) + bT(e_2) + cT(e_3) = a(1, 1) + b(3, 0) + c(4, -7)$, hence

$$T(a, b, c) = (a + 3b + 4c,\ a - 7c), \qquad T(1, -2, 3) = (1 - 6 + 12,\ 1 - 21) = (7,\ -20)$$

Q.5 Range, kernel, rank, nullity (maps of part (d), p. 53)

(1) $T : \mathbb{R}^2 \to \mathbb{R}^2$, $[T] = \begin{pmatrix} 2 & -1 \\ -8 & 4 \end{pmatrix}$.

- RREF of $[T]$: $\begin{pmatrix} 1 & -\frac{1}{2} \\ 0 & 0 \end{pmatrix}$; pivot columns: 1.
- **Basis of range (column space):** $\left\{ \begin{pmatrix} 2 \\ -8 \end{pmatrix} \right\}$.
- **Basis of kernel (null space):** $\left\{ \begin{pmatrix} \frac{1}{2} \\ 1 \end{pmatrix} \right\}$.
- **Rank** $= 1$, **nullity** $= 1$. Rank–Nullity: $1 + 1 = 2 = \dim \mathbb{R}^2$ ✓.
- (d) one-to-one: **no** (kernel has dimension 1); onto: **no** (rank $1 <$ dim codomain 2); not bijective.

(2) $T : \mathbb{R}^2 \to \mathbb{R}^2$, $[T] = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$.

- RREF of $[T]$: $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$; pivot columns: 1.
- **Basis of range (column space):** $\left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right\}$.
- **Basis of kernel (null space):** $\left\{ \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\}$.
- **Rank** $= 1$, **nullity** $= 1$. Rank–Nullity: $1 + 1 = 2 = \dim \mathbb{R}^2$ ✓.
- (d) one-to-one: **no** (kernel has dimension 1); onto: **no** (rank $1 <$ dim codomain 2); not bijective.

(3) $T : \mathbb{R}^2 \to \mathbb{R}^3$, $[T] = \begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{pmatrix}$.

- RREF of $[T]$: $\begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}$; pivot columns: 1, 2.
- **Basis of range (column space):** $\left\{ \begin{pmatrix} 1 \\ 3 \\ 5 \end{pmatrix}, \begin{pmatrix} 2 \\ 4 \\ 6 \end{pmatrix} \right\}$.
- **Kernel** $= \{\mathbf{0}\}$ (no basis vectors; $T\mathbf{x} = \mathbf{0}$ only for $\mathbf{x} = \mathbf{0}$).
- **Rank** $= 2$, **nullity** $= 0$. Rank–Nullity: $2 + 0 = 2 = \dim \mathbb{R}^2$ ✓.
- (d) one-to-one: **yes** (kernel is trivial); onto: **no** (rank 2 < dim codomain 3); not bijective.

(4) $T : \mathbb{R}^3 \to \mathbb{R}^3$, $[T] = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 4 & 9 \end{pmatrix}$.

- RREF of $[T]$: $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$; pivot columns: 1, 2, 3.
- **Basis of range (column space):** $\left\{ \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \\ 4 \end{pmatrix}, \begin{pmatrix} 1 \\ 3 \\ 9 \end{pmatrix} \right\}$.
- **Kernel** $= \{\mathbf{0}\}$ (no basis vectors; $T\mathbf{x} = \mathbf{0}$ only for $\mathbf{x} = \mathbf{0}$).
- **Rank** $= 3$, **nullity** $= 0$. Rank–Nullity: $3 + 0 = 3 = \dim \mathbb{R}^3$ ✓.
- (d) one-to-one: **yes** (kernel is trivial); onto: **yes** (rank 3 = dim codomain 3); **bijective**.

(5) $T : \mathbb{R}^3 \to \mathbb{R}^3$, $[T] = \begin{pmatrix} 1 & 2 & -1 \\ 0 & 1 & 1 \\ 1 & 1 & -2 \end{pmatrix}$.

- RREF of $[T]$: $\begin{pmatrix} 1 & 0 & -3 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix}$; pivot columns: 1, 2.
- **Basis of range (column space):** $\left\{ \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix} \right\}$.
- **Basis of kernel (null space):** $\left\{ \begin{pmatrix} 3 \\ -1 \\ 1 \end{pmatrix} \right\}$.
- **Rank** $= 2$, **nullity** $= 1$. Rank–Nullity: $2 + 1 = 3 = \dim \mathbb{R}^3$ ✓.
- (d) one-to-one: **no** (kernel has dimension 1); onto: **no** (rank 2 < dim codomain 3); not bijective.

(6) $T : \mathbb{R}^4 \to \mathbb{R}^3$, $[T] = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 5 & 5 & 5 & 6 \\ 3 & 3 & 3 & 3 \end{pmatrix}$.

- RREF of $[T]$: $\begin{pmatrix} 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$; pivot columns: 1, 4.
- **Basis of range (column space):** $\left\{ \begin{pmatrix} 1 \\ 5 \\ 3 \end{pmatrix}, \begin{pmatrix} 1 \\ 6 \\ 3 \end{pmatrix} \right\}$.
- **Basis of kernel (null space):** $\left\{ \begin{pmatrix} -1 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \\ 1 \\ 0 \end{pmatrix} \right\}$.
- **Rank** $= 2$, **nullity** $= 2$. Rank–Nullity: $2 + 2 = 4 = \dim \mathbb{R}^4$ ✓.
- (d) one-to-one: **no** (kernel has dimension 2); onto: **no** (rank $2 <$ dim codomain 3); not bijective.

(7) $T : \mathbb{R}^3 \to \mathbb{R}^4$, $[T] = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & -3 \\ 2 & 3 & -2 \\ 3 & 4 & -1 \end{pmatrix}$.

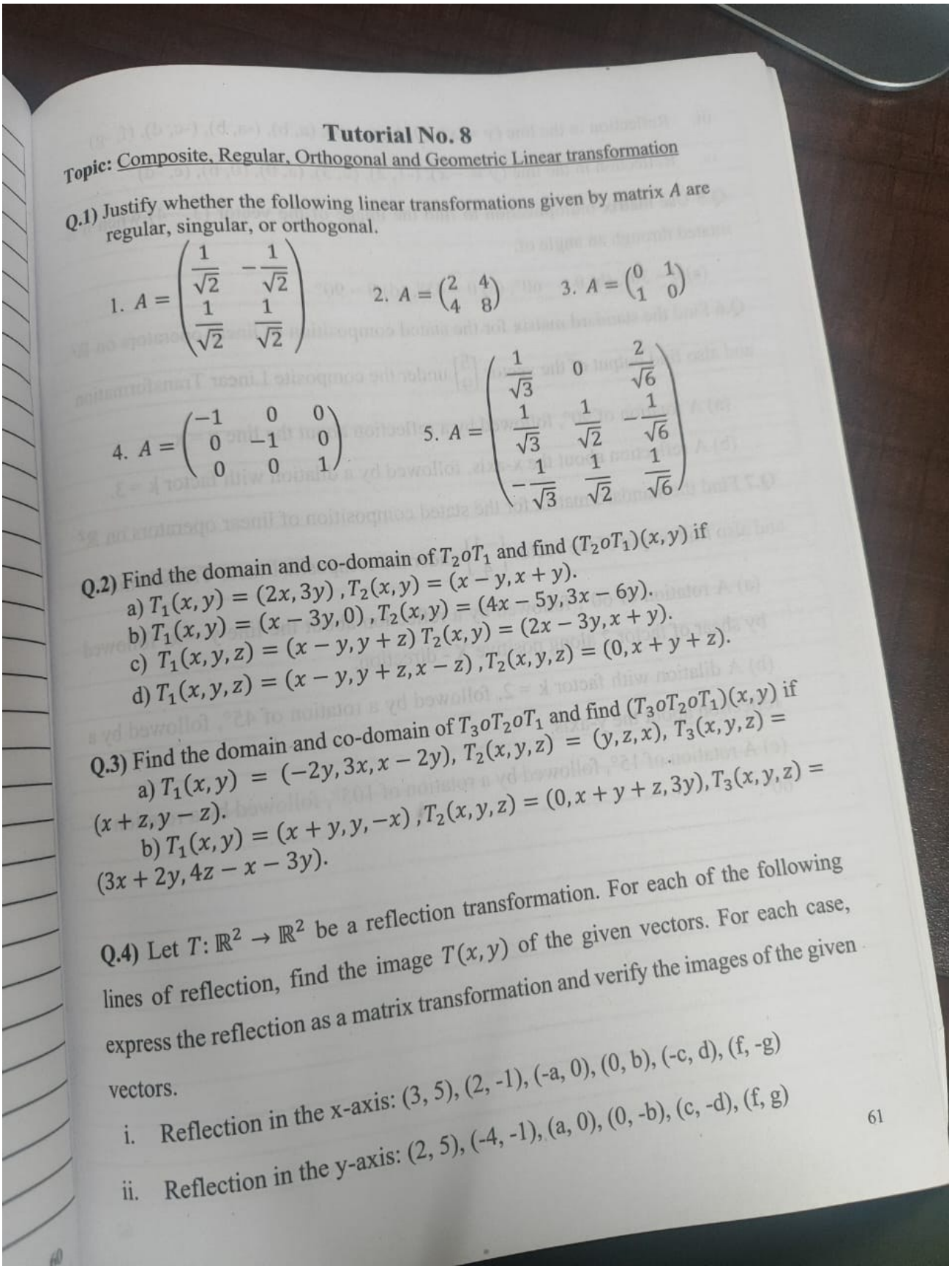
- RREF of $[T]$: $\begin{pmatrix} 1 & 0 & 5 \\ 0 & 1 & -4 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$; pivot columns: 1, 2.
- **Basis of range (column space):** $\left\{ \begin{pmatrix} 1 \\ 1 \\ 2 \\ 3 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \end{pmatrix} \right\}$.
- **Basis of kernel (null space):** $\left\{ \begin{pmatrix} -5 \\ 4 \\ 1 \end{pmatrix} \right\}$.
- **Rank** $= 2$, **nullity** $= 1$. Rank–Nullity: $2 + 1 = 3 = \dim \mathbb{R}^3$ ✓.
- (d) one-to-one: **no** (kernel has dimension 1); onto: **no** (rank $2 <$ dim codomain 4); not bijective.

(8) $T : \mathbb{R}^3 \to \mathbb{R}^3$, $[T] = \begin{pmatrix} 2 & 0 & -1 \\ 4 & 0 & -2 \\ 0 & 0 & 0 \end{pmatrix}$.

- RREF of $[T]$: $\begin{pmatrix} 1 & 0 & -\frac{1}{2} \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$; pivot columns: 1.
- **Basis of range (column space):** $\left\{ \begin{pmatrix} 2 \\ 4 \\ 0 \end{pmatrix} \right\}$.
- **Basis of kernel (null space):** $\left\{ \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} \frac{1}{2} \\ 0 \\ 1 \end{pmatrix} \right\}$.
- **Rank** $= 1$, **nullity** $= 2$. Rank–Nullity: $1 + 2 = 3 = \dim \mathbb{R}^3$ ✓.
- (d) one-to-one: **no** (kernel has dimension 2); onto: **no** (rank $1 <$ dim codomain 3); not bijective.

Tutorial No. 8: Composite, Regular, Orthogonal and Geometric Linear Transformations

Questions



Tutorial No. 8

Topic: Composite, Regular, Orthogonal and Geometric Linear transformation

Q.1) Justify whether the following linear transformations given by matrix A are regular, singular, or orthogonal.

1. $A = \begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}$ 2. $A = \begin{pmatrix} 2 & 4 \\ 4 & 8 \end{pmatrix}$ 3. $A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$

4. $A = \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ 5. $A = \begin{pmatrix} \frac{1}{\sqrt{3}} & 0 & \frac{2}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{6}} \\ -\frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \end{pmatrix}$

Q.2) Find the domain and co-domain of $T_2 o T_1$ and find $(T_2 o T_1)(x,y)$ if

a) $T_1(x,y) = (2x, 3y)$, $T_2(x,y) = (x - y, x + y)$.

b) $T_1(x,y) = (x - 3y, 0)$, $T_2(x,y) = (4x - 5y, 3x - 6y)$.

c) $T_1(x,y,z) = (x - y, y + z)$ $T_2(x,y) = (2x - 3y, x + y)$.

d) $T_1(x,y,z) = (x - y, y + z, x - z)$, $T_2(x,y,z) = (0, x + y + z)$.

Q.3) Find the domain and co-domain of $T_3 o T_2 o T_1$ and find $(T_3 o T_2 o T_1)(x,y)$ if

a) $T_1(x,y) = (-2y, 3x, x - 2y)$, $T_2(x,y,z) = (y,z,x)$, $T_3(x,y,z) = (x + z, y - z)$.

b) $T_1(x,y) = (x + y, y, -x)$, $T_2(x,y,z) = (0, x + y + z, 3y)$, $T_3(x,y,z) = (3x + 2y, 4z - x - 3y)$.

Q.4) Let $T: \mathbb{R}^2 \to \mathbb{R}^2$ be a reflection transformation. For each of the following lines of reflection, find the image $T(x,y)$ of the given vectors. For each case, express the reflection as a matrix transformation and verify the images of the given vectors.

i. Reflection in the x-axis: (3, 5), (2, -1), (-a, 0), (0, b), (-c, d), (f, -g)

ii. Reflection in the y-axis: (2, 5), (-4, -1), (a, 0), (0, -b), (c, -d), (f, g)

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iii. Reflection in the line $(y = x)$: (0, 1), (-1, 3), (a, b), (-a, b), (-c, d), (f, -g)

iv. Reflection in the line $(y = -x)$: (-1, 2), (2, 3), (a, 0), (0, b), (e, -d)

Q.5 Use matrix multiplication to find the image of the vector $(3, -4)$ when it is rotated through an angle of:

(a) $\theta = 30°$, (b) $\theta = -60°$, (c) $\theta = 45°$, (d) $\theta = 90°$

Q.6 Find the standard matrix for the stated composition of linear operators on $\mathbb{R}^2$ and also find output of the vector $\begin{bmatrix} 5 \\ 9 \end{bmatrix}$ under the composite Linear Transformation

(a) A rotation of 90°, followed by a reflection about the line y = x.

(b) A reflection about the x-axis, followed by a dilation with factor k = 3.

Q.7 Find the standard matrix for the stated composition of linear operators on $\mathbb{R}^2$ and also find output of the vector $\begin{bmatrix} -1 \\ 4 \end{bmatrix}$ under the composite Linear Transformation

(a) A rotation of 60°, followed by a reflection about the line y = x, followed by shear of factor 2 along positive X - direction.

(b) A dilation with factor k = 2, followed by a rotation of 45°, followed by a reflection about the y-axis.

(c) A rotation of 15°, followed by a rotation of 105°, followed by a rotation of 60°.

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Solutions — Tutorial No. 8

Q.1 Regular, singular, orthogonal

A matrix is *regular* (non-singular) if $\det A \neq 0$, *singular* if $\det A = 0$, and *orthogonal* if $A^T A = I$ (then $\det A = \pm 1$, so it is also regular).

(1) $A = \begin{pmatrix} \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix}$: $\det A = 1$, $A^T A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$. Since $A^T A = I$: **orthogonal** (and hence regular, $\det A = \pm 1$).

(2) $A = \begin{pmatrix} 2 & 4 \\ 4 & 8 \end{pmatrix}$: $\det A = 0$, $A^T A = \begin{pmatrix} 20 & 40 \\ 40 & 80 \end{pmatrix}$. Since $\det A = 0$: **singular** (not regular, not orthogonal).

(3) $A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$: $\det A = -1$, $A^T A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$. Since $A^T A = I$: **orthogonal** (and hence regular, $\det A = \pm 1$).

(4) $A = \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$: $\det A = 1$, $A^T A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$. Since $A^T A = I$: **orthogonal** (and hence regular, $\det A = \pm 1$).

(5) $A = \begin{pmatrix} \frac{\sqrt{3}}{3} & 0 & \frac{\sqrt{6}}{3} \\ \frac{\sqrt{3}}{3} & \frac{\sqrt{2}}{2} & -\frac{\sqrt{6}}{6} \\ -\frac{\sqrt{3}}{3} & \frac{\sqrt{2}}{2} & \frac{\sqrt{6}}{6} \end{pmatrix}$: $\det A = 1$, $A^T A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$. Since $A^T A = I$: **orthogonal** (and hence regular, $\det A = \pm 1$).

Q.2 Composition $T_2 \circ T_1$

$T_2 \circ T_1$ is defined when the co-domain of T_1 equals the domain of T_2; then domain = domain of T_1, co-domain = co-domain of T_2, and $(T_2 \circ T_1)(v) = T_2(T_1(v))$.

a) $T_1 : \mathbb{R}^2 \to \mathbb{R}^2$, $T_2 : \mathbb{R}^2 \to \mathbb{R}^2$. Domain of $T_2 \circ T_1$: $\mathbb{R}^2$; co-domain: $\mathbb{R}^2$.

$$(T_2 \circ T_1)(x, y) = T_2(2x,\, 3y) = (2x - 3y,\, 2x + 3y)$$

b) $T_1 : \mathbb{R}^2 \to \mathbb{R}^2$, $T_2 : \mathbb{R}^2 \to \mathbb{R}^2$. Domain of $T_2 \circ T_1$: $\mathbb{R}^2$; co-domain: $\mathbb{R}^2$.

$$(T_2 \circ T_1)(x, y) = T_2(x - 3y,\, 0) = (4x - 12y,\, 3x - 9y)$$

c) $T_1 : \mathbb{R}^3 \to \mathbb{R}^2$, $T_2 : \mathbb{R}^2 \to \mathbb{R}^2$. Domain of $T_2 \circ T_1$: $\mathbb{R}^3$; co-domain: $\mathbb{R}^2$.

$$(T_2 \circ T_1)(x, y, z) = T_2(x - y,\, y + z) = (2x - 5y - 3z,\, x + z)$$

d) $T_1 : \mathbb{R}^3 \to \mathbb{R}^3$, $T_2 : \mathbb{R}^3 \to \mathbb{R}^2$. Domain of $T_2 \circ T_1$: $\mathbb{R}^3$; co-domain: $\mathbb{R}^2$.

$$(T_2 \circ T_1)(x, y, z) = T_2(x - y,\, y + z,\, x - z) = (0,\, 2x)$$

Q.3 Composition $T_3 \circ T_2 \circ T_1$

a) $T_1 : \mathbb{R}^2 \to \mathbb{R}^3$, $T_2 : \mathbb{R}^3 \to \mathbb{R}^3$, $T_3 : \mathbb{R}^3 \to \mathbb{R}^2$; domain of $T_3 \circ T_2 \circ T_1$ is $\mathbb{R}^2$, co-domain $\mathbb{R}^2$.

$$T_1(x, y) = (-2y,\, 3x,\, x - 2y), \quad T_2 T_1(x, y) = (3x,\, x - 2y,\, -2y), \quad (T_3 T_2 T_1)(x, y) = (3x - 2y,\, x)$$

b) $T_1 : \mathbb{R}^2 \to \mathbb{R}^3$, $T_2 : \mathbb{R}^3 \to \mathbb{R}^3$, $T_3 : \mathbb{R}^3 \to \mathbb{R}^2$; domain of $T_3 \circ T_2 \circ T_1$ is $\mathbb{R}^2$, co-domain $\mathbb{R}^2$.

$$T_1(x, y) = (x + y,\, y,\, -x), \quad T_2 T_1(x, y) = (0,\, 2y,\, 3y), \quad (T_3 T_2 T_1)(x, y) = (4y,\, 6y)$$

Q.4 Reflections

Matrices: x-axis $\begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix}$, y-axis $\begin{pmatrix}-1 & 0\\ 0 & 1\end{pmatrix}$, line $y=x$: $\begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix}$, line $y=-x$: $\begin{pmatrix}0 & -1\\ -1 & 0\end{pmatrix}$; image $= M\,\mathbf{v}$ (matrix multiplication verifies each image).

i. Reflection in the x-axis, $M=\begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix}$: $T(3,5)=(3,-5)$; $T(2,-1)=(2,1)$; $T(-a,0)=(-a,0)$; $T(0,b)=(0,-b)$; $T(-c,d)=(-c,-d)$; $T(f,-g)=(f,g)$.

ii. Reflection in the y-axis, $M=\begin{pmatrix}-1 & 0\\ 0 & 1\end{pmatrix}$: $T(2,5)=(-2,5)$; $T(-4,-1)=(4,-1)$; $T(a,0)=(-a,0)$; $T(0,-b)=(0,-b)$; $T(c,-d)=(-c,-d)$; $T(f,g)=(-f,g)$.

iii. Reflection in $y=x$, $M=\begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix}$: $T(0,1)=(1,0)$; $T(-1,3)=(3,-1)$; $T(a,b)=(b,a)$; $T(-a,b)=(b,-a)$; $T(-c,d)=(d,-c)$; $T(f,-g)=(-g,f)$.

iv. Reflection in $y=-x$, $M=\begin{pmatrix}0 & -1\\ -1 & 0\end{pmatrix}$: $T(-1,2)=(-2,1)$; $T(2,3)=(-3,-2)$; $T(a,0)=(0,-a)$; $T(0,b)=(-b,0)$; $T(e,-d)=(d,-e)$.

Q.5 Rotation of $(3,-4)$

Rotation matrix $R_\theta=\begin{pmatrix}\cos\theta & -\sin\theta\\ \sin\theta & \cos\theta\end{pmatrix}$.

$$\text{(a) } \theta=30^\circ:\ R=\begin{pmatrix}\frac{\sqrt{3}}{2} & -\frac{1}{2}\\ \frac{1}{2} & \frac{\sqrt{3}}{2}\end{pmatrix},\ R\begin{pmatrix}3\\-4\end{pmatrix}=\begin{pmatrix}2+\frac{3\sqrt{3}}{2}\\ \frac{3}{2}-2\sqrt{3}\end{pmatrix}\approx\begin{pmatrix}4.5981\\-1.9641\end{pmatrix}$$

$$\text{(b) } \theta=-60^\circ:\ R=\begin{pmatrix}\frac{1}{2} & \frac{\sqrt{3}}{2}\\ -\frac{\sqrt{3}}{2} & \frac{1}{2}\end{pmatrix},\ R\begin{pmatrix}3\\-4\end{pmatrix}=\begin{pmatrix}\frac{3}{2}-2\sqrt{3}\\ -\frac{3\sqrt{3}}{2}-2\end{pmatrix}\approx\begin{pmatrix}-1.9641\\-4.5981\end{pmatrix}$$

$$\text{(c) } \theta=45^\circ:\ R=\begin{pmatrix}\frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2}\\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2}\end{pmatrix},\ R\begin{pmatrix}3\\-4\end{pmatrix}=\begin{pmatrix}\frac{7\sqrt{2}}{2}\\ -\frac{\sqrt{2}}{2}\end{pmatrix}\approx\begin{pmatrix}4.9497\\-0.7071\end{pmatrix}$$

$$\text{(d) } \theta=90^\circ:\ R=\begin{pmatrix}0 & -1\\ 1 & 0\end{pmatrix},\ R\begin{pmatrix}3\\-4\end{pmatrix}=\begin{pmatrix}4\\3\end{pmatrix}\approx\begin{pmatrix}4.0000\\3.0000\end{pmatrix}$$

Q.6 Composite operators on $\mathbb{R}^2$

If T_1 is applied first and T_2 second, the standard matrix is $[T_2][T_1]$ (later maps on the left).

$$\text{(a) } [T]=\begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix}\begin{pmatrix}0 & -1\\ 1 & 0\end{pmatrix}=\begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix},\quad [T]\begin{pmatrix}5\\9\end{pmatrix}=\begin{pmatrix}5\\-9\end{pmatrix}$$

$$\text{(b) } [T]=\begin{pmatrix}3 & 0\\ 0 & 3\end{pmatrix}\begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix}=\begin{pmatrix}3 & 0\\ 0 & -3\end{pmatrix},\quad [T]\begin{pmatrix}5\\9\end{pmatrix}=\begin{pmatrix}15\\-27\end{pmatrix}$$

Q.7

$$\text{(a) } [T]=S\,F\,R_{60^\circ}=\begin{pmatrix}1 & 2\\ 0 & 1\end{pmatrix}\begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix}\begin{pmatrix}\frac{1}{2} & -\frac{\sqrt{3}}{2}\\ \frac{\sqrt{3}}{2} & \frac{1}{2}\end{pmatrix}=\begin{pmatrix}\frac{\sqrt{3}}{2}+1 & \frac{1}{2}-\sqrt{3}\\ \frac{1}{2} & -\frac{\sqrt{3}}{2}\end{pmatrix}$$

$$[T]\begin{pmatrix}-1\\4\end{pmatrix}=\begin{pmatrix}1-\frac{9\sqrt{3}}{2}\\ -2\sqrt{3}-\frac{1}{2}\end{pmatrix}\approx\begin{pmatrix}-6.7942\\-3.9641\end{pmatrix}$$

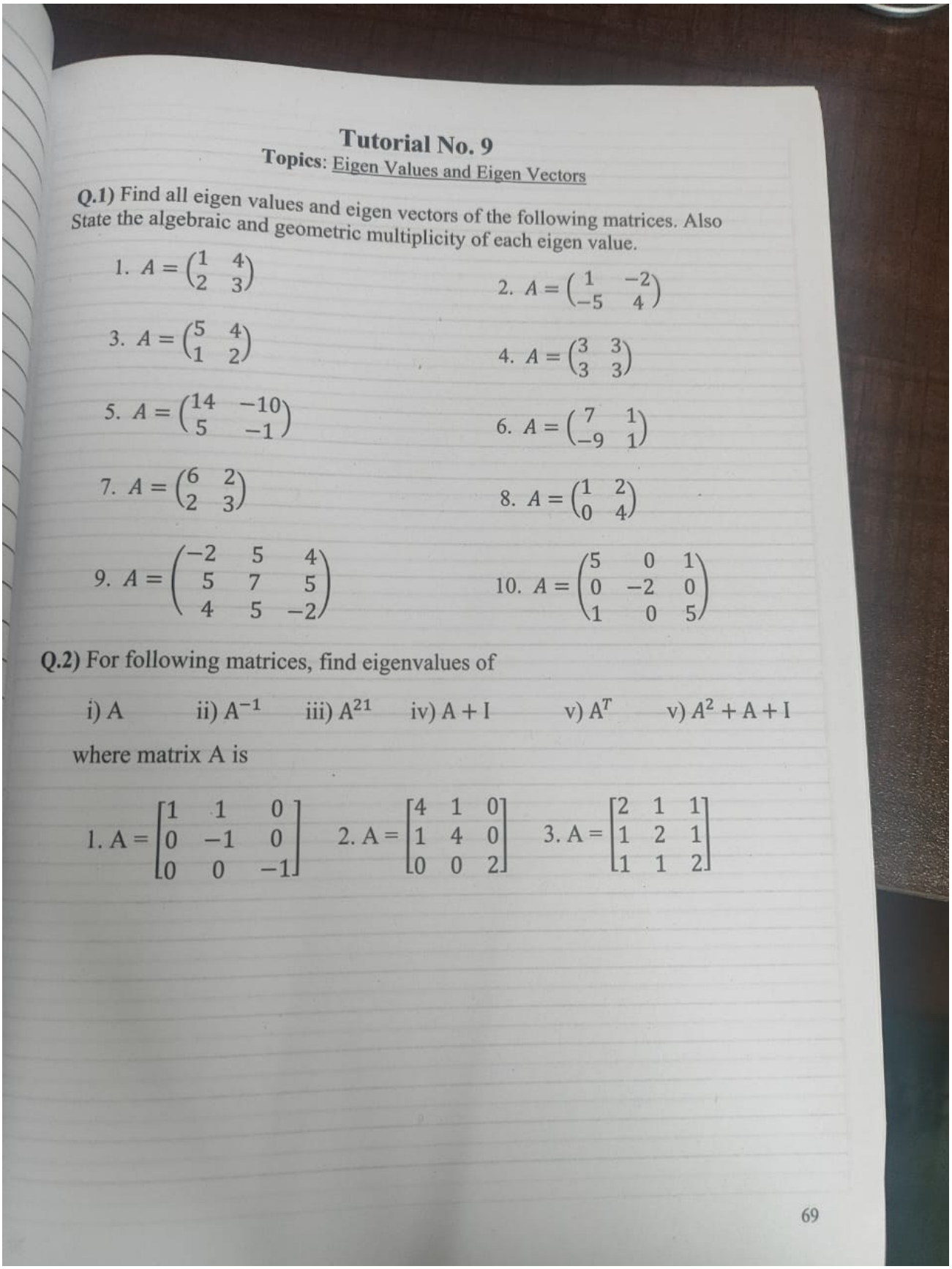
$$\text{(b) } [T]=R_{y\text{-axis}}\,R_{45^\circ}\,(2I)=\begin{pmatrix}-1 & 0\\ 0 & 1\end{pmatrix}\begin{pmatrix}\frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2}\\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2}\end{pmatrix}\begin{pmatrix}2 & 0\\ 0 & 2\end{pmatrix}=\begin{pmatrix}-\sqrt{2} & \sqrt{2}\\ \sqrt{2} & \sqrt{2}\end{pmatrix}$$

$$[T]\begin{pmatrix}-1\\4\end{pmatrix}=\begin{pmatrix}5\sqrt{2}\\ 3\sqrt{2}\end{pmatrix}\approx\begin{pmatrix}7.0711\\4.2426\end{pmatrix}$$

$$\text{(c) } [T]=R_{60^\circ}R_{105^\circ}R_{15^\circ}=R_{180^\circ}=\begin{pmatrix}-1 & 0\\ 0 & -1\end{pmatrix},\quad [T]\begin{pmatrix}-1\\4\end{pmatrix}=\begin{pmatrix}1\\-4\end{pmatrix}$$

Tutorial No. 9: Eigenvalues and Eigenvectors

Questions



Tutorial No. 9

Topics: Eigen Values and Eigen Vectors

Q.1) Find all eigen values and eigen vectors of the following matrices. Also State the algebraic and geometric multiplicity of each eigen value.

1. $A = \begin{pmatrix} 1 & 4 \\ 2 & 3 \end{pmatrix}$
2. $A = \begin{pmatrix} 1 & -2 \\ -5 & 4 \end{pmatrix}$
3. $A = \begin{pmatrix} 5 & 4 \\ 1 & 2 \end{pmatrix}$
4. $A = \begin{pmatrix} 3 & 3 \\ 3 & 3 \end{pmatrix}$
5. $A = \begin{pmatrix} 14 & -10 \\ 5 & -1 \end{pmatrix}$
6. $A = \begin{pmatrix} 7 & 1 \\ -9 & 1 \end{pmatrix}$
7. $A = \begin{pmatrix} 6 & 2 \\ 2 & 3 \end{pmatrix}$
8. $A = \begin{pmatrix} 1 & 2 \\ 0 & 4 \end{pmatrix}$
9. $A = \begin{pmatrix} -2 & 5 & 4 \\ 5 & 7 & 5 \\ 4 & 5 & -2 \end{pmatrix}$
10. $A = \begin{pmatrix} 5 & 0 & 1 \\ 0 & -2 & 0 \\ 1 & 0 & 5 \end{pmatrix}$

Q.2) For following matrices, find eigenvalues of

i) A ii) A^{-1} iii) A^{21} iv) A + I v) A^T v) $A^2 + A + I$

where matrix A is

1. $A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$
2. $A = \begin{bmatrix} 4 & 1 & 0 \\ 1 & 4 & 0 \\ 0 & 0 & 2 \end{bmatrix}$
3. $A = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix}$

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Solutions — Tutorial No. 9

Q.1 Eigenvalues, eigenvectors, multiplicities

AM = algebraic multiplicity (multiplicity as a root of $\det(A-\lambda I)$), GM = geometric multiplicity (= dim of the eigenspace).

(1) $A = \begin{pmatrix} 1 & 4 \\ 2 & 3 \end{pmatrix}$

$$\det(A-\lambda I) = (\lambda-5)(\lambda+1) = 0$$

$$\lambda = -1:\ \text{AM} = 1,\ \text{GM} = 1,\ \text{eigenvectors: } t\begin{pmatrix} -2 \\ 1 \end{pmatrix}\ (t \neq 0)$$

$$\lambda = 5:\ \text{AM} = 1,\ \text{GM} = 1,\ \text{eigenvectors: } t\begin{pmatrix} 1 \\ 1 \end{pmatrix}\ (t \neq 0)$$

(2) $A = \begin{pmatrix} 1 & -2 \\ -5 & 4 \end{pmatrix}$

$$\det(A-\lambda I) = (\lambda-6)(\lambda+1) = 0$$

$$\lambda = -1:\ \text{AM} = 1,\ \text{GM} = 1,\ \text{eigenvectors: } t\begin{pmatrix} 1 \\ 1 \end{pmatrix}\ (t \neq 0)$$

$$\lambda = 6:\ \text{AM} = 1,\ \text{GM} = 1,\ \text{eigenvectors: } t\begin{pmatrix} -\frac{2}{5} \\ 1 \end{pmatrix}\ (t \neq 0)$$

(3) $A = \begin{pmatrix} 5 & 4 \\ 1 & 2 \end{pmatrix}$

$$\det(A-\lambda I) = (\lambda-6)(\lambda-1) = 0$$

$$\lambda = 1:\ \text{AM} = 1,\ \text{GM} = 1,\ \text{eigenvectors: } t\begin{pmatrix} -1 \\ 1 \end{pmatrix}\ (t \neq 0)$$

$$\lambda = 6:\ \text{AM} = 1,\ \text{GM} = 1,\ \text{eigenvectors: } t\begin{pmatrix} 4 \\ 1 \end{pmatrix}\ (t \neq 0)$$

(4) $A = \begin{pmatrix} 3 & 3 \\ 3 & 3 \end{pmatrix}$

$$\det(A-\lambda I) = \lambda(\lambda-6) = 0$$

$$\lambda = 0:\ \text{AM} = 1,\ \text{GM} = 1,\ \text{eigenvectors: } t\begin{pmatrix} -1 \\ 1 \end{pmatrix}\ (t \neq 0)$$

$$\lambda = 6:\ \text{AM} = 1,\ \text{GM} = 1,\ \text{eigenvectors: } t\begin{pmatrix} 1 \\ 1 \end{pmatrix}\ (t \neq 0)$$

(5) $A = \begin{pmatrix} 14 & -10 \\ 5 & -1 \end{pmatrix}$

$$\det(A-\lambda I) = (\lambda-9)(\lambda-4) = 0$$

$$\lambda = 4:\ \text{AM} = 1,\ \text{GM} = 1,\ \text{eigenvectors: } t\begin{pmatrix} 1 \\ 1 \end{pmatrix}\ (t \neq 0)$$

$$\lambda = 9:\ \text{AM} = 1,\ \text{GM} = 1,\ \text{eigenvectors: } t\begin{pmatrix} 2 \\ 1 \end{pmatrix}\ (t \neq 0)$$

(6) $A = \begin{pmatrix} 7 & 1 \\ -9 & 1 \end{pmatrix}$

$$\det(A-\lambda I) = (\lambda-4)^2 = 0$$

$$\lambda = 4 : \ \text{AM} = 2, \ \text{GM} = 1, \ \text{eigenvectors: } t \begin{pmatrix} -\frac{1}{3} \\ 1 \end{pmatrix} \ (t \neq 0)$$

Since GM $<$ AM for some eigenvalue, A is *defective* (not diagonalizable).

(7) $A = \begin{pmatrix} 6 & 2 \\ 2 & 3 \end{pmatrix}$

$$\det(A - \lambda I) = (\lambda - 7)(\lambda - 2) = 0$$

$$\lambda = 2 : \ \text{AM} = 1, \ \text{GM} = 1, \ \text{eigenvectors: } t \begin{pmatrix} -\frac{1}{2} \\ 1 \end{pmatrix} \ (t \neq 0)$$

$$\lambda = 7 : \ \text{AM} = 1, \ \text{GM} = 1, \ \text{eigenvectors: } t \begin{pmatrix} 2 \\ 1 \end{pmatrix} \ (t \neq 0)$$

(8) $A = \begin{pmatrix} 1 & 2 \\ 0 & 4 \end{pmatrix}$

$$\det(A - \lambda I) = (\lambda - 4)(\lambda - 1) = 0$$

$$\lambda = 1 : \ \text{AM} = 1, \ \text{GM} = 1, \ \text{eigenvectors: } t \begin{pmatrix} 1 \\ 0 \end{pmatrix} \ (t \neq 0)$$

$$\lambda = 4 : \ \text{AM} = 1, \ \text{GM} = 1, \ \text{eigenvectors: } t \begin{pmatrix} \frac{2}{3} \\ 1 \end{pmatrix} \ (t \neq 0)$$

(9) $A = \begin{pmatrix} -2 & 5 & 4 \\ 5 & 7 & 5 \\ 4 & 5 & -2 \end{pmatrix}$

$$\det(A - \lambda I) = (\lambda - 12)(\lambda + 3)(\lambda + 6) = 0$$

$$\lambda = -6 : \ \text{AM} = 1, \ \text{GM} = 1, \ \text{eigenvectors: } t \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} \ (t \neq 0)$$

$$\lambda = -3 : \ \text{AM} = 1, \ \text{GM} = 1, \ \text{eigenvectors: } t \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} \ (t \neq 0)$$

$$\lambda = 12 : \ \text{AM} = 1, \ \text{GM} = 1, \ \text{eigenvectors: } t \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} \ (t \neq 0)$$

(10) $A = \begin{pmatrix} 5 & 0 & 1 \\ 0 & -2 & 0 \\ 1 & 0 & 5 \end{pmatrix}$

$$\det(A - \lambda I) = (\lambda - 6)(\lambda - 4)(\lambda + 2) = 0$$

$$\lambda = -2 : \ \text{AM} = 1, \ \text{GM} = 1, \ \text{eigenvectors: } t \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \ (t \neq 0)$$

$$\lambda = 4 : \ \text{AM} = 1, \ \text{GM} = 1, \ \text{eigenvectors: } t \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} \ (t \neq 0)$$

$$\lambda = 6 : \ \text{AM} = 1, \ \text{GM} = 1, \ \text{eigenvectors: } t \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \ (t \neq 0)$$

Q.2 Eigenvalues of related matrices

If $A\mathbf{x} = \lambda\mathbf{x}$ then $A^{-1}\mathbf{x} = \lambda^{-1}\mathbf{x}$ (if A is invertible), $A^{21}\mathbf{x} = \lambda^{21}\mathbf{x}$, $(A+I)\mathbf{x} = (\lambda+1)\mathbf{x}$, $(A^2+A+I)\mathbf{x} = (\lambda^2+\lambda+1)\mathbf{x}$, and A^T has the same eigenvalues as A (same characteristic polynomial).

(1) $A = \begin{pmatrix} 1 & 1 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$, $\det(A - \lambda I) = (\lambda - 1)(\lambda + 1)^2$.

matrix	λ_1	λ_2	λ_3
A	-1	-1	1
A^{-1}	-1	-1	1
A^{21}	-1	-1	1
$A+I$	0	0	2
A^T	-1	-1	1
A^2+A+I	1	1	3

(2) $A = \begin{pmatrix} 4 & 1 & 0 \\ 1 & 4 & 0 \\ 0 & 0 & 2 \end{pmatrix}$, $\det(A - \lambda I) = (\lambda - 5)(\lambda - 3)(\lambda - 2)$.

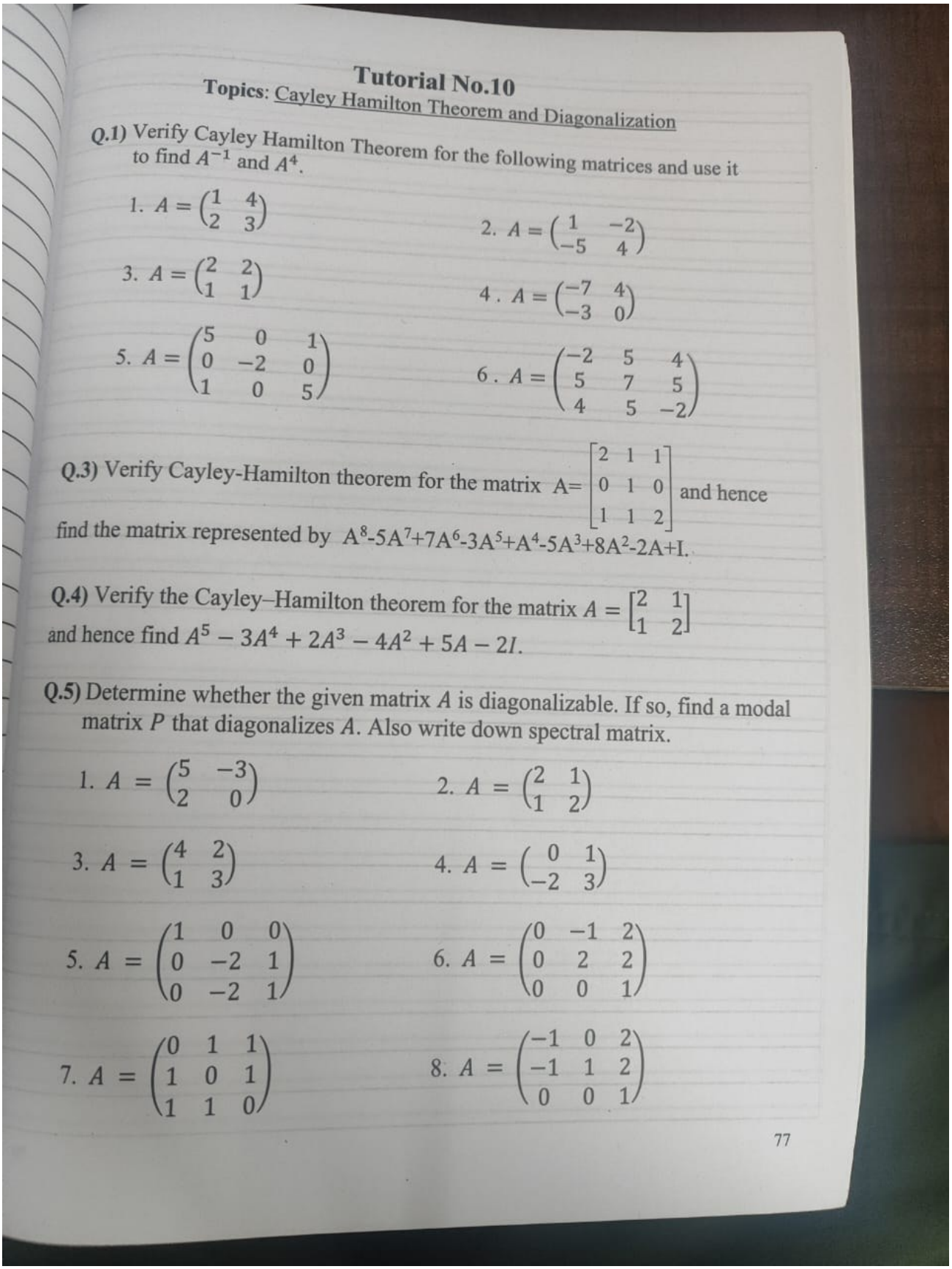
matrix	λ_1	λ_2	λ_3
A	2	3	5
A^{-1}	$\frac{1}{2}$	$\frac{1}{3}$	$\frac{1}{5}$
A^{21}	2097152	10460353203	476837158203125
$A+I$	3	4	6
A^T	2	3	5
A^2+A+I	7	13	31

(3) $A = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix}$, $\det(A - \lambda I) = (\lambda - 4)(\lambda - 1)^2$.

matrix	λ_1	λ_2	λ_3
A	1	1	4
A^{-1}	1	1	$\frac{1}{4}$
A^{21}	1	1	4398046511104
$A+I$	2	2	5
A^T	1	1	4
A^2+A+I	3	3	21

Tutorial No. 10: Cayley–Hamilton Theorem and Diagonalization

Questions



Tutorial No.10

Topics: Cayley Hamilton Theorem and Diagonalization

Q.1) Verify Cayley Hamilton Theorem for the following matrices and use it to find A^{-1} and A^4.

1. $A = \begin{pmatrix} 1 & 4 \\ 2 & 3 \end{pmatrix}$

2. $A = \begin{pmatrix} 1 & -2 \\ -5 & 4 \end{pmatrix}$

3. $A = \begin{pmatrix} 2 & 2 \\ 1 & 1 \end{pmatrix}$

4. $A = \begin{pmatrix} -7 & 4 \\ -3 & 0 \end{pmatrix}$

5. $A = \begin{pmatrix} 5 & 0 & 1 \\ 0 & -2 & 0 \\ 1 & 0 & 5 \end{pmatrix}$

6. $A = \begin{pmatrix} -2 & 5 & 4 \\ 5 & 7 & 5 \\ 4 & 5 & -2 \end{pmatrix}$

Q.3) Verify Cayley-Hamilton theorem for the matrix $A = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix}$ and hence find the matrix represented by $A^8-5A^7+7A^6-3A^5+A^4-5A^3+8A^2-2A+I$.

Q.4) Verify the Cayley–Hamilton theorem for the matrix $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ and hence find $A^5 - 3A^4 + 2A^3 - 4A^2 + 5A - 2I$.

Q.5) Determine whether the given matrix A is diagonalizable. If so, find a modal matrix P that diagonalizes A. Also write down spectral matrix.

1. $A = \begin{pmatrix} 5 & -3 \\ 2 & 0 \end{pmatrix}$

2. $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$

3. $A = \begin{pmatrix} 4 & 2 \\ 1 & 3 \end{pmatrix}$

4. $A = \begin{pmatrix} 0 & 1 \\ -2 & 3 \end{pmatrix}$

5. $A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -2 & 1 \\ 0 & -2 & 1 \end{pmatrix}$

6. $A = \begin{pmatrix} 0 & -1 & 2 \\ 0 & 2 & 2 \\ 0 & 0 & 1 \end{pmatrix}$

7. $A = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$

8. $A = \begin{pmatrix} -1 & 0 & 2 \\ -1 & 1 & 2 \\ 0 & 0 & 1 \end{pmatrix}$

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Solutions — Tutorial No. 10

Q.1 Cayley–Hamilton, A^{-1} and A^4

For $n \times n$ A with characteristic polynomial $p(\lambda) = \lambda^n + c_{n-1}\lambda^{n-1} + \cdots + c_0$ we have $p(A) = O$. Multiplying by A^{-1} gives $A^{-1} = -\frac{1}{c_0}(A^{n-1} + c_{n-1}A^{n-2} + \cdots + c_1 I)$, and A^4 is found by reducing X^4 modulo $p(X)$.

(1) $A = \begin{pmatrix} 1 & 4 \\ 2 & 3 \end{pmatrix}$

$$p(\lambda) = \det(\lambda I - A) = \lambda^2 - 4\lambda - 5$$

$$A^2 = \begin{pmatrix} 9 & 16 \\ 8 & 17 \end{pmatrix}$$

$$p(A) = A^2 - 4A - 5I = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} = O \checkmark$$

$$A^{-1} = -\frac{1}{-5}(A - 4I) = \begin{pmatrix} -\frac{3}{5} & \frac{4}{5} \\ \frac{2}{5} & -\frac{1}{5} \end{pmatrix}$$

$$X^4 \equiv 104X + 105 \pmod{p(X)} \;\Rightarrow\; A^4 = 104A + 105I = \begin{pmatrix} 209 & 416 \\ 208 & 417 \end{pmatrix}$$

(2) $A = \begin{pmatrix} 1 & -2 \\ -5 & 4 \end{pmatrix}$

$$p(\lambda) = \det(\lambda I - A) = \lambda^2 - 5\lambda - 6$$

$$A^2 = \begin{pmatrix} 11 & -10 \\ -25 & 26 \end{pmatrix}$$

$$p(A) = A^2 - 5A - 6I = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} = O \checkmark$$

$$A^{-1} = -\frac{1}{-6}(A - 5I) = \begin{pmatrix} -\frac{2}{3} & -\frac{1}{3} \\ -\frac{5}{6} & -\frac{1}{6} \end{pmatrix}$$

$$X^4 \equiv 185X + 186 \pmod{p(X)} \;\Rightarrow\; A^4 = 185A + 186I = \begin{pmatrix} 371 & -370 \\ -925 & 926 \end{pmatrix}$$

(3) $A = \begin{pmatrix} 2 & 2 \\ 1 & 1 \end{pmatrix}$

$$p(\lambda) = \det(\lambda I - A) = \lambda^2 - 3\lambda$$

$$A^2 = \begin{pmatrix} 6 & 6 \\ 3 & 3 \end{pmatrix}$$

$$p(A) = A^2 - 3A = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} = O \checkmark$$

$\det A = 0$, so A^{-1} does not exist.

$$X^4 \equiv 27X \pmod{p(X)} \;\Rightarrow\; A^4 = 27A = \begin{pmatrix} 54 & 54 \\ 27 & 27 \end{pmatrix}$$

(4) $A = \begin{pmatrix} -7 & 4 \\ -3 & 0 \end{pmatrix}$

$$p(\lambda) = \det(\lambda I - A) = \lambda^2 + 7\lambda + 12$$

$$A^2 = \begin{pmatrix} 37 & -28 \\ 21 & -12 \end{pmatrix}$$

$$p(A) = A^2 + 7A + 12I = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} = O \checkmark$$

$$A^{-1} = -\frac{1}{12}(A + 7I) = \begin{pmatrix} 0 & -\frac{1}{3} \\ \frac{1}{4} & -\frac{7}{12} \end{pmatrix}$$

$$X^4 \equiv -175X - 444 \pmod{p(X)} \;\Rightarrow\; A^4 = -175A - 444I = \begin{pmatrix} 781 & -700 \\ 525 & -444 \end{pmatrix}$$

(5) $A = \begin{pmatrix} 5 & 0 & 1 \\ 0 & -2 & 0 \\ 1 & 0 & 5 \end{pmatrix}$

$$p(\lambda) = \det(\lambda I - A) = \lambda^3 - 8\lambda^2 + 4\lambda + 48$$

$$A^2 = \begin{pmatrix} 26 & 0 & 10 \\ 0 & 4 & 0 \\ 10 & 0 & 26 \end{pmatrix}, \quad A^3 = \begin{pmatrix} 140 & 0 & 76 \\ 0 & -8 & 0 \\ 76 & 0 & 140 \end{pmatrix}$$

$$p(A) = A^3 - 8A^2 + 4A + 48I = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = O \checkmark$$

$$A^{-1} = -\frac{1}{48}\left(A^2 - 8A + 4I\right) = \begin{pmatrix} \frac{5}{24} & 0 & -\frac{1}{24} \\ 0 & -\frac{1}{2} & 0 \\ -\frac{1}{24} & 0 & \frac{5}{24} \end{pmatrix}$$

$$X^4 \equiv 60X^2 - 80X - 384 \pmod{p(X)} \;\Rightarrow\; A^4 = 60A^2 - 80A - 384I = \begin{pmatrix} 776 & 0 & 520 \\ 0 & 16 & 0 \\ 520 & 0 & 776 \end{pmatrix}$$

(6) $A = \begin{pmatrix} -2 & 5 & 4 \\ 5 & 7 & 5 \\ 4 & 5 & -2 \end{pmatrix}$

$$p(\lambda) = \det(\lambda I - A) = \lambda^3 - 3\lambda^2 - 90\lambda - 216$$

$$A^2 = \begin{pmatrix} 45 & 45 & 9 \\ 45 & 99 & 45 \\ 9 & 45 & 45 \end{pmatrix}, \quad A^3 = \begin{pmatrix} 171 & 585 & 387 \\ 585 & 1143 & 585 \\ 387 & 585 & 171 \end{pmatrix}$$

$$p(A) = A^3 - 3A^2 - 90A - 216I = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = O \checkmark$$

$$A^{-1} = -\frac{1}{-216}\left(A^2 - 3A - 90I\right) = \begin{pmatrix} -\frac{13}{72} & \frac{5}{36} & -\frac{1}{72} \\ \frac{5}{36} & -\frac{1}{18} & \frac{5}{36} \\ -\frac{1}{72} & \frac{5}{36} & -\frac{13}{72} \end{pmatrix}$$

$$X^4 \equiv 99X^2 + 486X + 648 \pmod{p(X)} \;\Rightarrow\; A^4 = 99A^2 + 486A + 648I = \begin{pmatrix} 4131 & 6885 & 2835 \\ 6885 & 13851 & 6885 \\ 2835 & 6885 & 4131 \end{pmatrix}$$

Q.3

$A = \begin{pmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{pmatrix}$.

$$p(\lambda) = \det(\lambda I - A) = \lambda^3 - 5\lambda^2 + 7\lambda - 3$$

$$A^2 = \begin{pmatrix} 5 & 4 & 4 \\ 0 & 1 & 0 \\ 4 & 4 & 5 \end{pmatrix}, \quad A^3 = \begin{pmatrix} 14 & 13 & 13 \\ 0 & 1 & 0 \\ 13 & 13 & 14 \end{pmatrix}$$

$$p(A) = A^3 - 5A^2 + 7A - 3I = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = O \checkmark$$

$$A^{-1} = -\frac{1}{-3}\left(A^2 - 5A + 7I\right) = \begin{pmatrix} \frac{2}{3} & -\frac{1}{3} & -\frac{1}{3} \\ 0 & 1 & 0 \\ -\frac{1}{3} & -\frac{1}{3} & \frac{2}{3} \end{pmatrix}$$

$$X^4 \equiv 18X^2 - 32X + 15 \pmod{p(X)} \;\Rightarrow\; A^4 = 18A^2 - 32A + 15I = \begin{pmatrix} 41 & 40 & 40 \\ 0 & 1 & 0 \\ 40 & 40 & 41 \end{pmatrix}$$

$p(X) = X^3 - 5X^2 + 7X - 3$. Dividing $f(X) = X^8 - 5X^7 + 7X^6 - 3X^5 + X^4 - 5X^3 + 8X^2 - 2X + 1$ by $p(X)$: quotient $q(X) = X^5 + X$, remainder $r(X) = X^2 + X + 1$. Since $p(A) = O$, $f(A) = q(A)p(A) + r(A) = r(A)$.

$$f(A) = r(A) = \begin{pmatrix} 8 & 5 & 5 \\ 0 & 3 & 0 \\ 5 & 5 & 8 \end{pmatrix}$$

Q.4

$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$.

$$p(\lambda) = \det(\lambda I - A) = \lambda^2 - 4\lambda + 3$$

$$A^2 = \begin{pmatrix} 5 & 4 \\ 4 & 5 \end{pmatrix}$$

$$p(A) = A^2 - 4A + 3I = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} = O \checkmark$$

$$A^{-1} = -\frac{1}{3}\left(A - 4I\right) = \begin{pmatrix} \frac{2}{3} & -\frac{1}{3} \\ -\frac{1}{3} & \frac{2}{3} \end{pmatrix}$$

$$X^4 \equiv 40X - 39 \pmod{p(X)} \;\Rightarrow\; A^4 = 40A - 39I = \begin{pmatrix} 41 & 40 \\ 40 & 41 \end{pmatrix}$$

$p(X) = X^2 - 4X + 3$. Dividing $f(X) = X^5 - 3X^4 + 2X^3 - 4X^2 + 5X - 2$ by $p(X)$: quotient $q(X) = X^3 + X^2 + 3X + 5$, remainder $r(X) = 16X - 17$. Since $p(A) = O$, $f(A) = q(A)p(A) + r(A) = r(A)$.

$$f(A) = r(A) = \begin{pmatrix} 15 & 16 \\ 16 & 15 \end{pmatrix}$$

Q.5 Diagonalization

A is diagonalizable iff it has n linearly independent eigenvectors. Then $P = [\mathbf{v}_1 \cdots \mathbf{v}_n]$ (modal matrix) and the spectral matrix is $D = P^{-1}AP = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$.

(1) $A = \begin{pmatrix} 5 & -3 \\ 2 & 0 \end{pmatrix}$, $\det(A - \lambda I) = (\lambda - 3)(\lambda - 2)$.

- Eigenvectors: $\lambda = 2$ (AM = 1, GM = 1): $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$; $\lambda = 3$ (AM = 1, GM = 1): $\begin{pmatrix} \frac{3}{2} \\ 1 \end{pmatrix}$.
- GM = AM for every eigenvalue, so A is **diagonalizable**. Modal matrix $P = \begin{pmatrix} 1 & \frac{3}{2} \\ 1 & 1 \end{pmatrix}$, spectral matrix $D = P^{-1}AP = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$.

(2) $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$, $\det(A - \lambda I) = (\lambda - 3)(\lambda - 1)$.

- Eigenvectors: $\lambda = 1$ (AM = 1, GM = 1): $\begin{pmatrix} -1 \\ 1 \end{pmatrix}$; $\lambda = 3$ (AM = 1, GM = 1): $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$.
- GM = AM for every eigenvalue, so A is **diagonalizable**. Modal matrix $P = \begin{pmatrix} -1 & 1 \\ 1 & 1 \end{pmatrix}$, spectral matrix $D = P^{-1}AP = \begin{pmatrix} 1 & 0 \\ 0 & 3 \end{pmatrix}$.

(3) $A = \begin{pmatrix} 4 & 2 \\ 1 & 3 \end{pmatrix}$, $\det(A - \lambda I) = (\lambda - 5)(\lambda - 2)$.

- Eigenvectors: $\lambda = 2$ (AM = 1, GM = 1): $\begin{pmatrix} -1 \\ 1 \end{pmatrix}$; $\lambda = 5$ (AM = 1, GM = 1): $\begin{pmatrix} 2 \\ 1 \end{pmatrix}$.
- GM = AM for every eigenvalue, so A is **diagonalizable**. Modal matrix $P = \begin{pmatrix} -1 & 2 \\ 1 & 1 \end{pmatrix}$, spectral matrix $D = P^{-1}AP = \begin{pmatrix} 2 & 0 \\ 0 & 5 \end{pmatrix}$.

(4) $A = \begin{pmatrix} 0 & 1 \\ -2 & 3 \end{pmatrix}$, $\det(A - \lambda I) = (\lambda - 2)(\lambda - 1)$.

- Eigenvectors: $\lambda = 1$ (AM = 1, GM = 1): $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$; $\lambda = 2$ (AM = 1, GM = 1): $\begin{pmatrix} \frac{1}{2} \\ 1 \end{pmatrix}$.
- GM = AM for every eigenvalue, so A is **diagonalizable**. Modal matrix $P = \begin{pmatrix} 1 & \frac{1}{2} \\ 1 & 1 \end{pmatrix}$, spectral matrix $D = P^{-1}AP = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$.

(5) $A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -2 & 1 \\ 0 & -2 & 1 \end{pmatrix}$, $\det(A - \lambda I) = \lambda(\lambda - 1)(\lambda + 1)$.

- Eigenvectors: $\lambda = -1$ (AM = 1, GM = 1): $\begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$; $\lambda = 0$ (AM = 1, GM = 1): $\begin{pmatrix} 0 \\ \frac{1}{2} \\ 1 \end{pmatrix}$; $\lambda = 1$ (AM = 1, GM = 1): $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$.
- GM = AM for every eigenvalue, so A is **diagonalizable**. Modal matrix $P = \begin{pmatrix} 0 & 0 & 1 \\ 1 & \frac{1}{2} & 0 \\ 1 & 1 & 0 \end{pmatrix}$, spectral matrix $D = P^{-1}AP = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$.

(6) $A = \begin{pmatrix} 0 & -1 & 2 \\ 0 & 2 & 2 \\ 0 & 0 & 1 \end{pmatrix}$, $\det(A - \lambda I) = \lambda(\lambda - 2)(\lambda - 1)$.

- Eigenvectors: $\lambda = 0$ (AM = 1, GM = 1): $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$; $\lambda = 1$ (AM = 1, GM = 1): $\begin{pmatrix} 4 \\ -2 \\ 1 \end{pmatrix}$; $\lambda = 2$ (AM = 1, GM = 1): $\begin{pmatrix} -\frac{1}{2} \\ 1 \\ 0 \end{pmatrix}$.
- GM = AM for every eigenvalue, so A is **diagonalizable**. Modal matrix $P = \begin{pmatrix} 1 & 4 & -\frac{1}{2} \\ 0 & -2 & 1 \\ 0 & 1 & 0 \end{pmatrix}$, spectral matrix $D = P^{-1}AP = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix}$.

(7) $A = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$, $\det(A - \lambda I) = (\lambda - 2)(\lambda + 1)^2$.

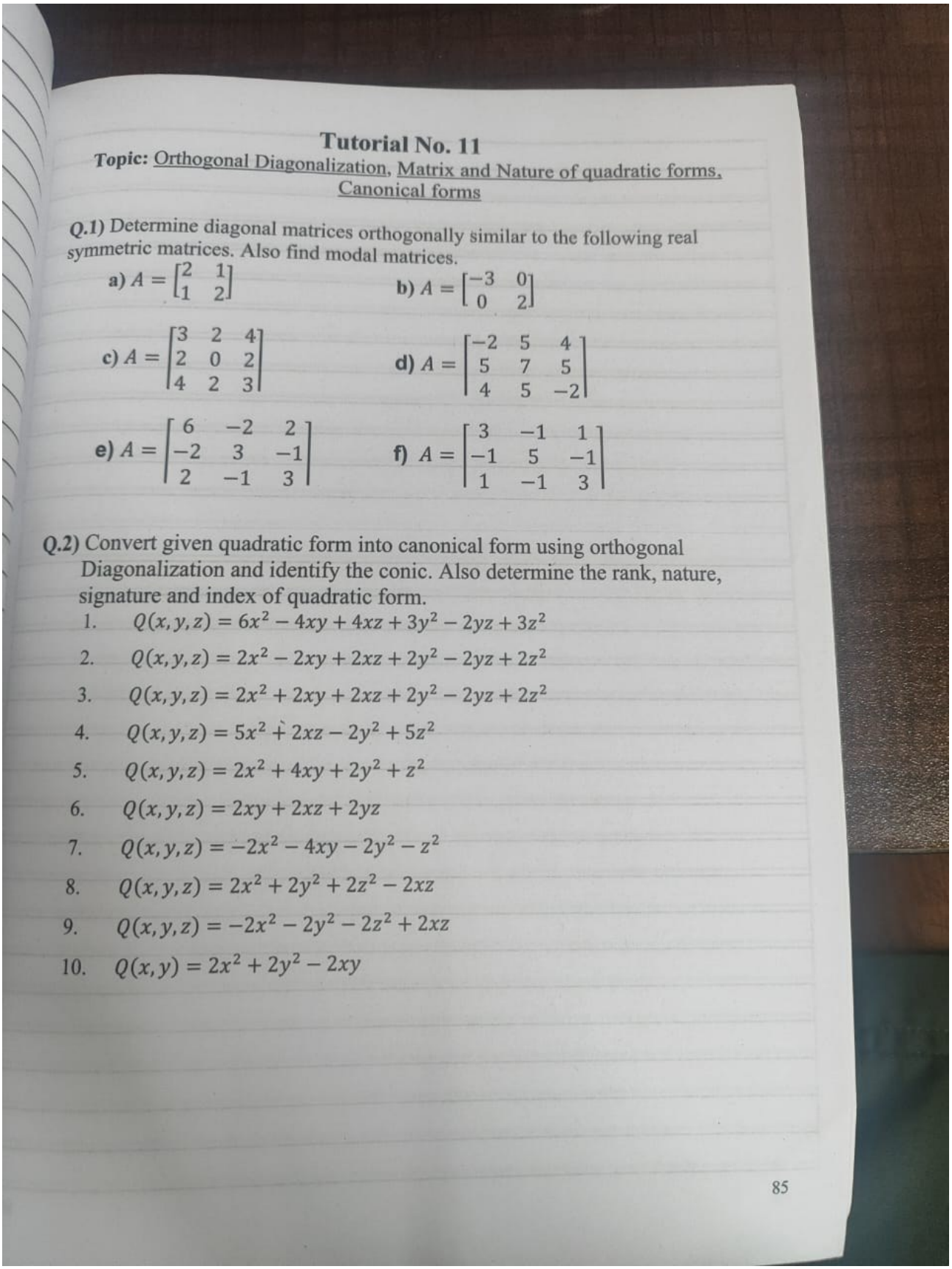
- Eigenvectors: $\lambda = -1$ (AM = 2, GM = 2): $\begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}$, $\begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$; $\lambda = 2$ (AM = 1, GM = 1): $\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$.
- GM = AM for every eigenvalue, so A is **diagonalizable**. Modal matrix $P = \begin{pmatrix} -1 & -1 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}$, spectral matrix $D = P^{-1}AP = \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 2 \end{pmatrix}$.

(8) $A = \begin{pmatrix} -1 & 0 & 2 \\ -1 & 1 & 2 \\ 0 & 0 & 1 \end{pmatrix}$, $\det(A - \lambda I) = (\lambda - 1)^2(\lambda + 1)$.

- Eigenvectors: $\lambda = -1$ (AM = 1, GM = 1): $\begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}$; $\lambda = 1$ (AM = 2, GM = 1): $\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$.
- Only 2 independent eigenvector(s) for a 3×3 matrix (GM $<$ AM), so A is **not diagonalizable**.

Tutorial No. 11: Orthogonal Diagonalization, Quadratic Forms and Canonical Forms

Questions



Tutorial No. 11

Topic: Orthogonal Diagonalization, Matrix and Nature of quadratic forms, Canonical forms

Q.1) Determine diagonal matrices orthogonally similar to the following real symmetric matrices. Also find modal matrices.

a) $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ **b)** $A = \begin{bmatrix} -3 & 0 \\ 0 & 2 \end{bmatrix}$

c) $A = \begin{bmatrix} 3 & 2 & 4 \\ 2 & 0 & 2 \\ 4 & 2 & 3 \end{bmatrix}$ **d)** $A = \begin{bmatrix} -2 & 5 & 4 \\ 5 & 7 & 5 \\ 4 & 5 & -2 \end{bmatrix}$

e) $A = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$ **f)** $A = \begin{bmatrix} 3 & -1 & 1 \\ -1 & 5 & -1 \\ 1 & -1 & 3 \end{bmatrix}$

Q.2) Convert given quadratic form into canonical form using orthogonal Diagonalization and identify the conic. Also determine the rank, nature, signature and index of quadratic form.

1. $Q(x,y,z) = 6x^2 - 4xy + 4xz + 3y^2 - 2yz + 3z^2$
2. $Q(x,y,z) = 2x^2 - 2xy + 2xz + 2y^2 - 2yz + 2z^2$
3. $Q(x,y,z) = 2x^2 + 2xy + 2xz + 2y^2 - 2yz + 2z^2$
4. $Q(x,y,z) = 5x^2 + 2xz - 2y^2 + 5z^2$
5. $Q(x,y,z) = 2x^2 + 4xy + 2y^2 + z^2$
6. $Q(x,y,z) = 2xy + 2xz + 2yz$
7. $Q(x,y,z) = -2x^2 - 4xy - 2y^2 - z^2$
8. $Q(x,y,z) = 2x^2 + 2y^2 + 2z^2 - 2xz$
9. $Q(x,y,z) = -2x^2 - 2y^2 - 2z^2 + 2xz$
10. $Q(x,y) = 2x^2 + 2y^2 - 2xy$

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Solutions — Tutorial No. 11

Q.1 Orthogonal diagonalization of symmetric matrices

For real symmetric A: eigenvectors of distinct eigenvalues are orthogonal; inside an eigenspace use Gram–Schmidt; normalise. The modal matrix P (orthonormal eigenvectors as columns) is orthogonal and $D = P^T AP = P^{-1}AP$.

a) $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$, $\det(A - \lambda I) = (\lambda - 3)(\lambda - 1)$, eigenvalues 1 (mult. 1), 3 (mult. 1).

- Orthonormal eigenvectors: $\lambda = 1$: $\begin{pmatrix} -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} \end{pmatrix}$; $\lambda = 3$: $\begin{pmatrix} \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} \end{pmatrix}$.
- Modal matrix $P = \begin{pmatrix} -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix}$, and $P^T AP = D = \begin{pmatrix} 1 & 0 \\ 0 & 3 \end{pmatrix}$.

b) $A = \begin{pmatrix} -3 & 0 \\ 0 & 2 \end{pmatrix}$, $\det(A - \lambda I) = (\lambda - 2)(\lambda + 3)$, eigenvalues -3 (mult. 1), 2 (mult. 1).

- Orthonormal eigenvectors: $\lambda = -3$: $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$; $\lambda = 2$: $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$.
- Modal matrix $P = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, and $P^T AP = D = \begin{pmatrix} -3 & 0 \\ 0 & 2 \end{pmatrix}$.

c) $A = \begin{pmatrix} 3 & 2 & 4 \\ 2 & 0 & 2 \\ 4 & 2 & 3 \end{pmatrix}$, $\det(A - \lambda I) = (\lambda - 8)(\lambda + 1)^2$, eigenvalues -1 (mult. 2), 8 (mult. 1).

- Orthonormal eigenvectors: $\lambda = -1$: $\begin{pmatrix} -\frac{\sqrt{5}}{5} \\ \frac{2\sqrt{5}}{5} \\ 0 \end{pmatrix}$, $\begin{pmatrix} -\frac{4\sqrt{5}}{15} \\ -\frac{2\sqrt{5}}{15} \\ \frac{\sqrt{5}}{3} \end{pmatrix}$; $\lambda = 8$: $\begin{pmatrix} \frac{2}{3} \\ \frac{1}{3} \\ \frac{2}{3} \end{pmatrix}$.
- Modal matrix $P = \begin{pmatrix} -\frac{\sqrt{5}}{5} & -\frac{4\sqrt{5}}{15} & \frac{2}{3} \\ \frac{2\sqrt{5}}{5} & -\frac{2\sqrt{5}}{15} & \frac{1}{3} \\ 0 & \frac{\sqrt{5}}{3} & \frac{2}{3} \end{pmatrix}$, and $P^T AP = D = \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 8 \end{pmatrix}$.

d) $A = \begin{pmatrix} -2 & 5 & 4 \\ 5 & 7 & 5 \\ 4 & 5 & -2 \end{pmatrix}$, $\det(A - \lambda I) = (\lambda - 12)(\lambda + 3)(\lambda + 6)$, eigenvalues -6 (mult. 1), -3 (mult. 1), 12 (mult. 1).

- Orthonormal eigenvectors: $\lambda = -6$: $\begin{pmatrix} -\frac{\sqrt{2}}{2} \\ 0 \\ \frac{\sqrt{2}}{2} \end{pmatrix}$; $\lambda = -3$: $\begin{pmatrix} \frac{\sqrt{3}}{3} \\ -\frac{\sqrt{3}}{3} \\ \frac{\sqrt{3}}{3} \end{pmatrix}$; $\lambda = 12$: $\begin{pmatrix} \frac{\sqrt{6}}{6} \\ \frac{\sqrt{6}}{3} \\ \frac{\sqrt{6}}{6} \end{pmatrix}$.
- Modal matrix $P = \begin{pmatrix} -\frac{\sqrt{2}}{2} & \frac{\sqrt{3}}{3} & \frac{\sqrt{6}}{6} \\ 0 & -\frac{\sqrt{3}}{3} & \frac{\sqrt{6}}{3} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{3}}{3} & \frac{\sqrt{6}}{6} \end{pmatrix}$, and $P^T AP = D = \begin{pmatrix} -6 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & 12 \end{pmatrix}$.

e) $A = \begin{pmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{pmatrix}$, $\det(A - \lambda I) = (\lambda - 8)(\lambda - 2)^2$, eigenvalues 2 (mult. 2), 8 (mult. 1).

- Orthonormal eigenvectors: $\lambda = 2$: $\begin{pmatrix} \frac{\sqrt{5}}{5} \\ \frac{2\sqrt{5}}{5} \\ 0 \end{pmatrix}$, $\begin{pmatrix} -\frac{\sqrt{30}}{15} \\ \frac{\sqrt{30}}{30} \\ \frac{\sqrt{30}}{6} \end{pmatrix}$; $\lambda = 8$: $\begin{pmatrix} \frac{\sqrt{6}}{3} \\ -\frac{\sqrt{6}}{6} \\ \frac{\sqrt{6}}{6} \end{pmatrix}$.
- Modal matrix $P = \begin{pmatrix} \frac{\sqrt{5}}{5} & -\frac{\sqrt{30}}{15} & \frac{\sqrt{6}}{3} \\ \frac{2\sqrt{5}}{5} & \frac{\sqrt{30}}{30} & -\frac{\sqrt{6}}{6} \\ 0 & \frac{\sqrt{30}}{6} & \frac{\sqrt{6}}{6} \end{pmatrix}$, and $P^T AP = D = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 8 \end{pmatrix}$.

f) $A = \begin{pmatrix} 3 & -1 & 1 \\ -1 & 5 & -1 \\ 1 & -1 & 3 \end{pmatrix}$, $\det(A - \lambda I) = (\lambda - 6)(\lambda - 3)(\lambda - 2)$, eigenvalues 2 (mult. 1), 3 (mult. 1), 6 (mult. 1).

- Orthonormal eigenvectors: $\lambda = 2$: $\begin{pmatrix} -\frac{\sqrt{2}}{2} \\ 0 \\ \frac{\sqrt{2}}{2} \end{pmatrix}$; $\lambda = 3$: $\begin{pmatrix} \frac{\sqrt{3}}{3} \\ \frac{\sqrt{3}}{3} \\ \frac{\sqrt{3}}{3} \end{pmatrix}$; $\lambda = 6$: $\begin{pmatrix} \frac{\sqrt{6}}{6} \\ -\frac{\sqrt{6}}{3} \\ \frac{\sqrt{6}}{6} \end{pmatrix}$.
- Modal matrix $P = \begin{pmatrix} -\frac{\sqrt{2}}{2} & \frac{\sqrt{3}}{3} & \frac{\sqrt{6}}{6} \\ 0 & \frac{\sqrt{3}}{3} & -\frac{\sqrt{6}}{3} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{3}}{3} & \frac{\sqrt{6}}{6} \end{pmatrix}$, and $P^T AP = D = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 6 \end{pmatrix}$.

Q.2 Canonical forms of quadratic forms

Write $Q = \mathbf{x}^T A\mathbf{x}$ with A symmetric; with $\mathbf{x} = P\mathbf{y}$ (P orthogonal) $Q = \lambda_1 y_1^2 + \lambda_2 y_2^2 + \ldots$. Rank r = number of non-zero λ_i; index p = number of positive λ_i; signature $s = p - (r - p) = 2p - r$. The surface type is stated for $Q = 1$.

1. $Q = 6x^2 - 4xy + 4xz + 3y^2 - 2yz + 3z^2$, $A = \begin{pmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{pmatrix}$.

- Eigenvalues: 2, 2, 8; orthonormal eigenvectors (columns of P): $P = \begin{pmatrix} \frac{\sqrt{5}}{5} & -\frac{\sqrt{30}}{15} & \frac{\sqrt{6}}{3} \\ \frac{2\sqrt{5}}{5} & \frac{\sqrt{30}}{30} & -\frac{\sqrt{6}}{6} \\ 0 & \frac{\sqrt{30}}{6} & \frac{\sqrt{6}}{6} \end{pmatrix}$.
- Canonical form: $Q = 2y_1^2 + 2y_2^2 + 8y_3^2$.
- Rank = 3, index = 3, signature = 3; nature: **positive definite**. Type ($Q = 1$): ellipsoid.

2. $Q = 2x^2 - 2xy + 2xz + 2y^2 - 2yz + 2z^2$, $A = \begin{pmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{pmatrix}$.

- Eigenvalues: 1, 1, 4; orthonormal eigenvectors (columns of P): $P = \begin{pmatrix} \frac{\sqrt{2}}{2} & -\frac{\sqrt{6}}{6} & \frac{\sqrt{3}}{3} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{6}}{6} & -\frac{\sqrt{3}}{3} \\ 0 & \frac{\sqrt{6}}{3} & \frac{\sqrt{3}}{3} \end{pmatrix}$.
- Canonical form: $Q = y_1^2 + y_2^2 + 4y_3^2$.
- Rank = 3, index = 3, signature = 3; nature: **positive definite**. Type ($Q = 1$): ellipsoid.

3. $Q = 2x^2 + 2xy + 2xz + 2y^2 - 2yz + 2z^2$, $A = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & -1 \\ 1 & -1 & 2 \end{pmatrix}$.

- Eigenvalues: 0, 3, 3; orthonormal eigenvectors (columns of P): $P = \begin{pmatrix} -\frac{\sqrt{3}}{3} & \frac{\sqrt{2}}{2} & \frac{\sqrt{6}}{6} \\ \frac{\sqrt{3}}{3} & \frac{\sqrt{2}}{2} & -\frac{\sqrt{6}}{6} \\ \frac{\sqrt{3}}{3} & 0 & \frac{\sqrt{6}}{3} \end{pmatrix}$.
- Canonical form: $Q = 3y_2^2 + 3y_3^2$.
- Rank = 2, index = 2, signature = 2; nature: **positive semi-definite**. Type ($Q = 1$): elliptic cylinder.

4. $Q = 5x^2 + 2xz - 2y^2 + 5z^2$, $A = \begin{pmatrix} 5 & 0 & 1 \\ 0 & -2 & 0 \\ 1 & 0 & 5 \end{pmatrix}$.

- Eigenvalues: -2, 4, 6; orthonormal eigenvectors (columns of P): $P = \begin{pmatrix} 0 & -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ 1 & 0 & 0 \\ 0 & \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix}$.
- Canonical form: $Q = -2y_1^2 + 4y_2^2 + 6y_3^2$.
- Rank = 3, index = 2, signature = 1; nature: **indefinite**. Type ($Q = 1$): hyperboloid of one sheet.

5. $Q = 2x^2 + 4xy + 2y^2 + z^2$, $A = \begin{pmatrix} 2 & 2 & 0 \\ 2 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix}$.

- Eigenvalues: 0, 1, 4; orthonormal eigenvectors (columns of P): $P = \begin{pmatrix} -\frac{\sqrt{2}}{2} & 0 & \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & 0 & \frac{\sqrt{2}}{2} \\ 0 & 1 & 0 \end{pmatrix}$.
- Canonical form: $Q = y_2^2 + 4y_3^2$.
- Rank = 2, index = 2, signature = 2; nature: **positive semi-definite**. Type ($Q = 1$): elliptic cylinder.

6. $Q = 2xy + 2xz + 2yz$, $A = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$.

- Eigenvalues: -1, -1, 2; orthonormal eigenvectors (columns of P): $P = \begin{pmatrix} -\frac{\sqrt{2}}{2} & -\frac{\sqrt{6}}{6} & \frac{\sqrt{3}}{3} \\ \frac{\sqrt{2}}{2} & -\frac{\sqrt{6}}{6} & \frac{\sqrt{3}}{3} \\ 0 & \frac{\sqrt{6}}{3} & \frac{\sqrt{3}}{3} \end{pmatrix}$.
- Canonical form: $Q = -y_1^2 - y_2^2 + 2y_3^2$.
- Rank = 3, index = 1, signature = -1; nature: **indefinite**. Type ($Q = 1$): hyperboloid of two sheets.

7. $Q = -2x^2 - 4xy - 2y^2 - z^2$, $A = \begin{pmatrix} -2 & -2 & 0 \\ -2 & -2 & 0 \\ 0 & 0 & -1 \end{pmatrix}$.

- Eigenvalues: -4, -1, 0; orthonormal eigenvectors (columns of P): $P = \begin{pmatrix} \frac{\sqrt{2}}{2} & 0 & -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & 0 & \frac{\sqrt{2}}{2} \\ 0 & 1 & 0 \end{pmatrix}$.

- Canonical form: $Q = -4y_1^2 - y_2^2$.
- Rank = 2, index = 0, signature = −2; nature: **negative semi-definite**. Type ($Q = 1$): no real points; $Q = -1$ is an elliptic cylinder.

8. $Q = 2x^2 - 2xz + 2y^2 + 2z^2$, $A = \begin{pmatrix} 2 & 0 & -1 \\ 0 & 2 & 0 \\ -1 & 0 & 2 \end{pmatrix}$.

- Eigenvalues: 1, 2, 3; orthonormal eigenvectors (columns of P): $P = \begin{pmatrix} \frac{\sqrt{2}}{2} & 0 & -\frac{\sqrt{2}}{2} \\ 0 & 1 & 0 \\ \frac{\sqrt{2}}{2} & 0 & \frac{\sqrt{2}}{2} \end{pmatrix}$.
- Canonical form: $Q = y_1^2 + 2y_2^2 + 3y_3^2$.
- Rank = 3, index = 3, signature = 3; nature: **positive definite**. Type ($Q = 1$): ellipsoid.

9. $Q = -2x^2 + 2xz - 2y^2 - 2z^2$, $A = \begin{pmatrix} -2 & 0 & 1 \\ 0 & -2 & 0 \\ 1 & 0 & -2 \end{pmatrix}$.

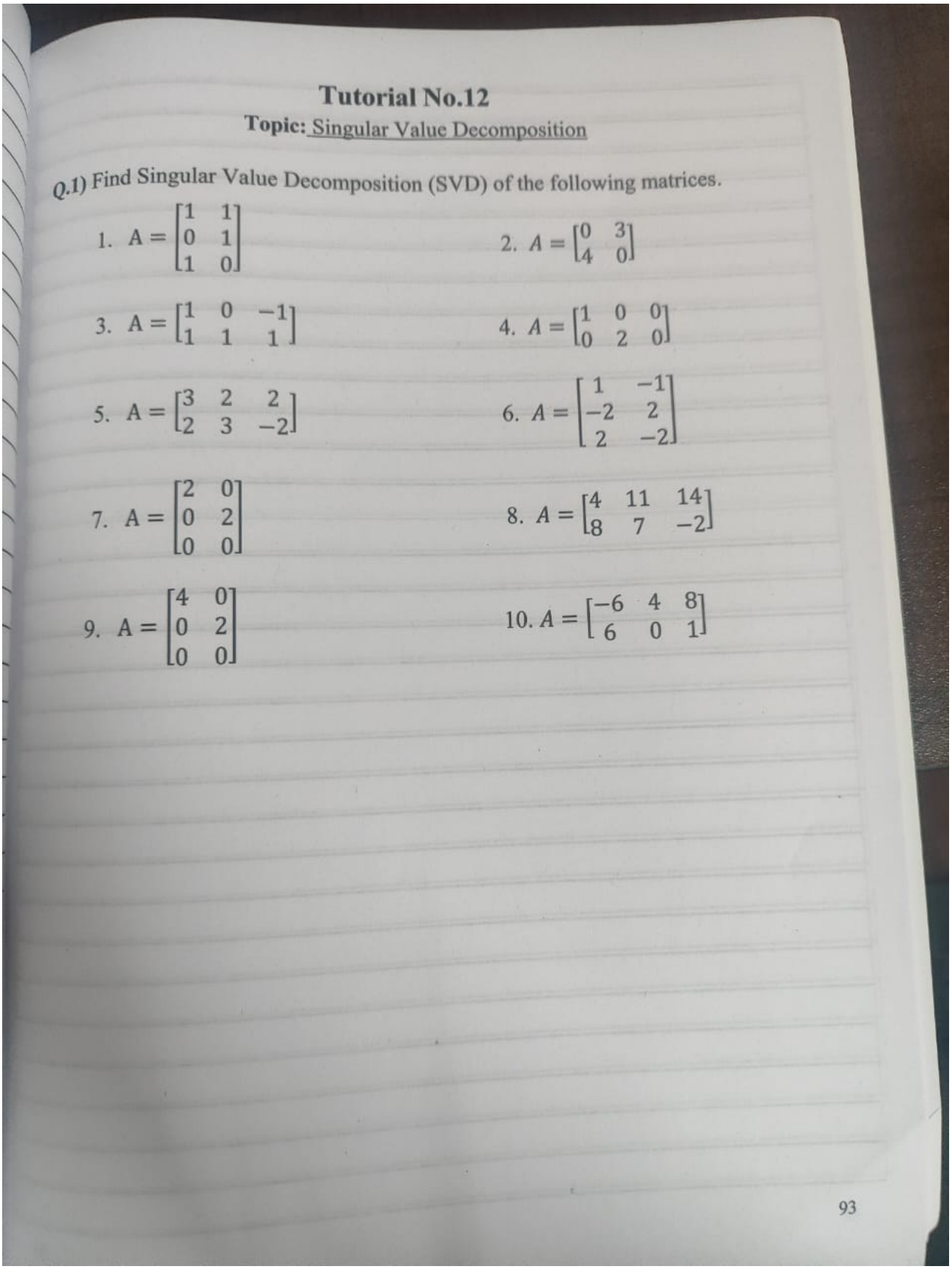
- Eigenvalues: −3, −2, −1; orthonormal eigenvectors (columns of P): $P = \begin{pmatrix} -\frac{\sqrt{2}}{2} & 0 & \frac{\sqrt{2}}{2} \\ 0 & 1 & 0 \\ \frac{\sqrt{2}}{2} & 0 & \frac{\sqrt{2}}{2} \end{pmatrix}$.
- Canonical form: $Q = -3y_1^2 - 2y_2^2 - y_3^2$.
- Rank = 3, index = 0, signature = −3; nature: **negative definite**. Type ($Q = 1$): no real points (imaginary ellipsoid); $Q = -1$ is an ellipsoid.

10. $Q = 2x^2 - 2xy + 2y^2$, $A = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$.

- Eigenvalues: 1, 3; orthonormal eigenvectors (columns of P): $P = \begin{pmatrix} \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix}$.
- Canonical form: $Q = y_1^2 + 3y_2^2$.
- Rank = 2, index = 2, signature = 2; nature: **positive definite**. Type ($Q = 1$): ellipse.

Tutorial No. 12: Singular Value Decomposition

Questions



Tutorial No.12

Topic: Singular Value Decomposition

Q.1) Find Singular Value Decomposition (SVD) of the following matrices.

1. $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}$
2. $A = \begin{bmatrix} 0 & 3 \\ 4 & 0 \end{bmatrix}$
3. $A = \begin{bmatrix} 1 & 0 & -1 \\ 1 & 1 & 1 \end{bmatrix}$
4. $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \end{bmatrix}$
5. $A = \begin{bmatrix} 3 & 2 & 2 \\ 2 & 3 & -2 \end{bmatrix}$
6. $A = \begin{bmatrix} 1 & -1 \\ -2 & 2 \\ 2 & -2 \end{bmatrix}$
7. $A = \begin{bmatrix} 2 & 0 \\ 0 & 2 \\ 0 & 0 \end{bmatrix}$
8. $A = \begin{bmatrix} 4 & 11 & 14 \\ 8 & 7 & -2 \end{bmatrix}$
9. $A = \begin{bmatrix} 4 & 0 \\ 0 & 2 \\ 0 & 0 \end{bmatrix}$
10. $A = \begin{bmatrix} -6 & 4 & 8 \\ 6 & 0 & 1 \end{bmatrix}$

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Solutions — Tutorial No. 12

Q.1 Singular value decomposition $A = U\Sigma V^T$

Method: eigen-decompose A^TA (or AA^T if that is smaller): its eigenvalues are σ_i^2; orthonormal eigenvectors give V (columns $\mathbf{v}_i$); $\mathbf{u}_i = A\mathbf{v}_i/\sigma_i$ for $\sigma_i > 0$, completed to an orthonormal basis of $\mathbb{R}^m$ for the remaining columns of U. Σ has $\sigma_1 \geq \sigma_2 \geq \ldots$ on its diagonal.

(1) $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \\ 1 & 0 \end{pmatrix}$

- $A^TA = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ has eigenvalues 3, 1; singular values $\sigma_i = \sqrt{\lambda_i}$: $\sqrt{3}$, 1.
- $V = \begin{pmatrix} \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix}$, $U = \begin{pmatrix} \frac{\sqrt{6}}{3} & 0 & -\frac{\sqrt{3}}{3} \\ \frac{\sqrt{6}}{6} & \frac{\sqrt{2}}{2} & \frac{\sqrt{3}}{3} \\ \frac{\sqrt{6}}{6} & -\frac{\sqrt{2}}{2} & \frac{\sqrt{3}}{3} \end{pmatrix}$, $\Sigma = \begin{pmatrix} \sqrt{3} & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}$.
- $A = U\Sigma V^T$ (checked by multiplying out).

(2) $A = \begin{pmatrix} 0 & 3 \\ 4 & 0 \end{pmatrix}$

- $A^TA = \begin{pmatrix} 16 & 0 \\ 0 & 9 \end{pmatrix}$ has eigenvalues 16, 9; singular values $\sigma_i = \sqrt{\lambda_i}$: 4, 3.
- $V = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, $U = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, $\Sigma = \begin{pmatrix} 4 & 0 \\ 0 & 3 \end{pmatrix}$.
- $A = U\Sigma V^T$ (checked by multiplying out).

(3) $A = \begin{pmatrix} 1 & 0 & -1 \\ 1 & 1 & 1 \end{pmatrix}$

- $AA^T = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$ has eigenvalues 3, 2; singular values $\sigma_i = \sqrt{\lambda_i}$: $\sqrt{3}$, $\sqrt{2}$.
- $V = \begin{pmatrix} \frac{\sqrt{3}}{3} & \frac{\sqrt{2}}{2} & \frac{\sqrt{6}}{6} \\ \frac{\sqrt{3}}{3} & 0 & -\frac{\sqrt{6}}{3} \\ \frac{\sqrt{3}}{3} & -\frac{\sqrt{2}}{2} & \frac{\sqrt{6}}{6} \end{pmatrix}$, $U = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, $\Sigma = \begin{pmatrix} \sqrt{3} & 0 & 0 \\ 0 & \sqrt{2} & 0 \end{pmatrix}$.
- $A = U\Sigma V^T$ (checked by multiplying out).

(4) $A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \end{pmatrix}$

- $AA^T = \begin{pmatrix} 1 & 0 \\ 0 & 4 \end{pmatrix}$ has eigenvalues 4, 1; singular values $\sigma_i = \sqrt{\lambda_i}$: 2, 1.
- $V = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$, $U = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, $\Sigma = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$.
- $A = U\Sigma V^T$ (checked by multiplying out).

(5) $A = \begin{pmatrix} 3 & 2 & 2 \\ 2 & 3 & -2 \end{pmatrix}$

- $AA^T = \begin{pmatrix} 17 & 8 \\ 8 & 17 \end{pmatrix}$ has eigenvalues 25, 9; singular values $\sigma_i = \sqrt{\lambda_i}$: 5, 3.

- $V = \begin{pmatrix} \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{6} & -\frac{2}{3} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{6} & \frac{2}{3} \\ 0 & -\frac{2\sqrt{2}}{3} & \frac{1}{3} \end{pmatrix}$, $U = \begin{pmatrix} \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix}$, $\Sigma = \begin{pmatrix} 5 & 0 & 0 \\ 0 & 3 & 0 \end{pmatrix}$.

- $A = U\Sigma V^T$ (checked by multiplying out).

(6) $A = \begin{pmatrix} 1 & -1 \\ -2 & 2 \\ 2 & -2 \end{pmatrix}$

- $A^T A = \begin{pmatrix} 9 & -9 \\ -9 & 9 \end{pmatrix}$ has eigenvalues 18, 0; singular values $\sigma_i = \sqrt{\lambda_i}$: $3\sqrt{2}$, 0.

- $V = \begin{pmatrix} -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix}$, $U = \begin{pmatrix} -\frac{1}{3} & \frac{2\sqrt{5}}{5} & -\frac{2\sqrt{5}}{15} \\ \frac{2}{3} & \frac{\sqrt{5}}{5} & \frac{4\sqrt{5}}{15} \\ -\frac{2}{3} & 0 & \frac{\sqrt{5}}{3} \end{pmatrix}$, $\Sigma = \begin{pmatrix} 3\sqrt{2} & 0 \\ 0 & 0 \\ 0 & 0 \end{pmatrix}$.

- $A = U\Sigma V^T$ (checked by multiplying out).

(7) $A = \begin{pmatrix} 2 & 0 \\ 0 & 2 \\ 0 & 0 \end{pmatrix}$

- $A^T A = \begin{pmatrix} 4 & 0 \\ 0 & 4 \end{pmatrix}$ has eigenvalues 4, 4; singular values $\sigma_i = \sqrt{\lambda_i}$: 2, 2.

- $V = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, $U = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$, $\Sigma = \begin{pmatrix} 2 & 0 \\ 0 & 2 \\ 0 & 0 \end{pmatrix}$.

- $A = U\Sigma V^T$ (checked by multiplying out).

(8) $A = \begin{pmatrix} 4 & 11 & 14 \\ 8 & 7 & -2 \end{pmatrix}$

- $AA^T = \begin{pmatrix} 333 & 81 \\ 81 & 117 \end{pmatrix}$ has eigenvalues 360, 90; singular values $\sigma_i = \sqrt{\lambda_i}$: $6\sqrt{10}$, $3\sqrt{10}$.

- $V = \begin{pmatrix} \frac{1}{3} & \frac{2}{3} & \frac{2}{3} \\ \frac{2}{3} & \frac{1}{3} & -\frac{2}{3} \\ \frac{2}{3} & -\frac{2}{3} & \frac{1}{3} \end{pmatrix}$, $U = \begin{pmatrix} \frac{3\sqrt{10}}{10} & -\frac{\sqrt{10}}{10} \\ \frac{\sqrt{10}}{10} & \frac{3\sqrt{10}}{10} \end{pmatrix}$, $\Sigma = \begin{pmatrix} 6\sqrt{10} & 0 & 0 \\ 0 & 3\sqrt{10} & 0 \end{pmatrix}$.

- $A = U\Sigma V^T$ (checked by multiplying out).

(9) $A = \begin{pmatrix} 4 & 0 \\ 0 & 2 \\ 0 & 0 \end{pmatrix}$

- $A^T A = \begin{pmatrix} 16 & 0 \\ 0 & 4 \end{pmatrix}$ has eigenvalues 16, 4; singular values $\sigma_i = \sqrt{\lambda_i}$: 4, 2.

- $V = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, $U = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$, $\Sigma = \begin{pmatrix} 4 & 0 \\ 0 & 2 \\ 0 & 0 \end{pmatrix}$.

- $A = U\Sigma V^T$ (checked by multiplying out).

(10) $A = \begin{pmatrix} -6 & 4 & 8 \\ 6 & 0 & 1 \end{pmatrix}$

- $AA^T = \begin{pmatrix} 116 & -28 \\ -28 & 37 \end{pmatrix}$ has eigenvalues $\lambda = 124.917, 28.0825$ (roots of $\lambda^2 - 153\lambda + 3508 = 0$ – not perfect squares), so the singular values are irrational: $\sigma_1 \approx 11.1766,\ \sigma_2 \approx 5.2993$.
- Numerically (4 d.p.), $A = U\Sigma V^T$ with

$$U \approx \begin{pmatrix} -0.9528 & 0.3035 \\ 0.3035 & 0.9528 \end{pmatrix}, \quad \Sigma \approx \begin{pmatrix} 11.1766 & 0.0 & 0.0 \\ 0.0 & 5.2993 & 0.0 \end{pmatrix}, \quad V \approx \begin{pmatrix} 0.6744 & 0.7352 & -0.0675 \\ -0.341 & 0.2291 & -0.9117 \\ -0.6549 & 0.6379 & 0.4052 \end{pmatrix}$$